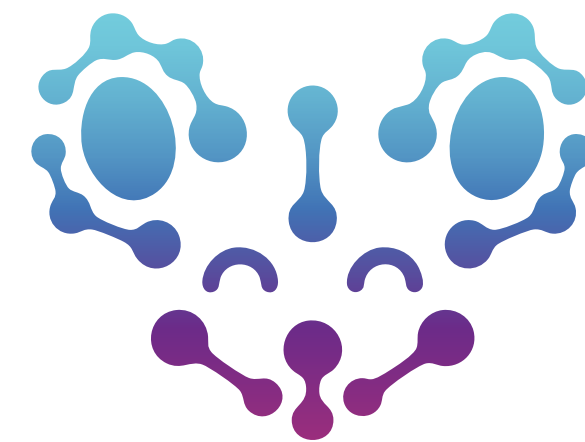


# Measuring Progress in Neuro Foundation Models with BrainWideBench

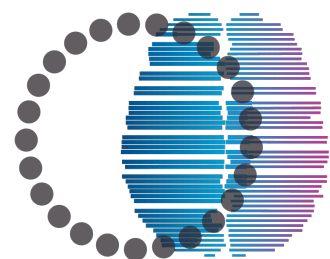


**Alexandre Andre & Shivashriganesh Mahato**

NerDS Lab, University of Pennsylvania

October 5, 2026

IBL Lab Meeting



INTERNATIONAL  
**BRAIN**  
LABORATORY



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- Do the learned representations reflect biological properties?

# Toward Broader Neural Models



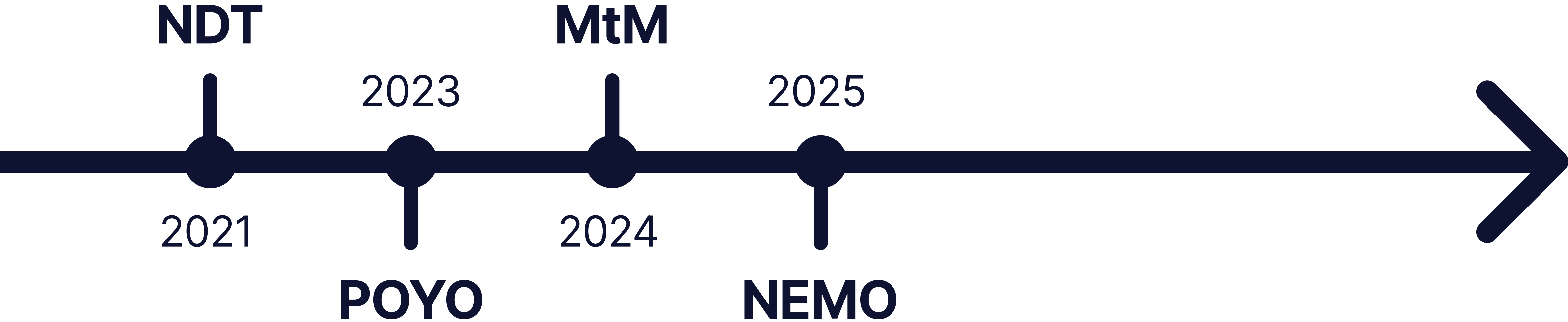
# Toward Broader Neural Models



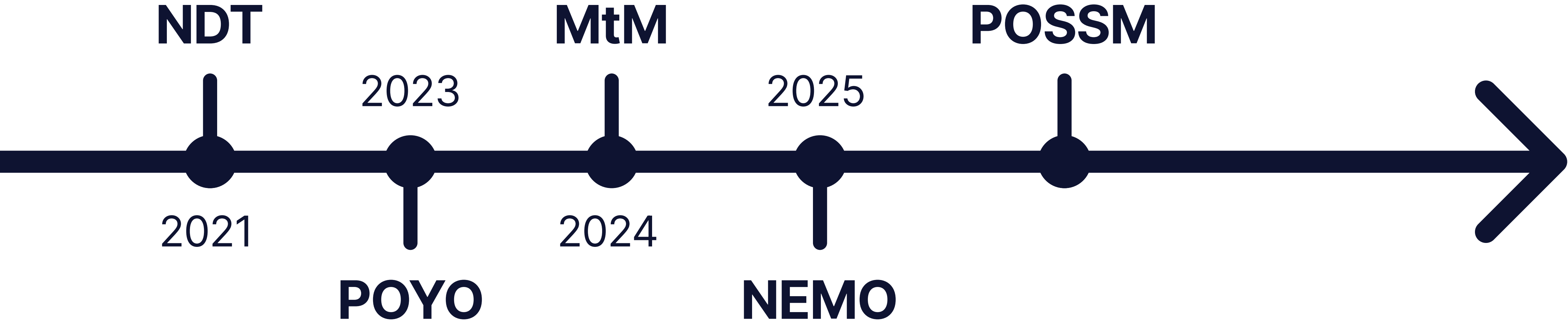
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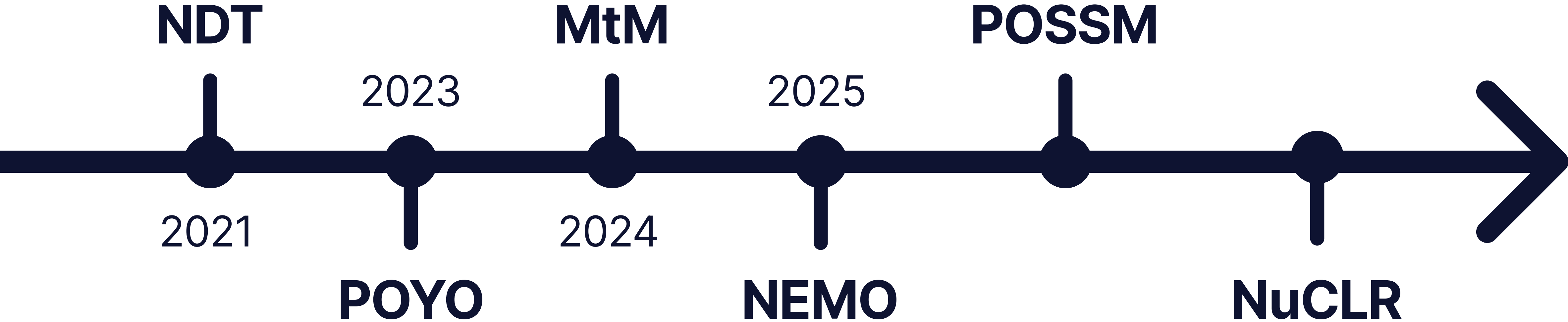
# Toward Broader Neural Models



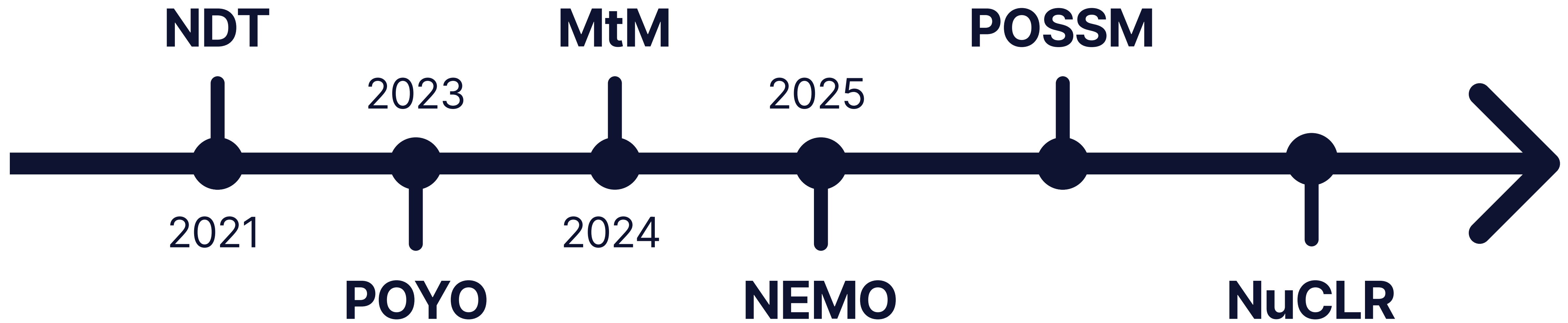
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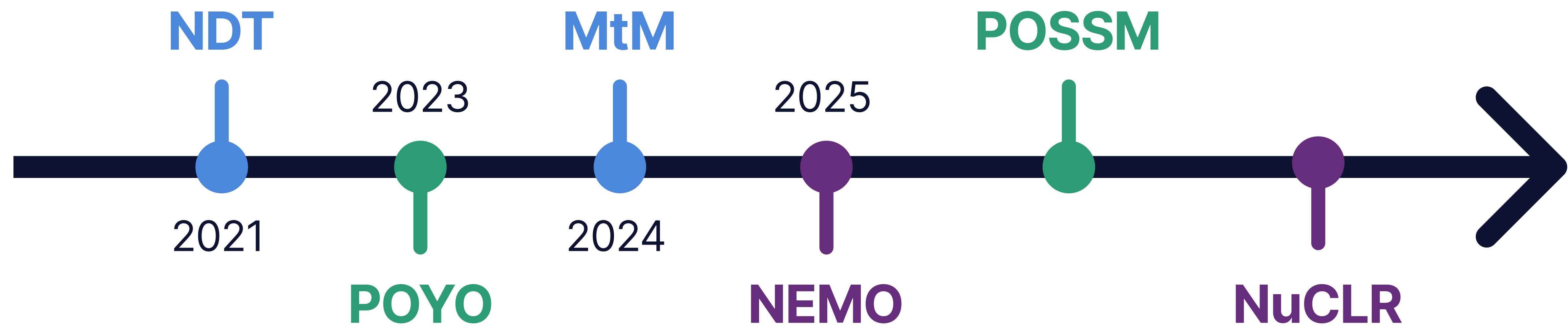


# Toward Broader Neural Models



**Diverse goals:** decoding, forecasting, cell-type & brain region classification, real-time inference

# Models Capture Different Aspects





Dynamics



Behavior



Anatomy

# A General Model Should Do All Three



Dynamics



Behavior



Anatomy



# Measuring Progress

**Problem:** no shared evaluation

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- **Strong baselines:** well-tuned single-session and pretrained models
- **Open & easy to use:** plug in a new model with minimal effort

# BrainWideBench

## Dataset

**139** mice  
**452** sessions

**276** brain regions  
**830k+** unit-hours

## Baseline

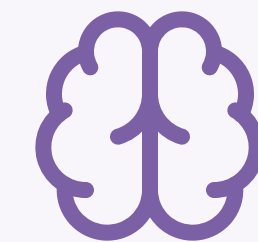
**18** single-session  
**6** pretrained



Dynamics



Behavior

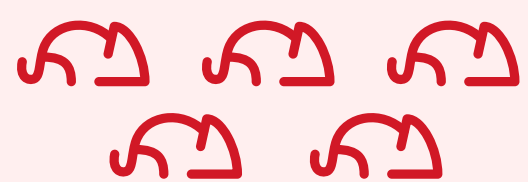


Anatomy

# Common Evaluation Framework

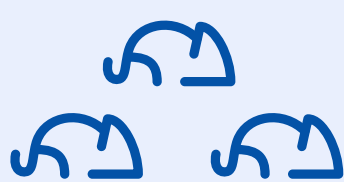
## Pretrain

(126 Mice · 423 Sessions)



## Eval

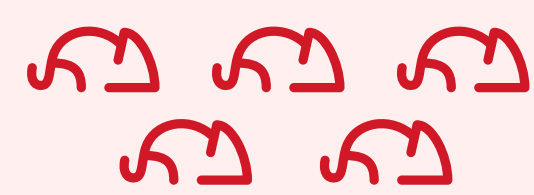
(13 Mice · 29 Sessions)



# Common Evaluation Framework

## Pretrain

(126 Mice · 423 Sessions)



Model

···· Optional



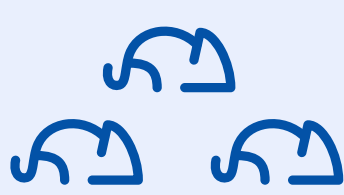
Input



Target

## Eval

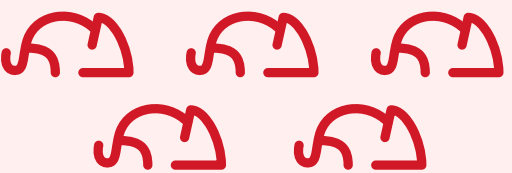
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# Common Evaluation Framework

## Pretrain

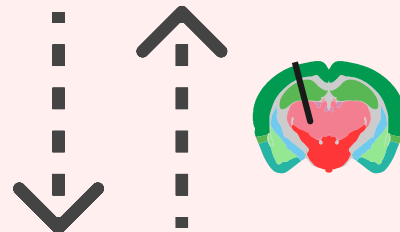
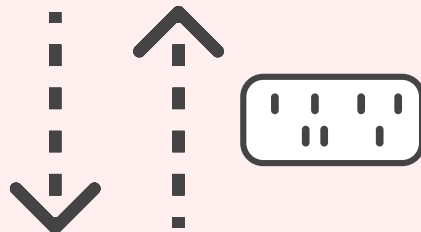
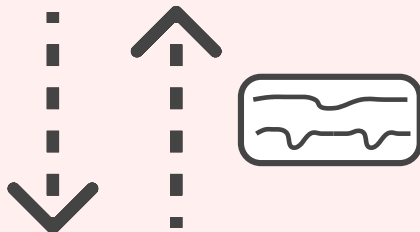
(126 Mice · 423 Sessions)



Behavior

Spikes

Anatomy

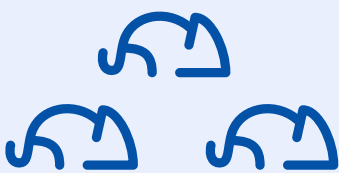


Model

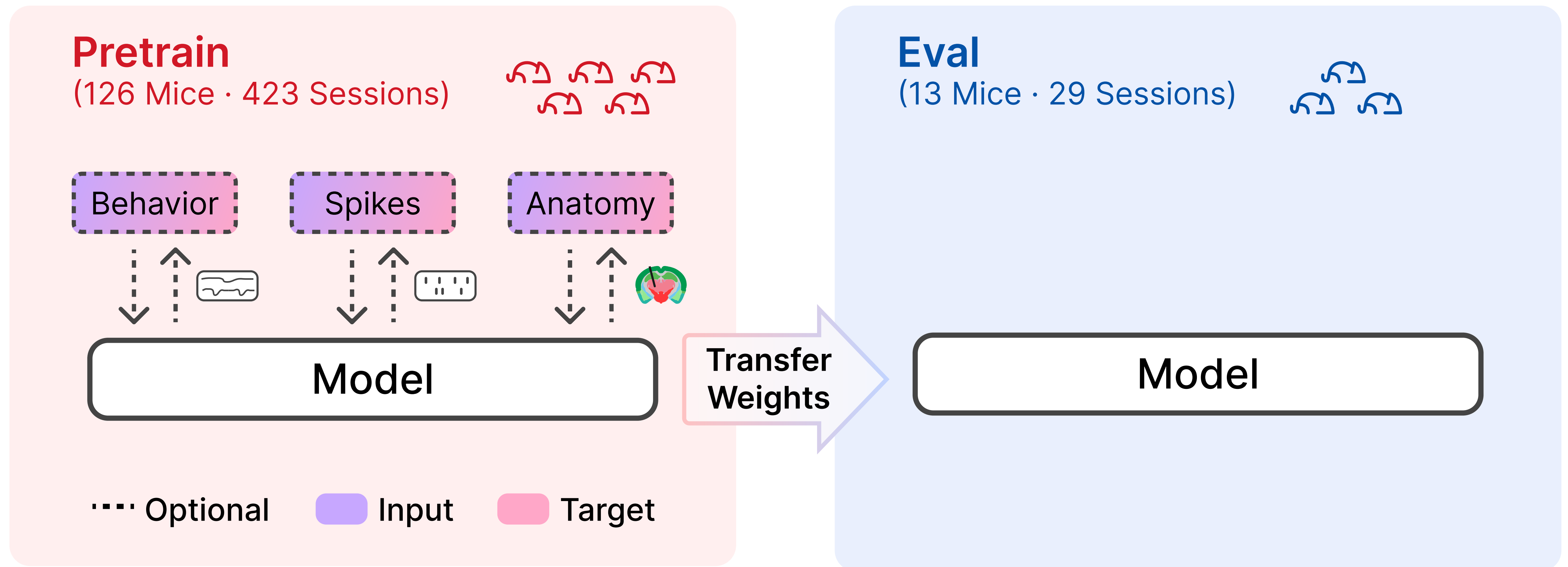
····· Optional      Input      Target

## Eval

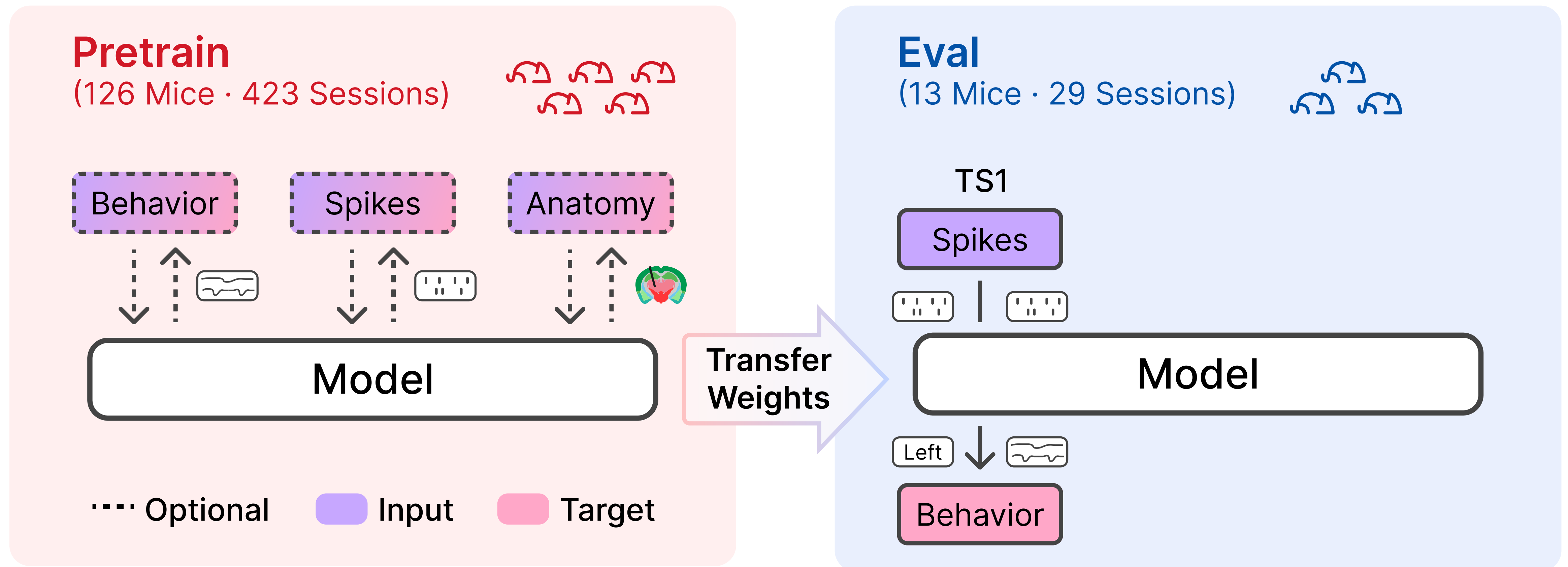
(13 Mice · 29 Sessions)



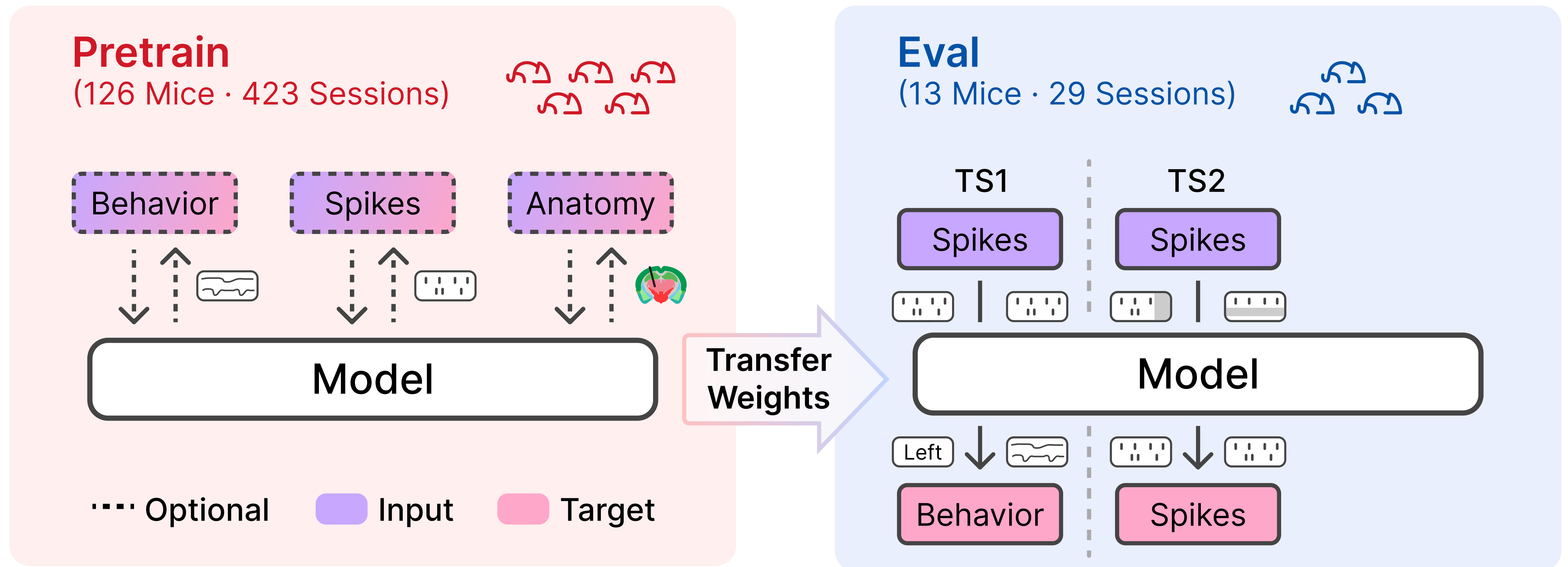
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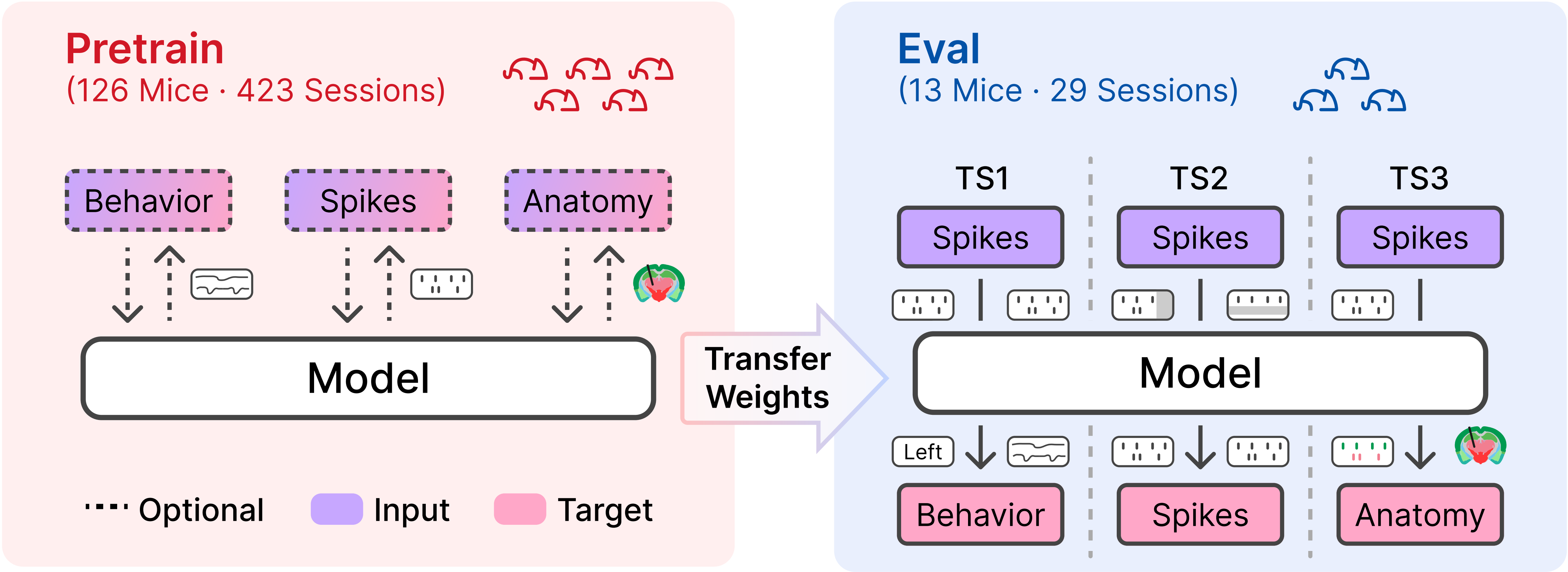
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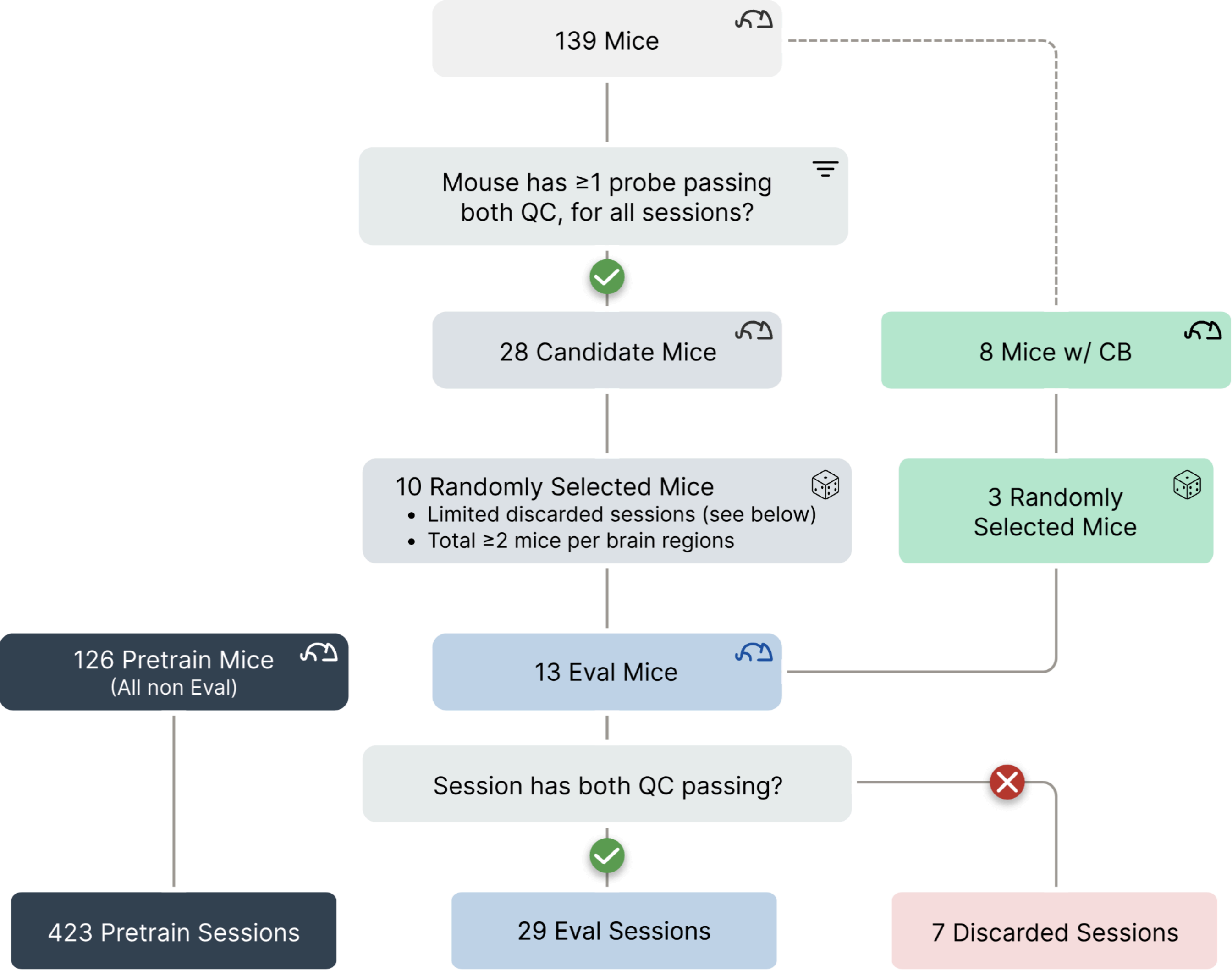


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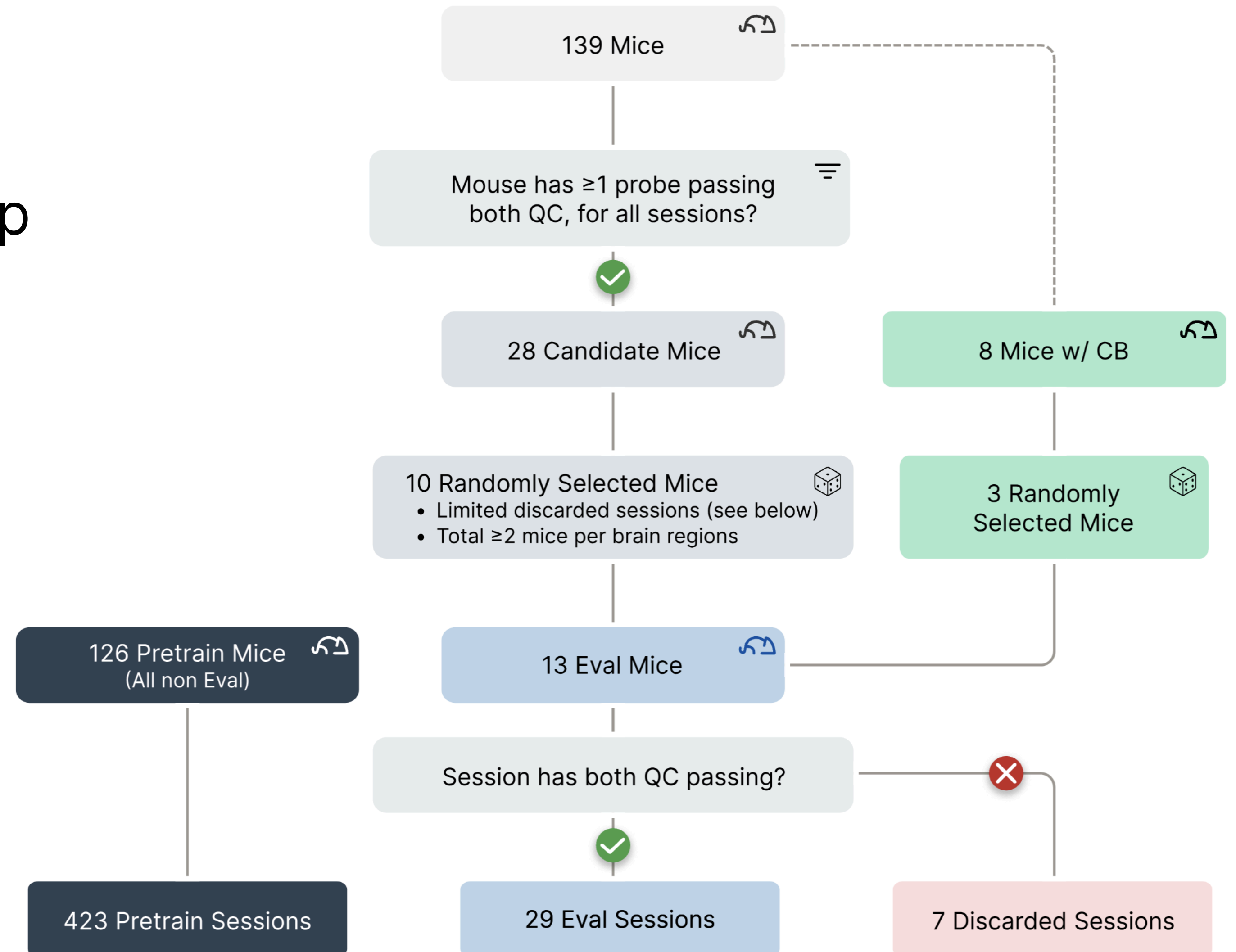
## Curated evaluation set:



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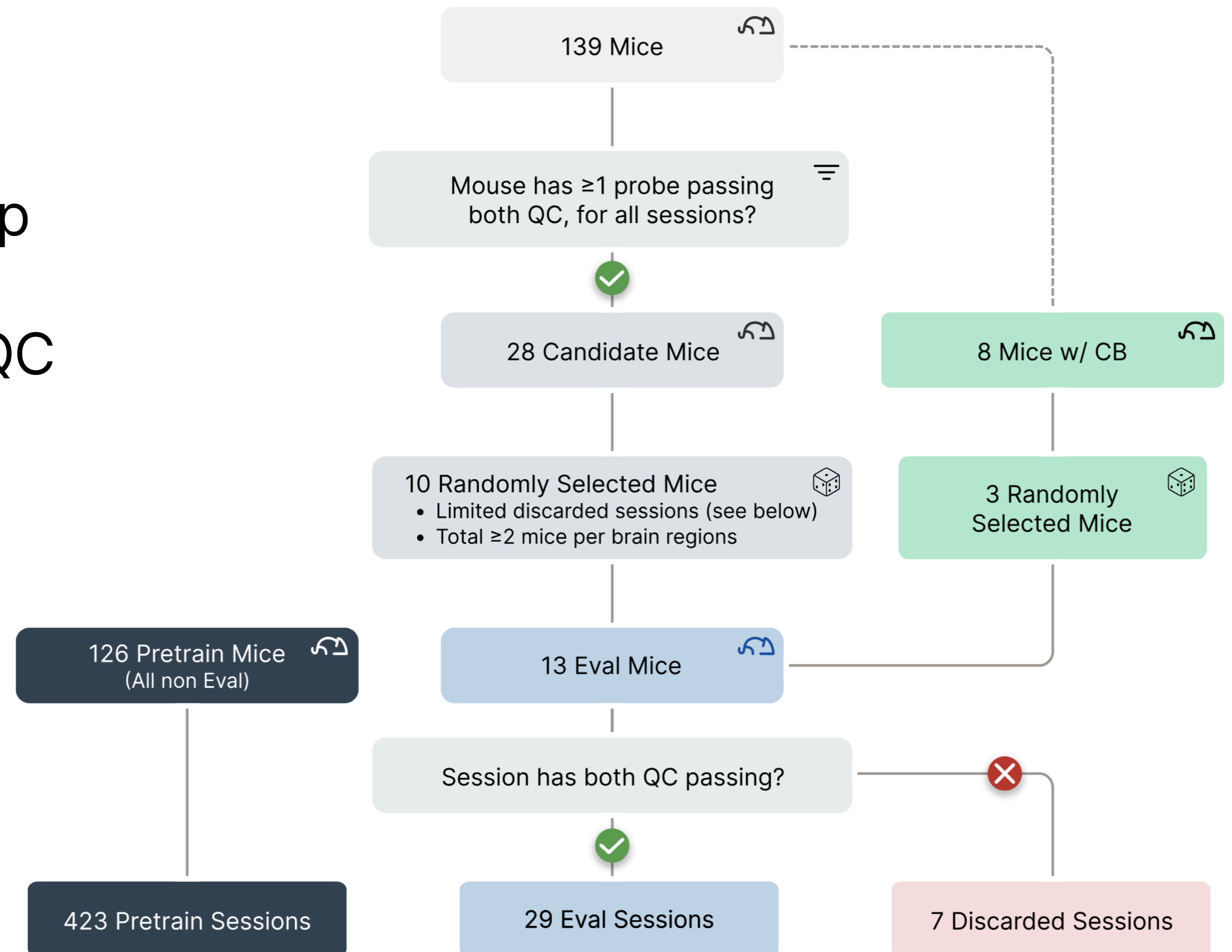
- QC labels
  - neural analysis working group
  - ephys atlas task force



# Common Evaluation Framework

## Curated evaluation set:

- QC labels
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  - ephys atlas task force
  - session-, trial-, probe-level QC



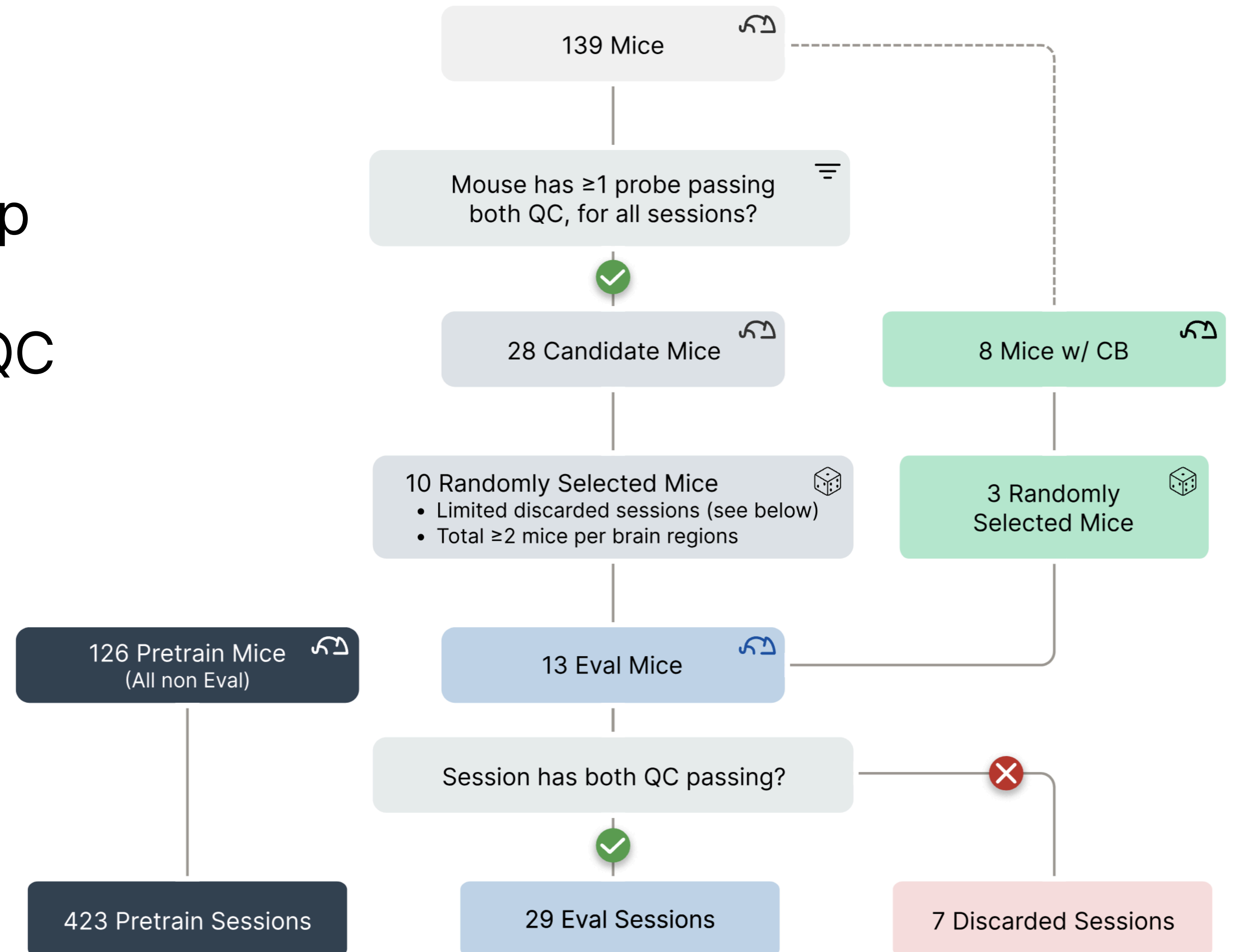
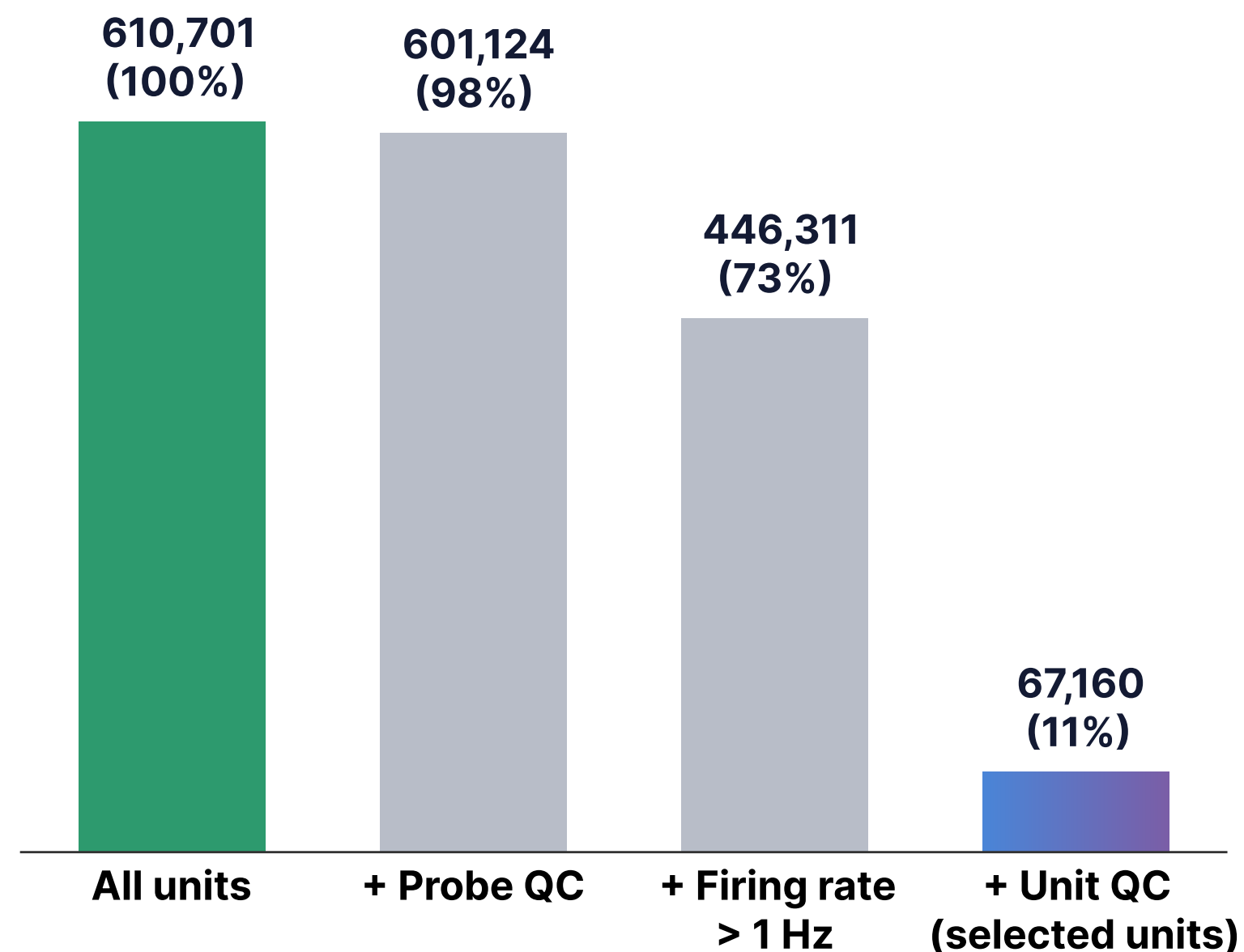
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TS1 uses all units

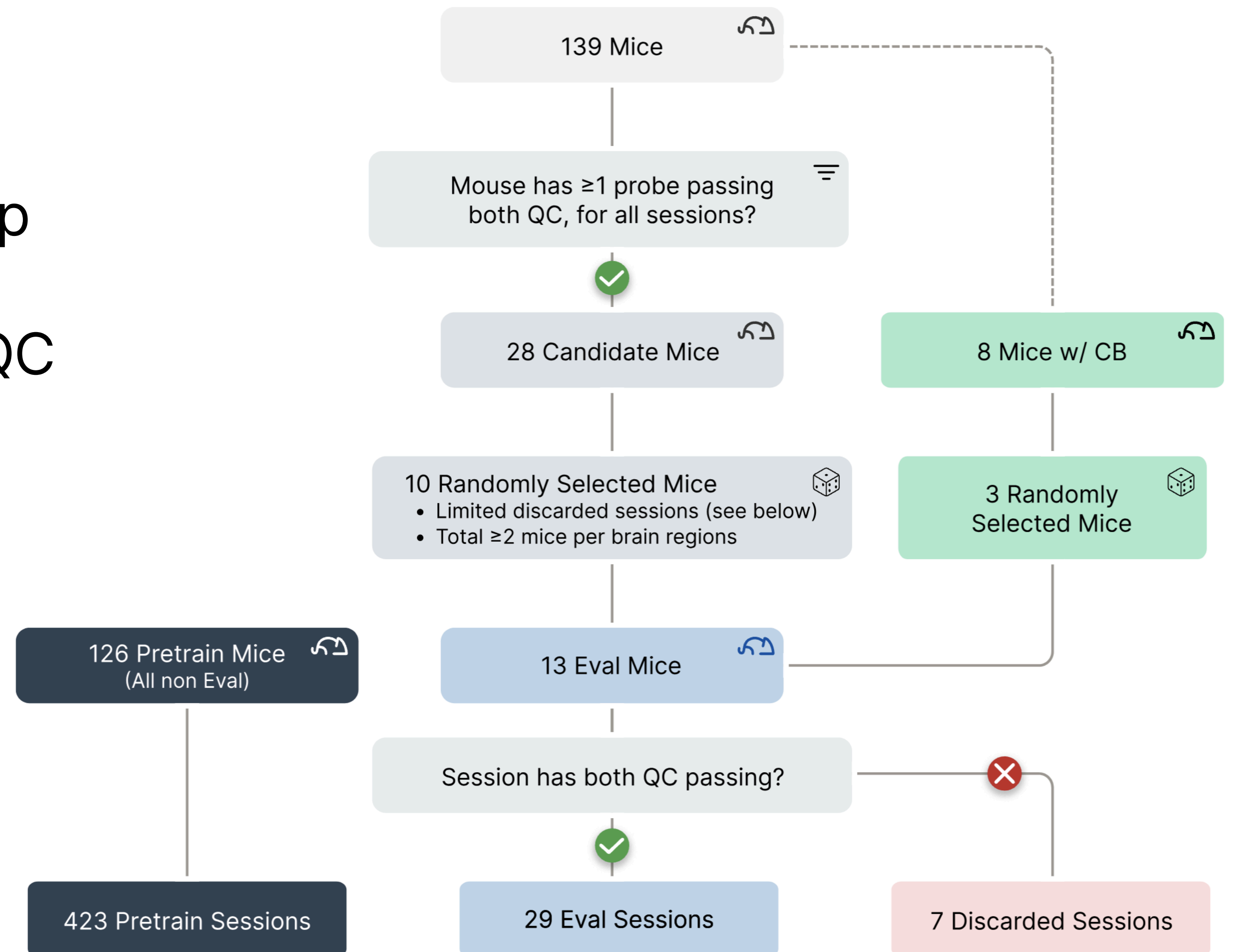
TS2 / TS3  
use selected units only



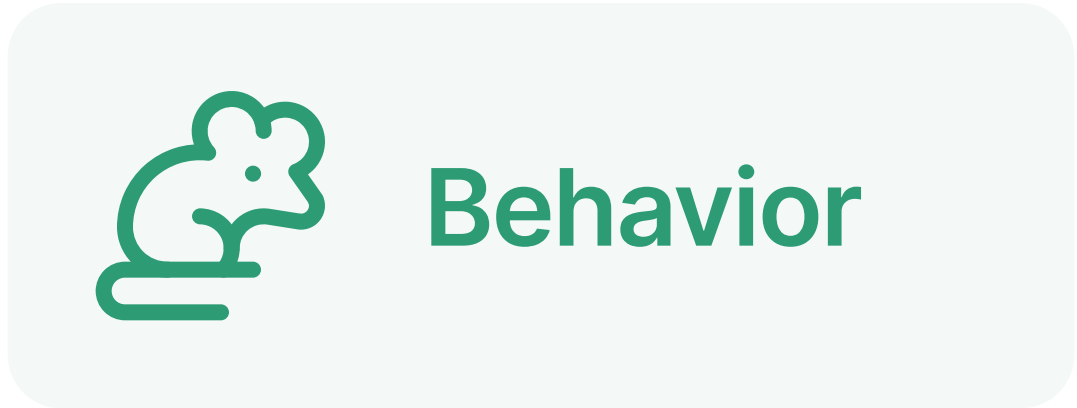
# Common Evaluation Framework

## Curated evaluation set:

- QC labels
  - neural analysis working group
  - ephys atlas task force
  - session-, trial-, probe-level QC
- region distribution

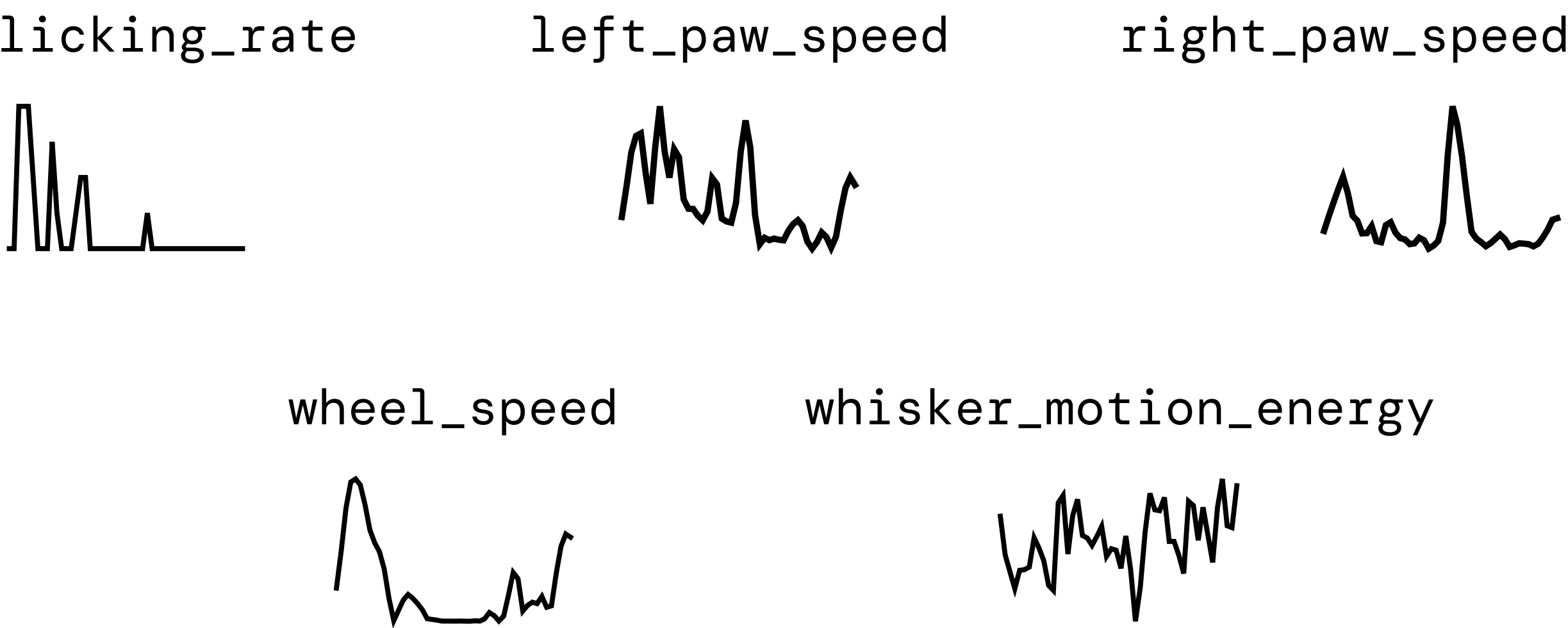


# Evaluation Suites: Behavior Decoding (TS1)



**8 tasks** spanning frame-level regression and sequence-level classification

## TIMESTEP



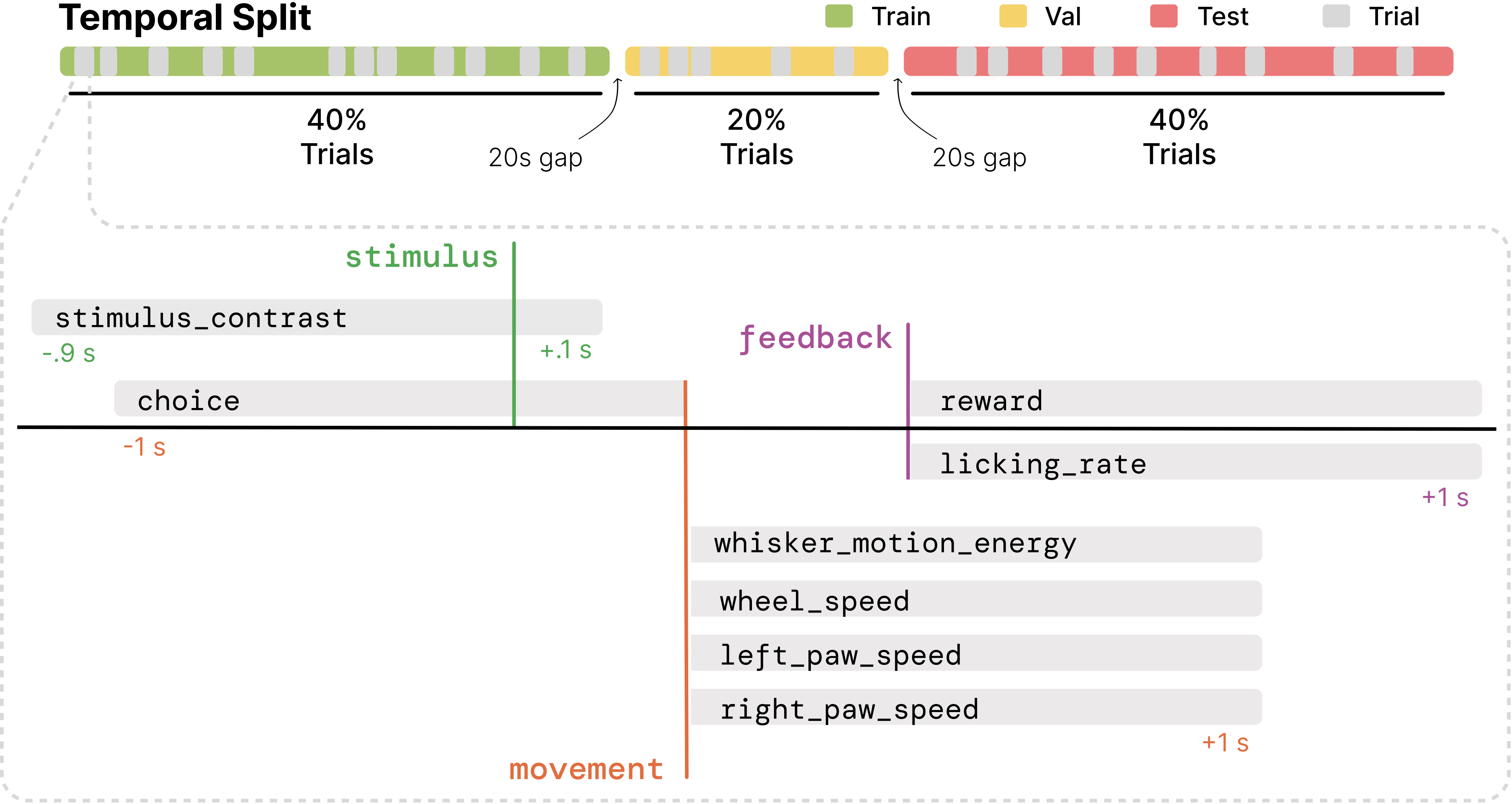
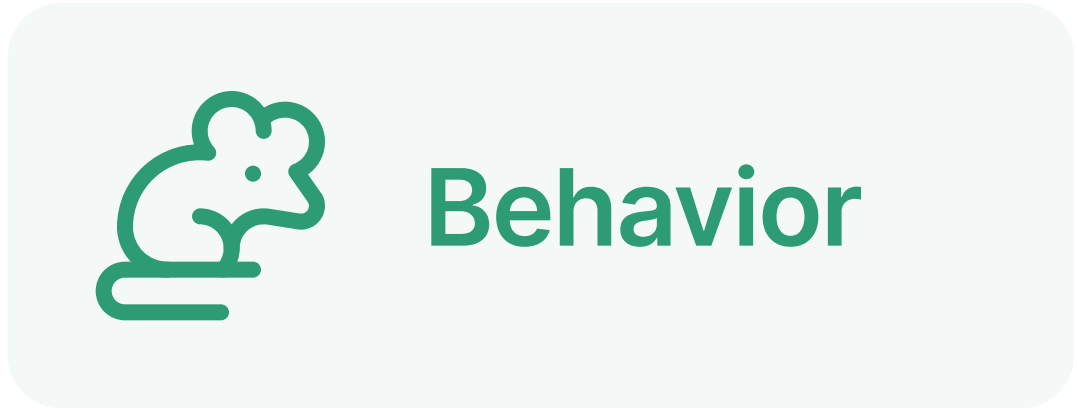
## SEQUENCE

| reward |              | choice |       |
|--------|--------------|--------|-------|
| 0:     | Not rewarded | 0:     | Left  |
| 1:     | Rewarded     | 1:     | Right |

| stimulus_contrast |       |    |       |    |      |
|-------------------|-------|----|-------|----|------|
| 0:                | 0%    | 2: | 12.5% | 4: | 100% |
| 1:                | 6.25% | 3: | 25%   |    |      |

# Evaluation Suites: Behavior Decoding (TS1)



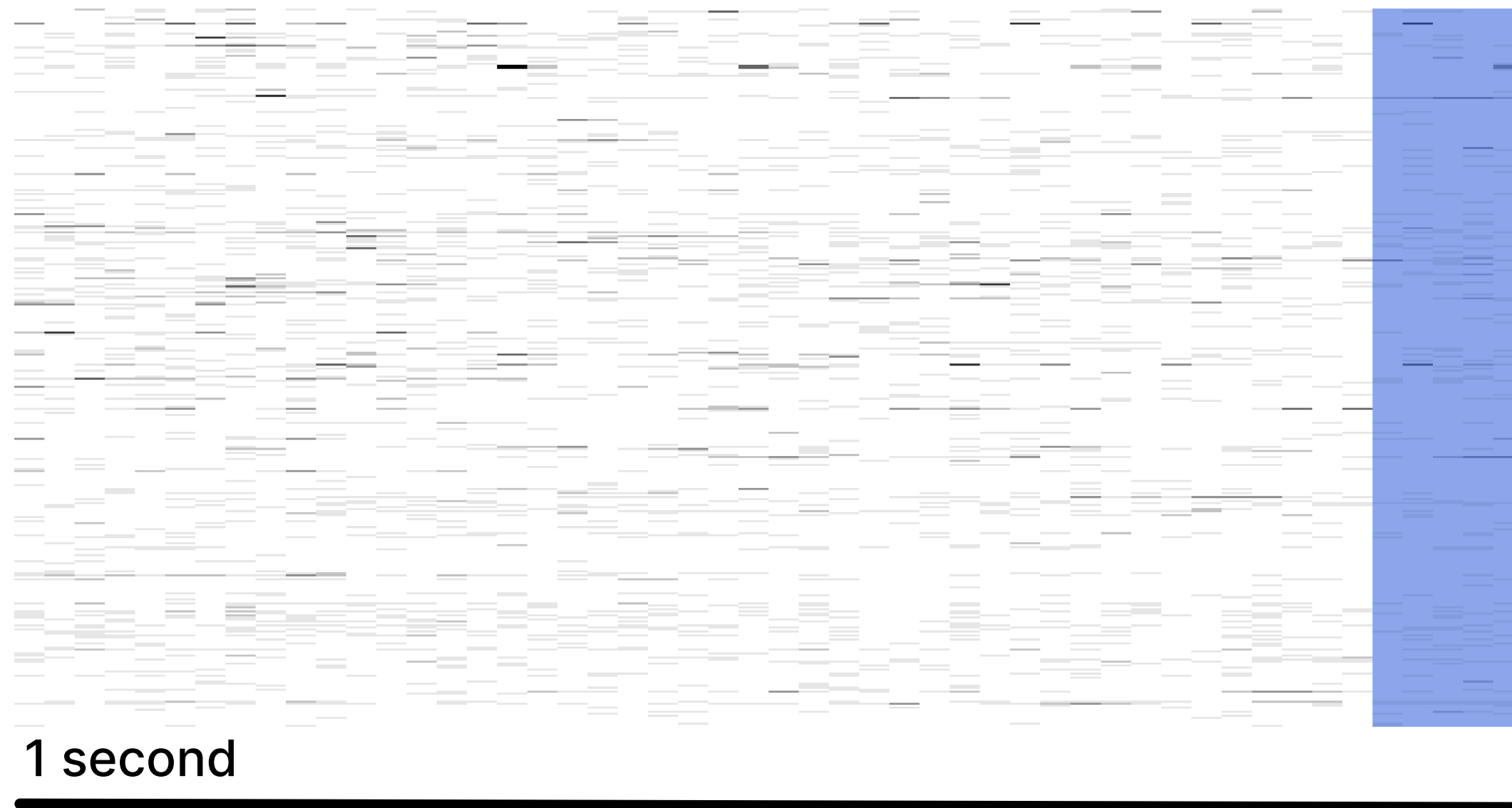
# Evaluation Suites: Neural Activity Prediction (TS2)



**2 tasks** spanning prediction of neural activity across time (**forecasting**) and space (**co-smoothing**)

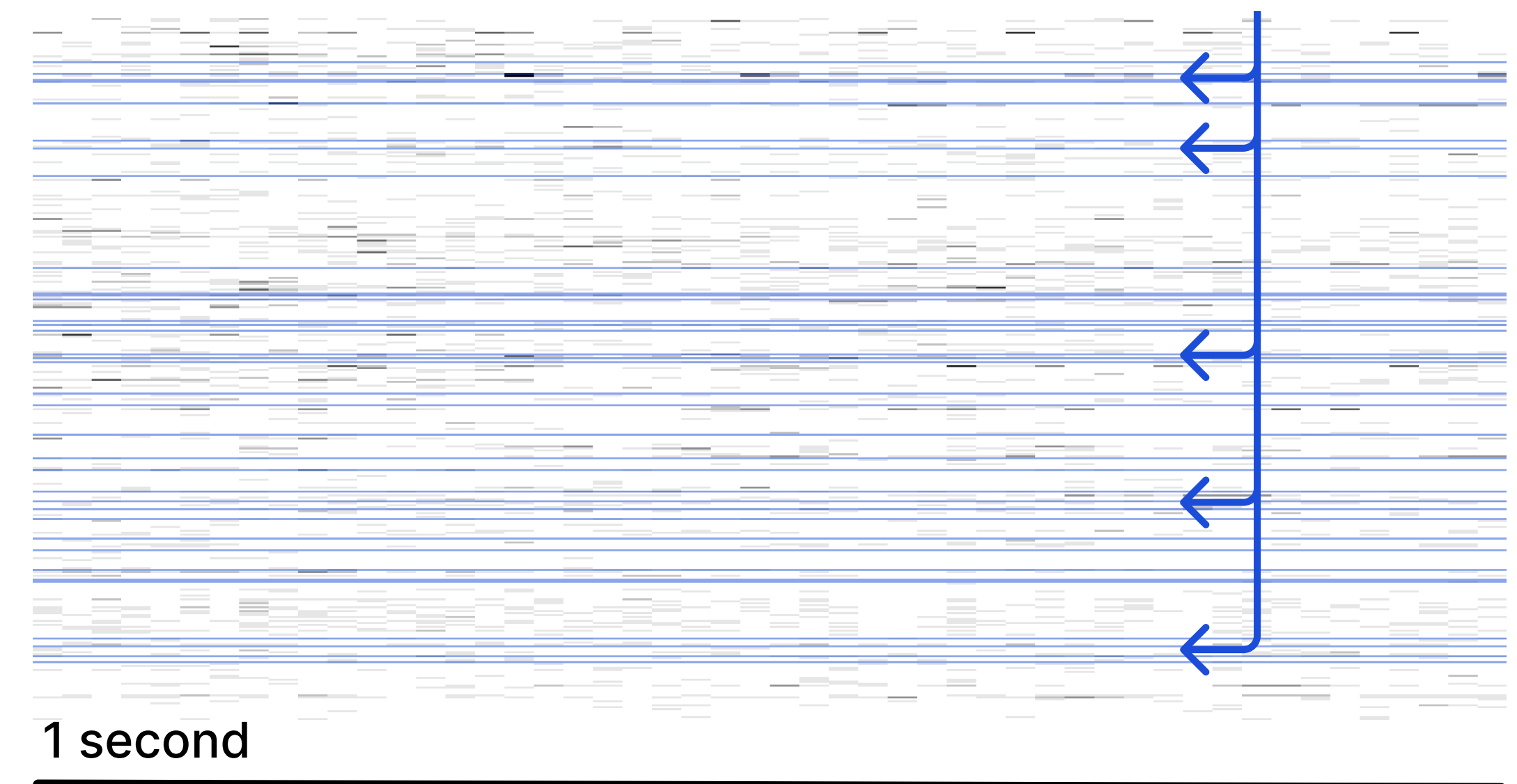
forecasting

10% time = 100 ms



co\_smoothing

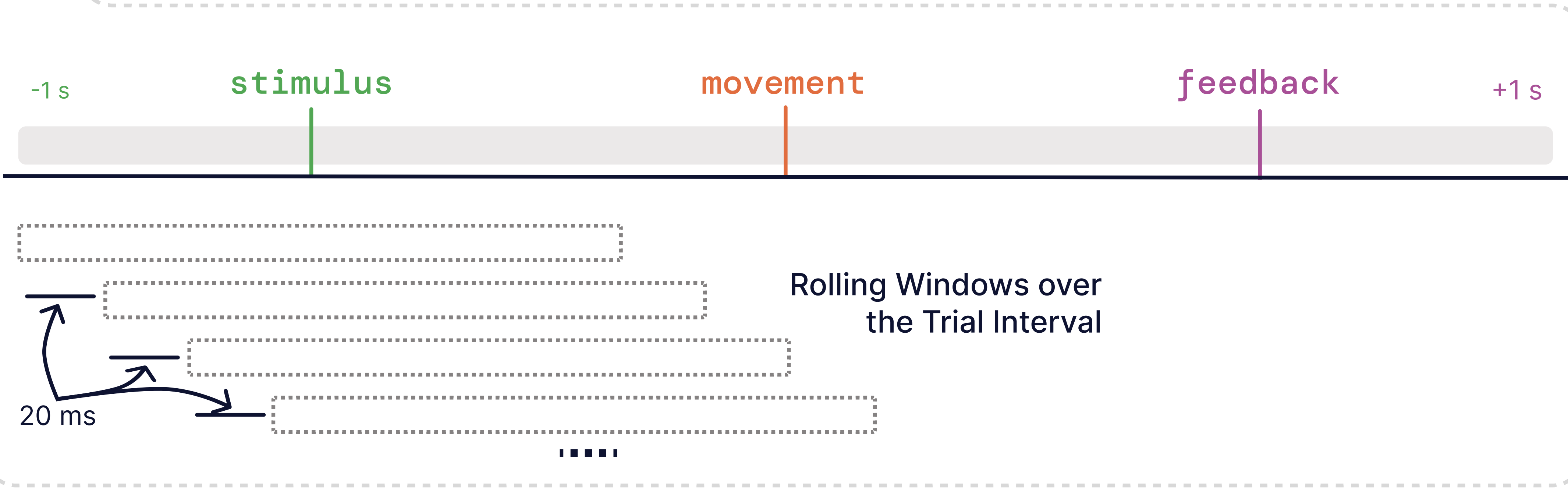
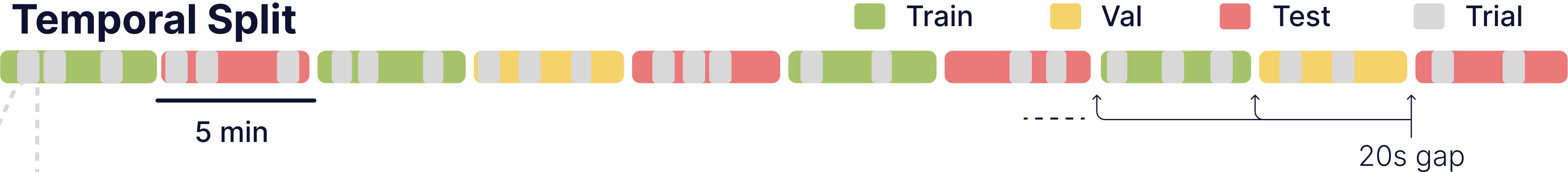
10% units



# Evaluation Suites: Neural Activity Prediction (TS2)



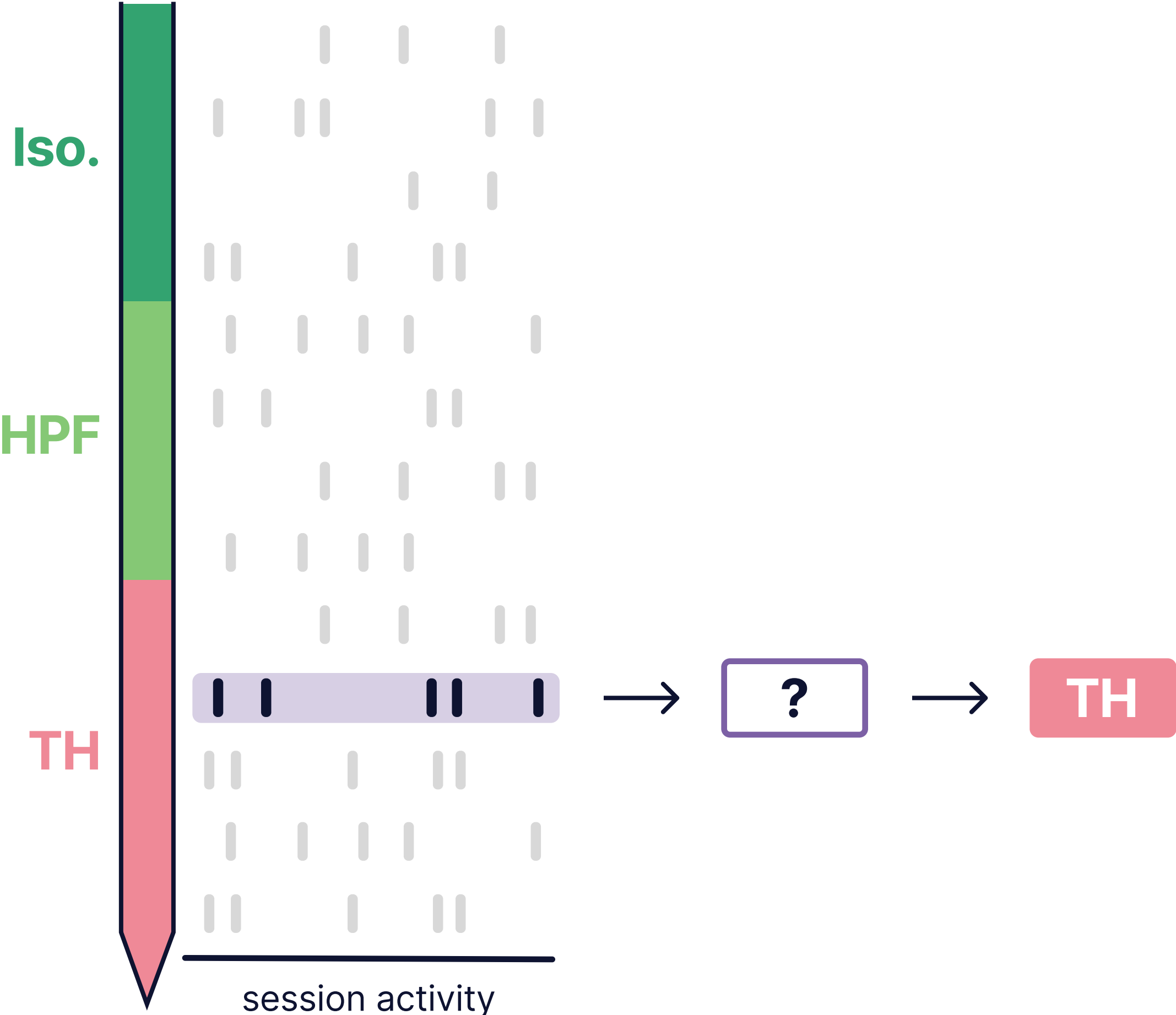
## Temporal Split



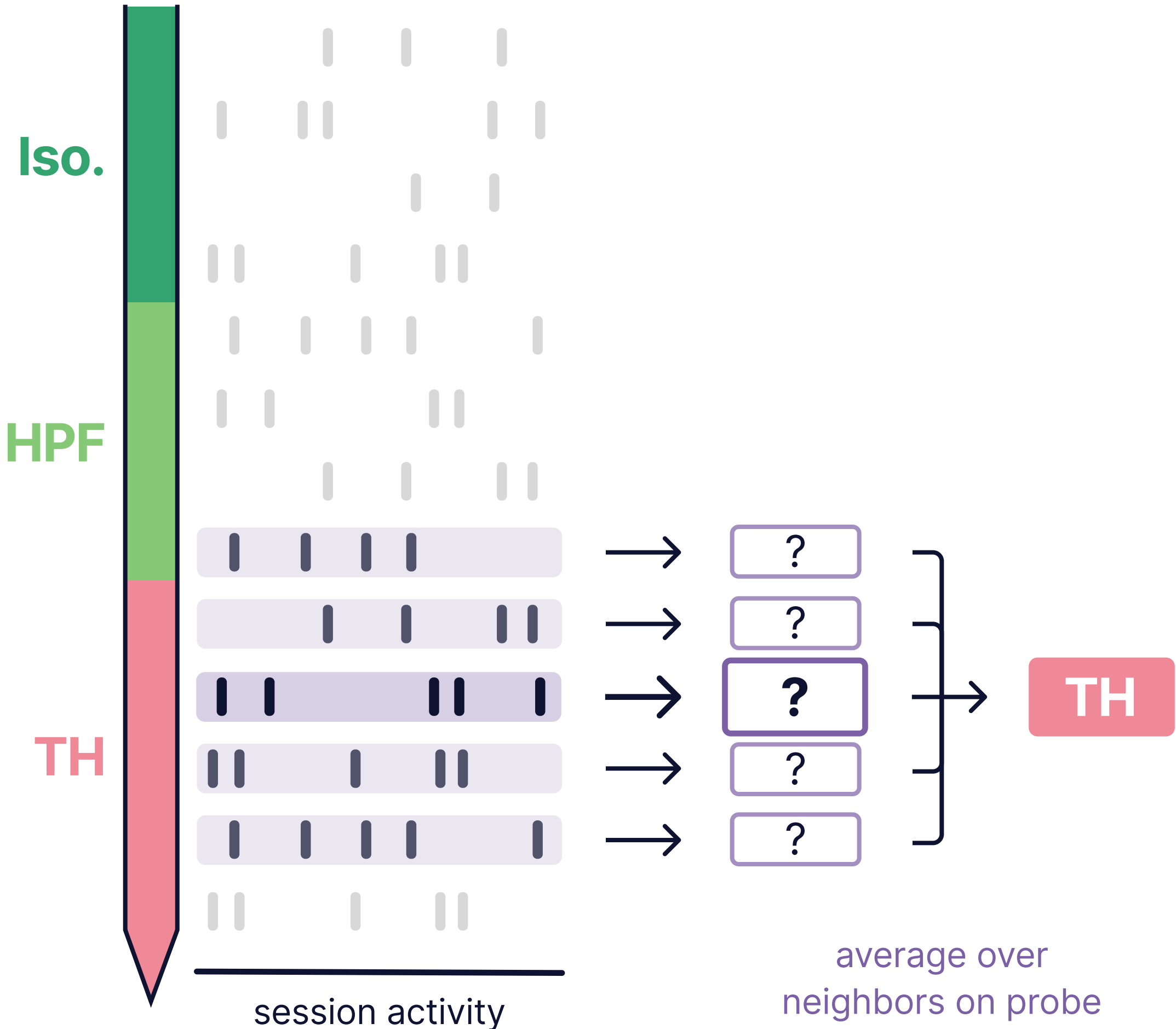
# Evaluation Suites: Brain Region Identity (TS3)



single\_unit



multi\_unit



# Pretrained Baselines Within the Benchmark Framework

 Model Input     Model Target

NDT



POYO · POSSM



MtM



NuCLR · NEMO



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POYO · POSSM



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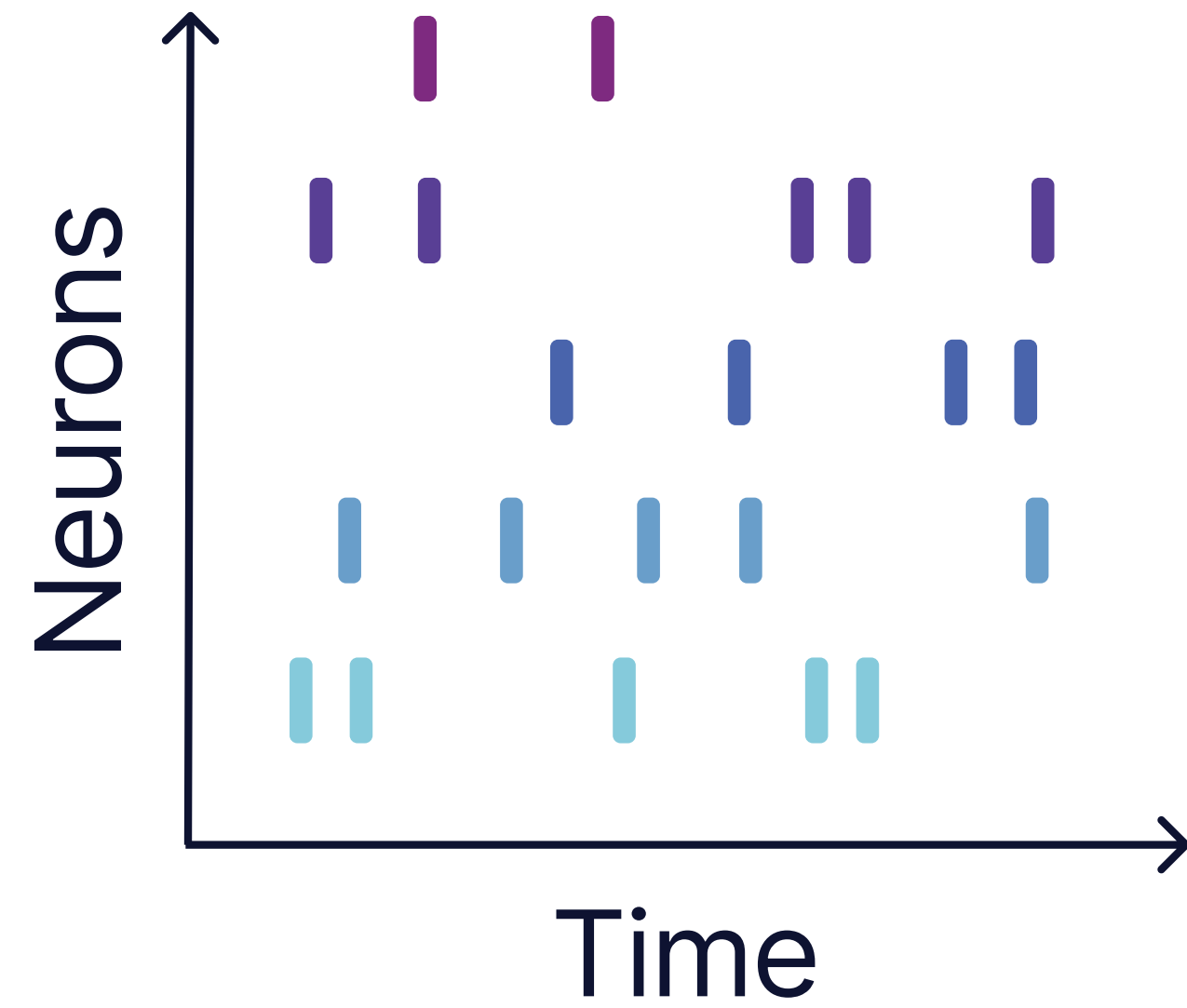


NuCLR · NEMO



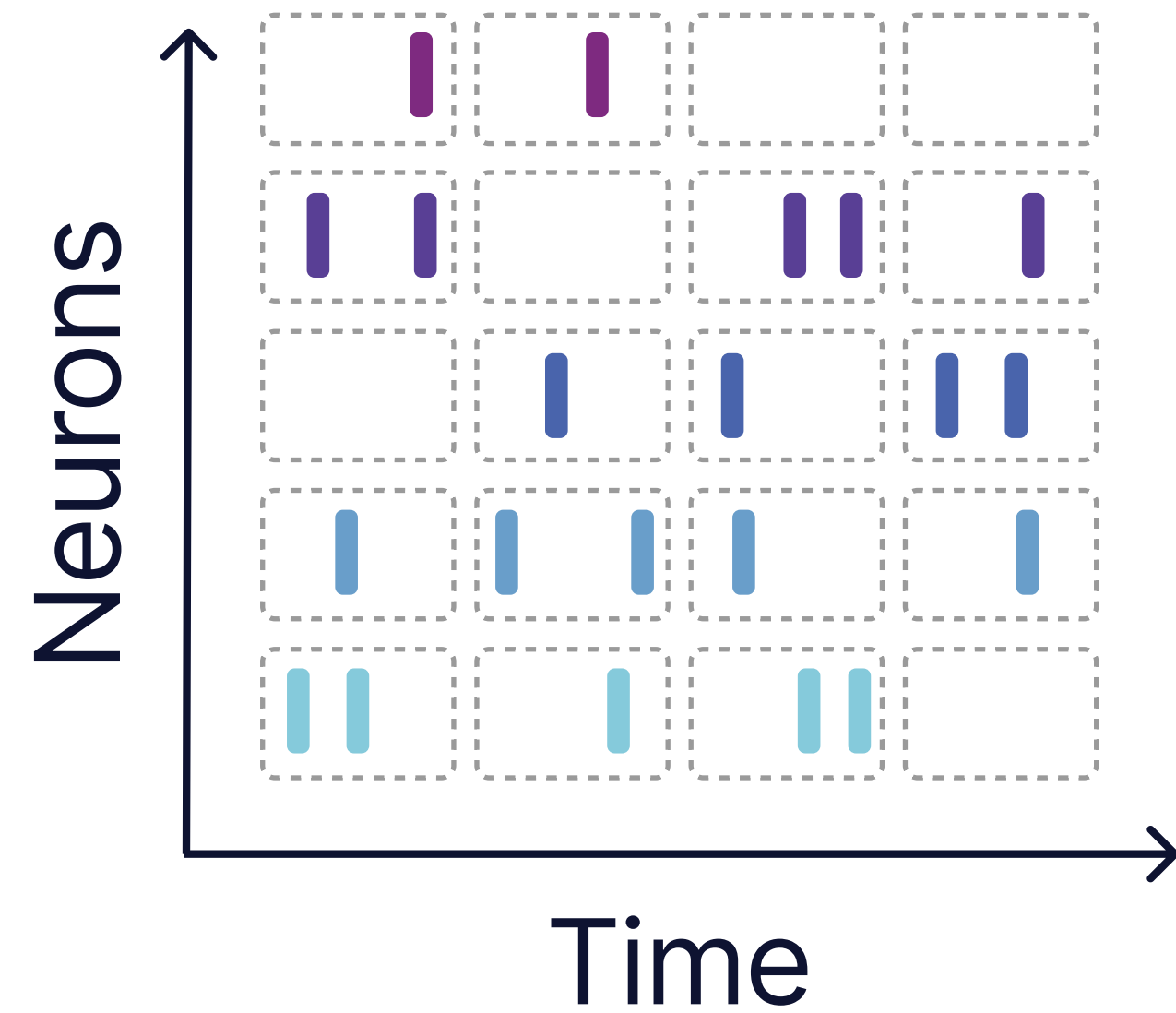
# NDT (*Ye & Pandarinath, 2021*)

Transformer on binned spikes



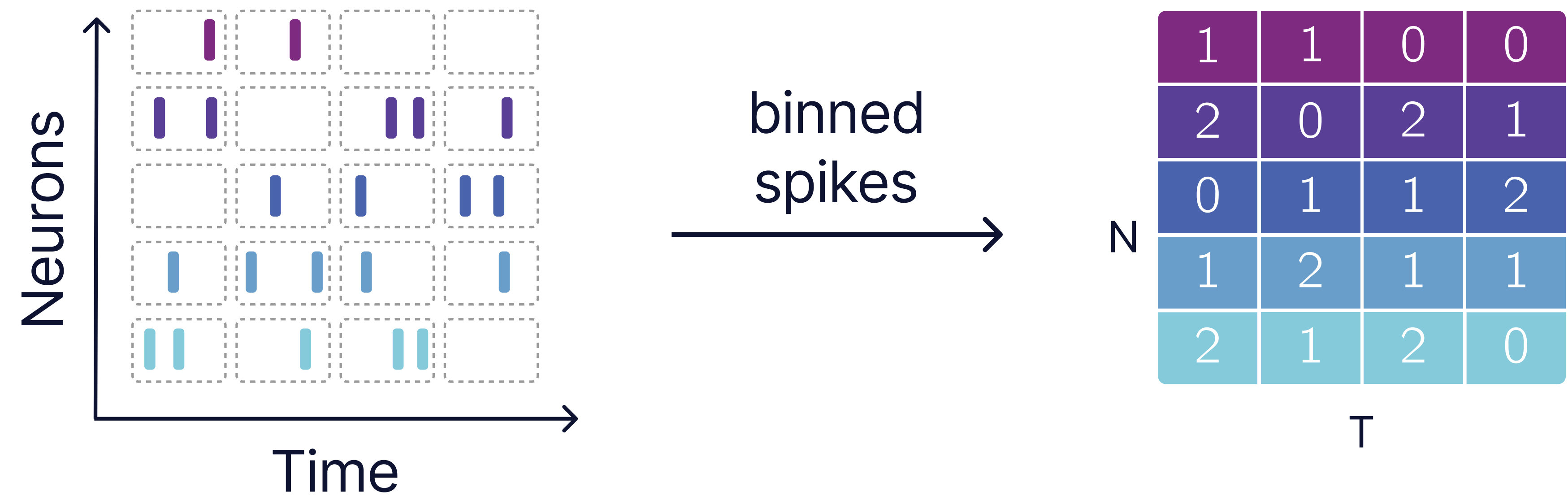
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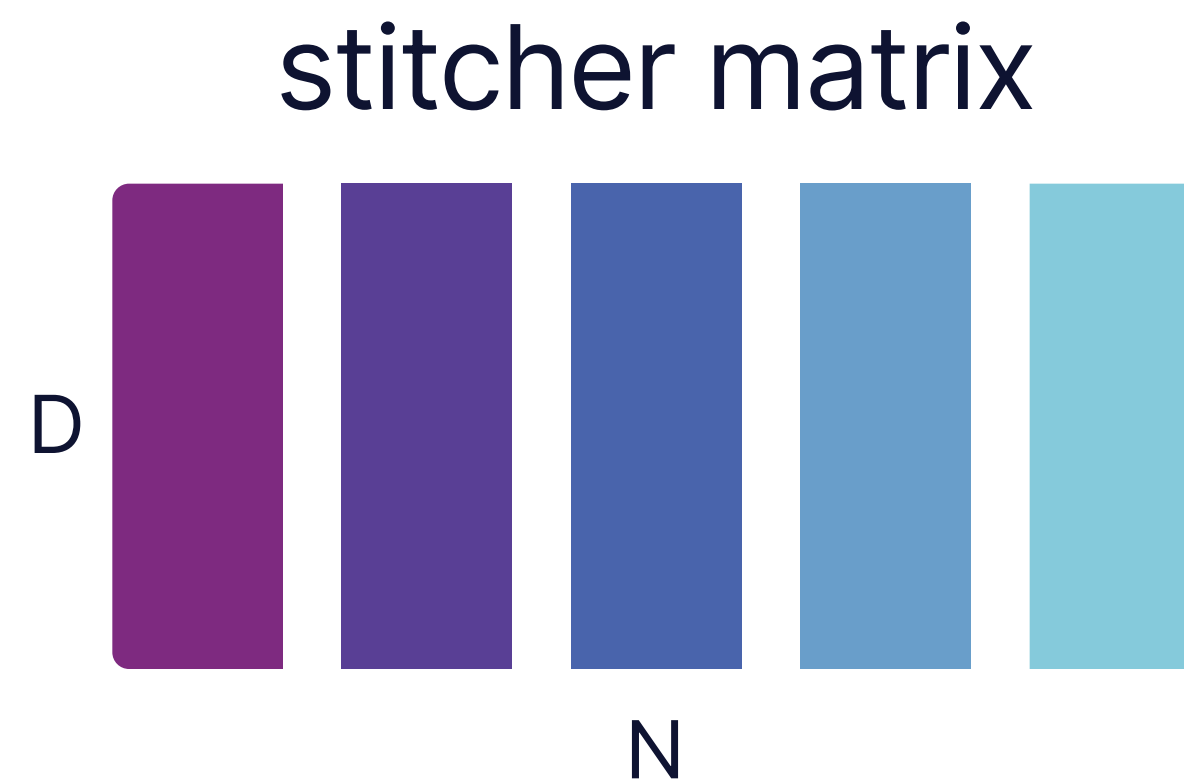
# NDT (*Ye & Pandarinath, 2021*)

Per-session stitcher → multi-session pretraining

**stitcher matrix  $\times$  binned spikes = tokens**

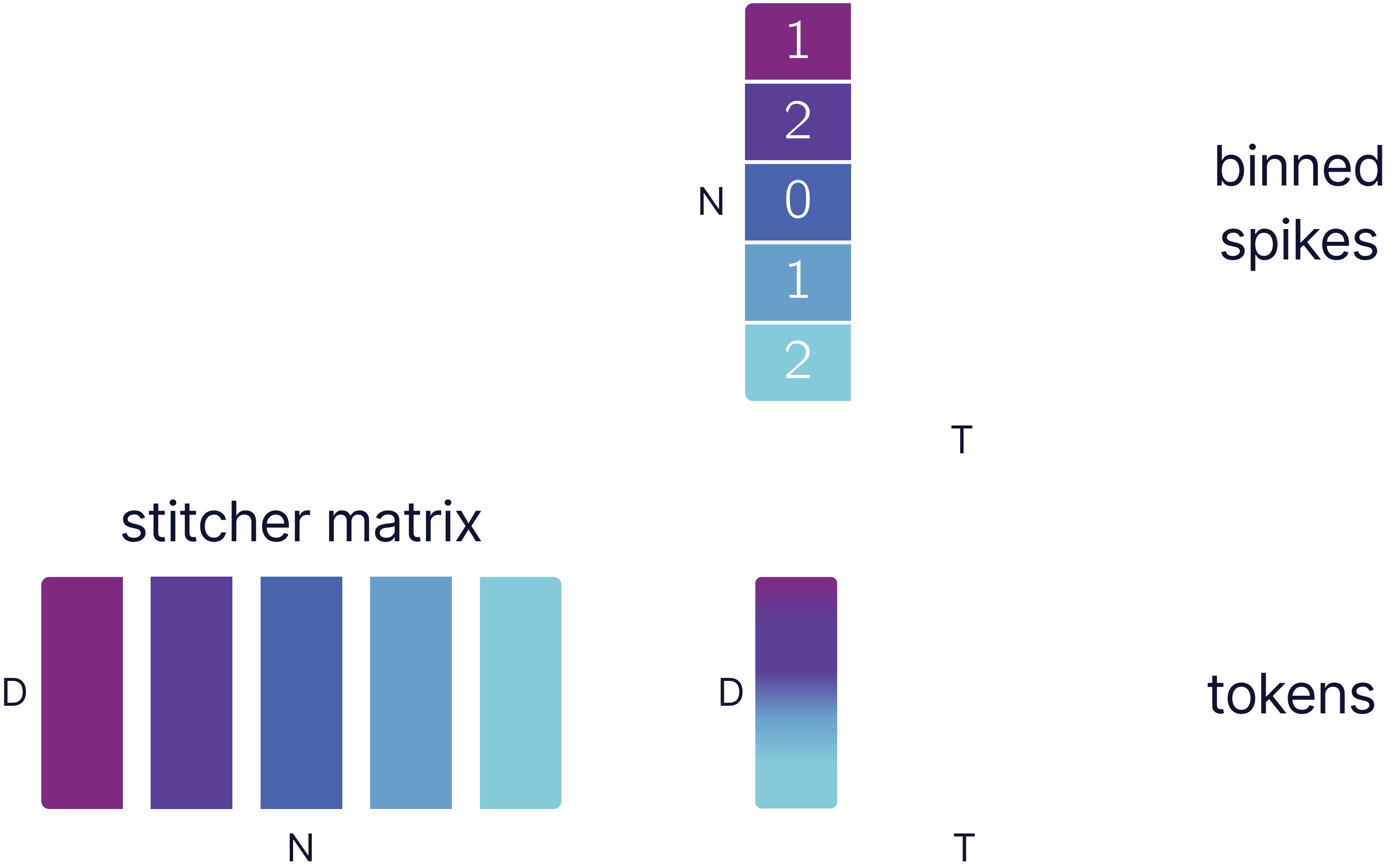
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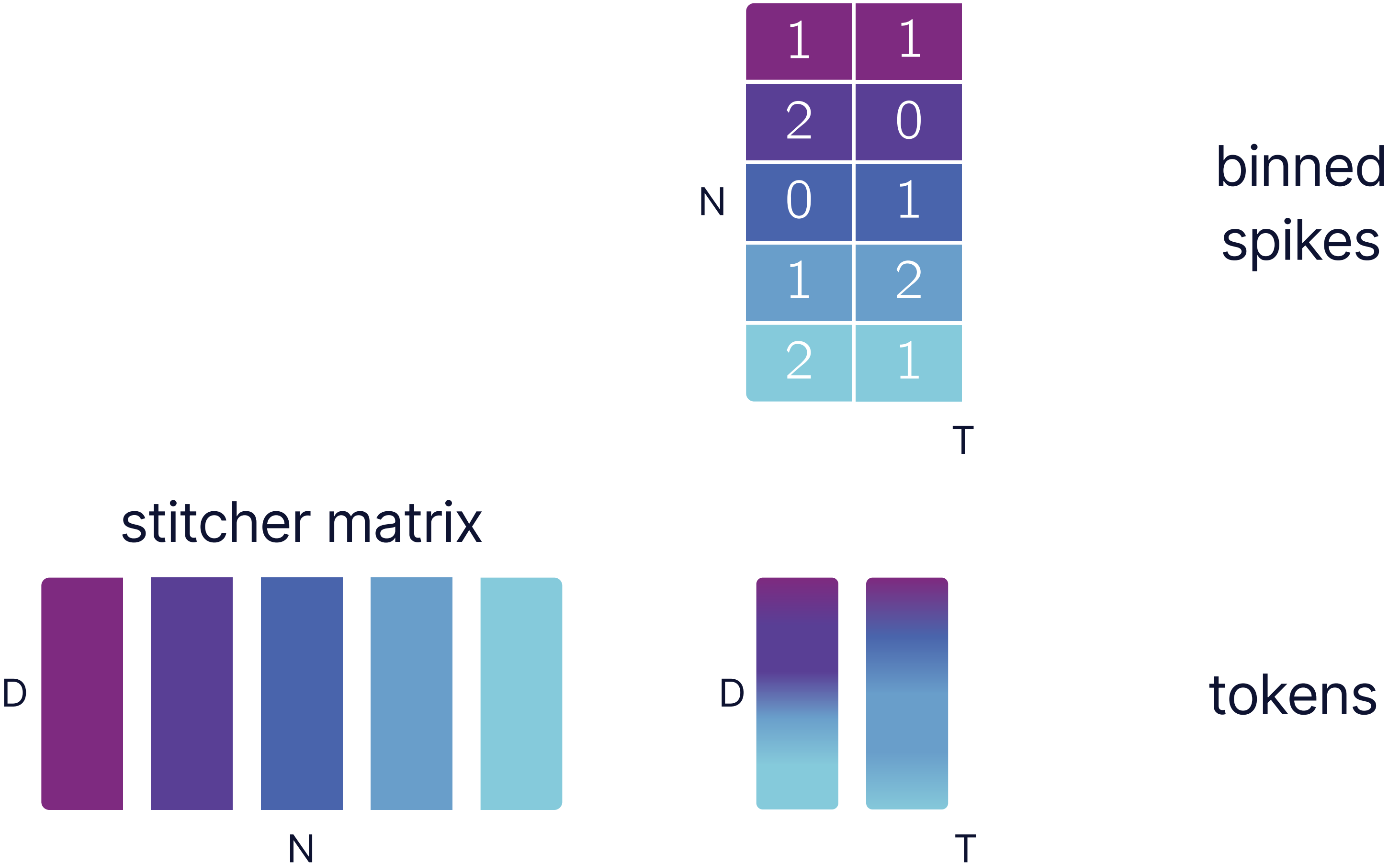
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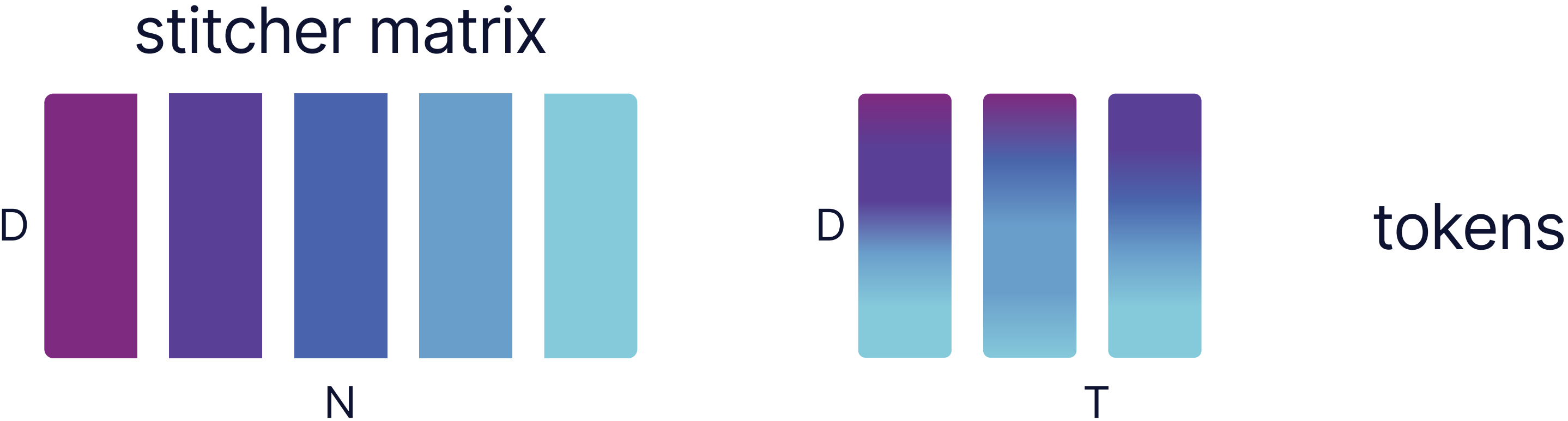
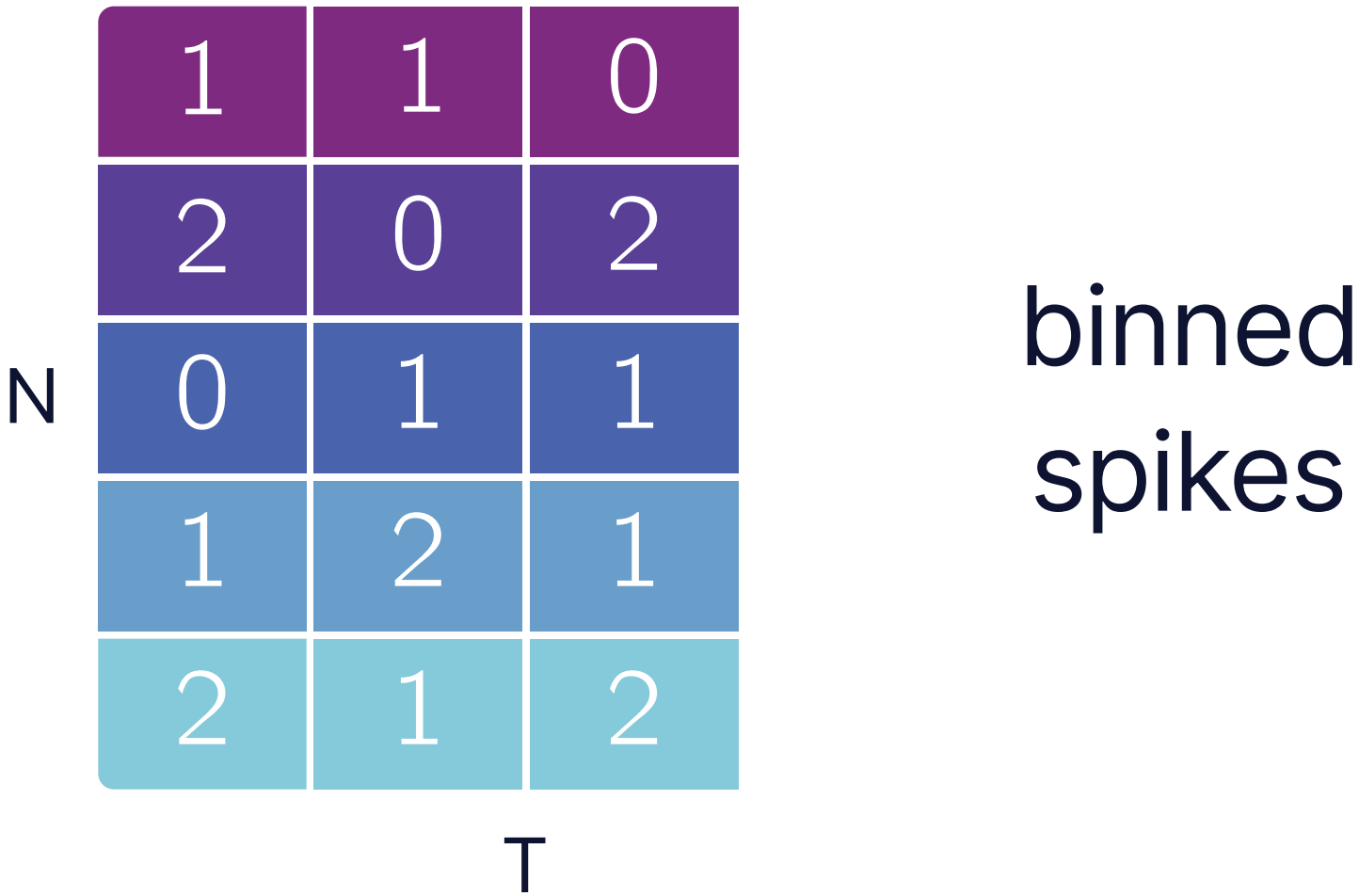
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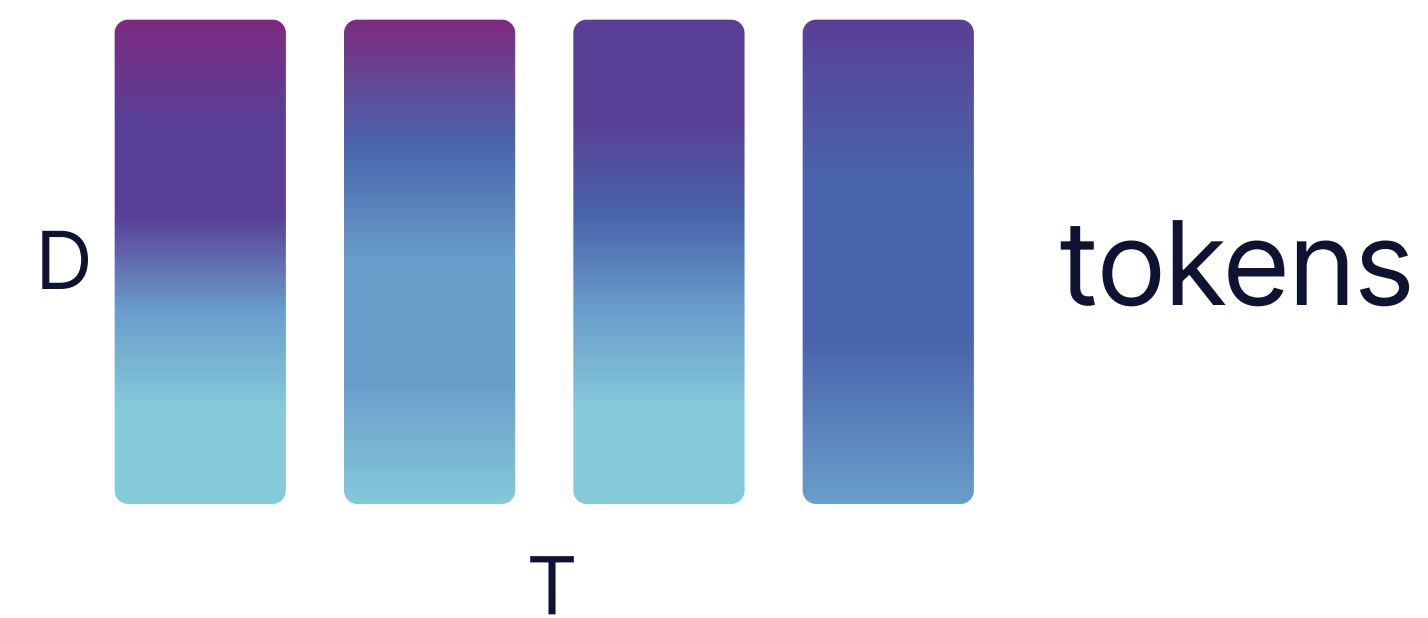
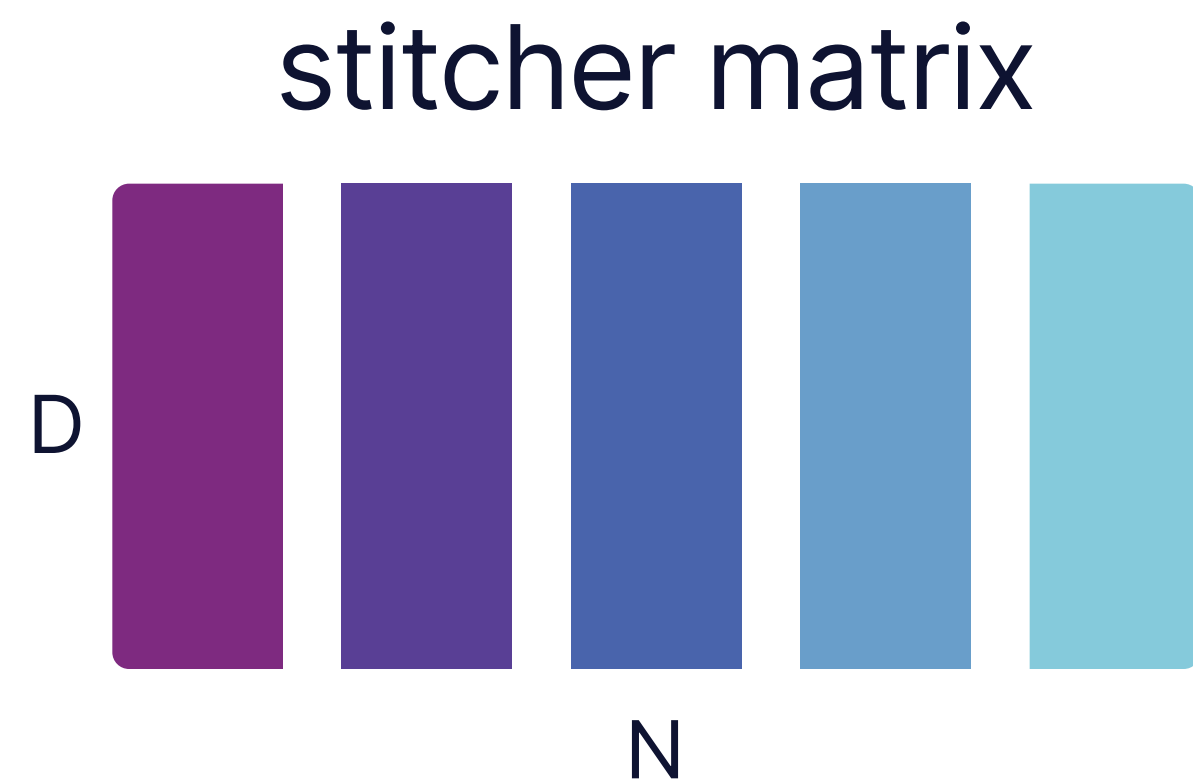
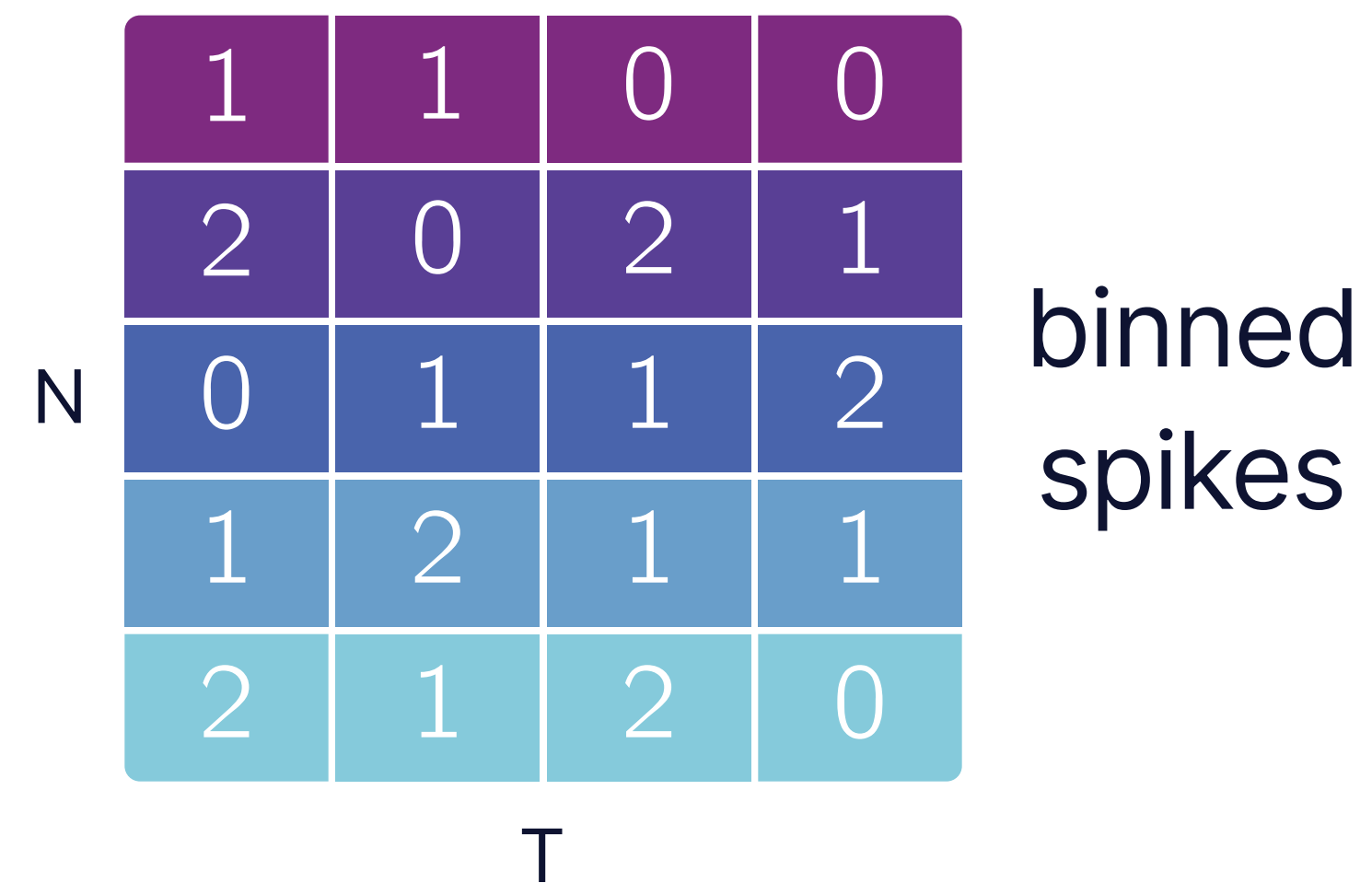
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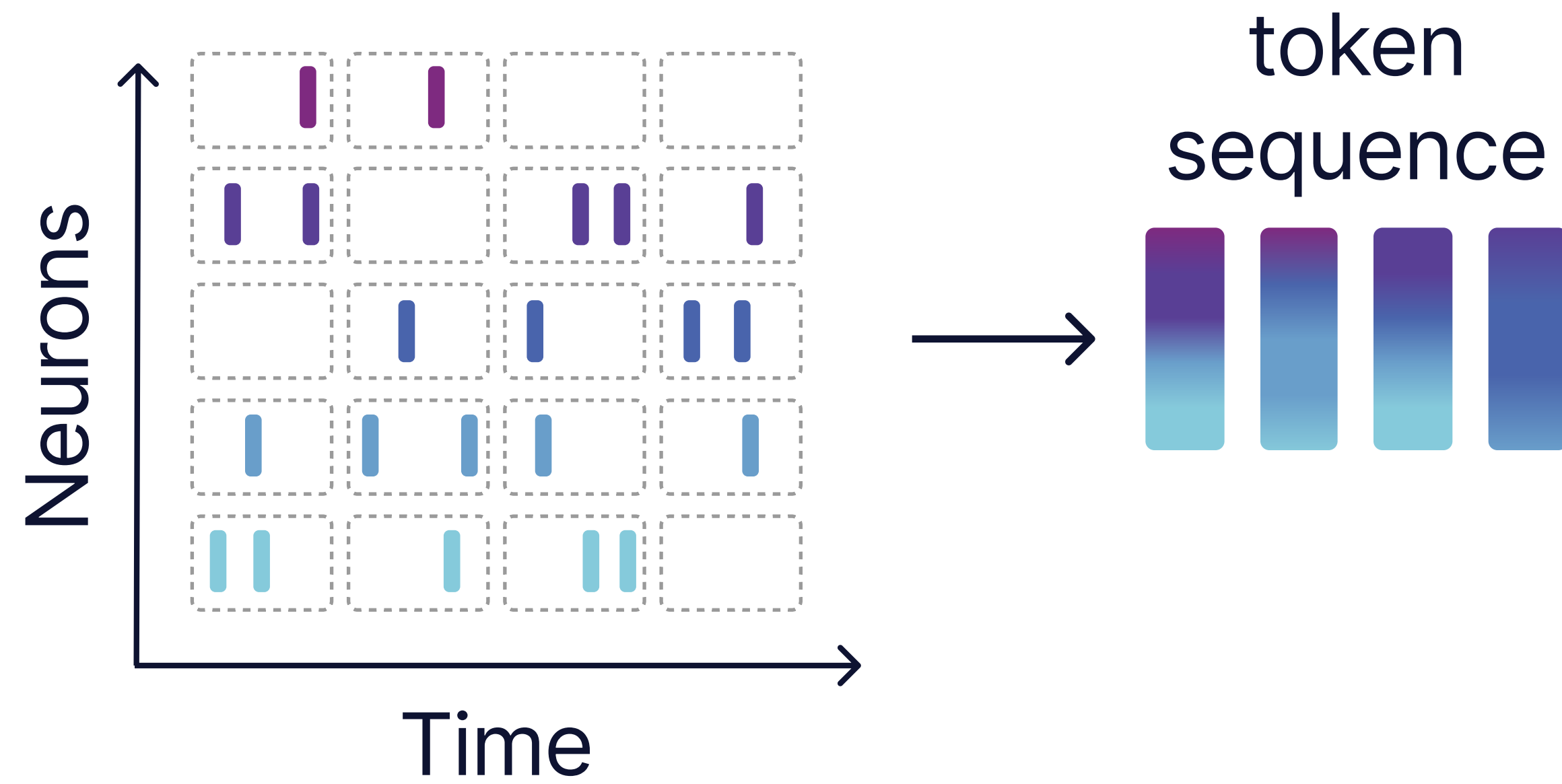
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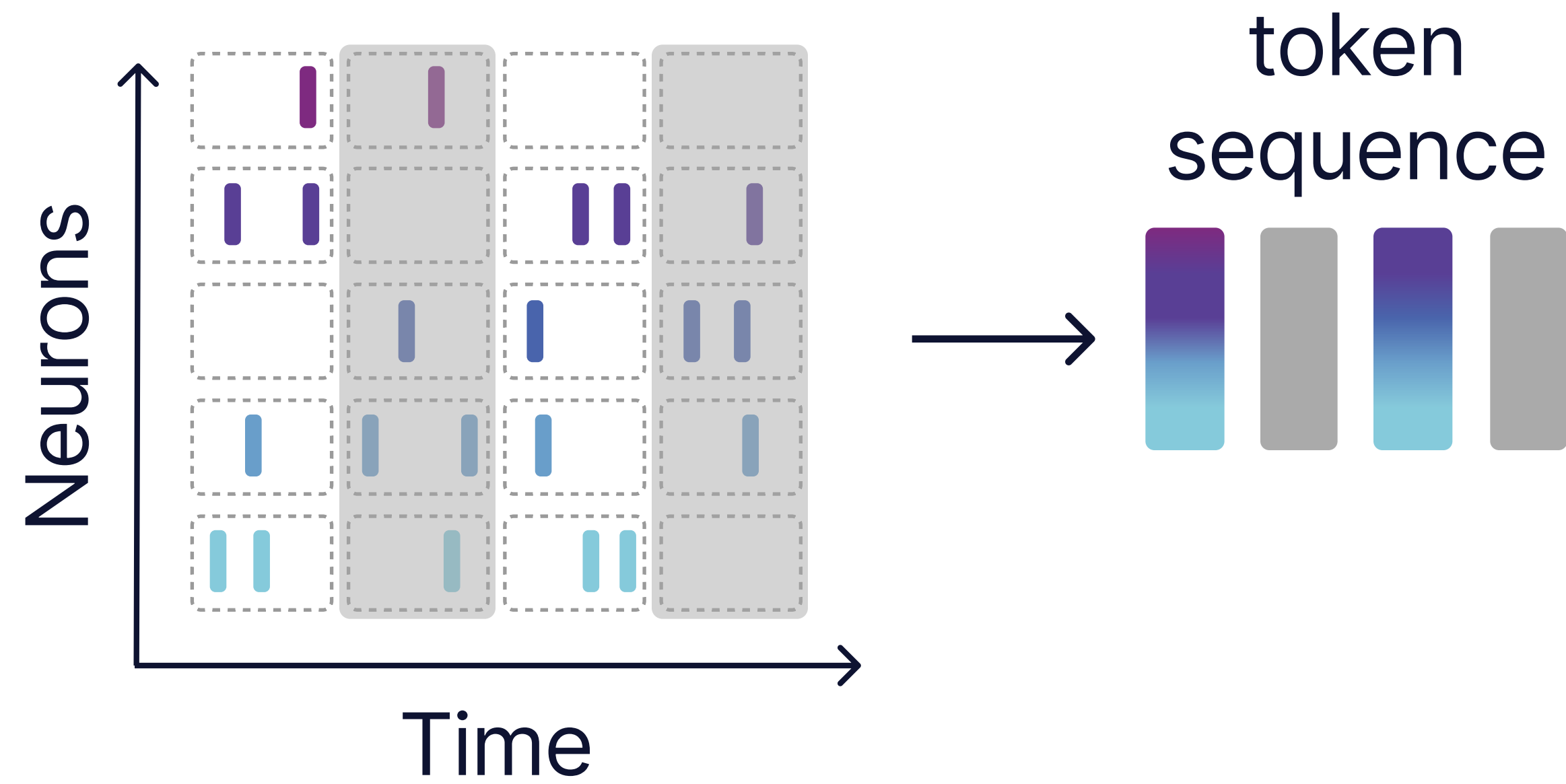
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Masked reconstruction (BERT-style)



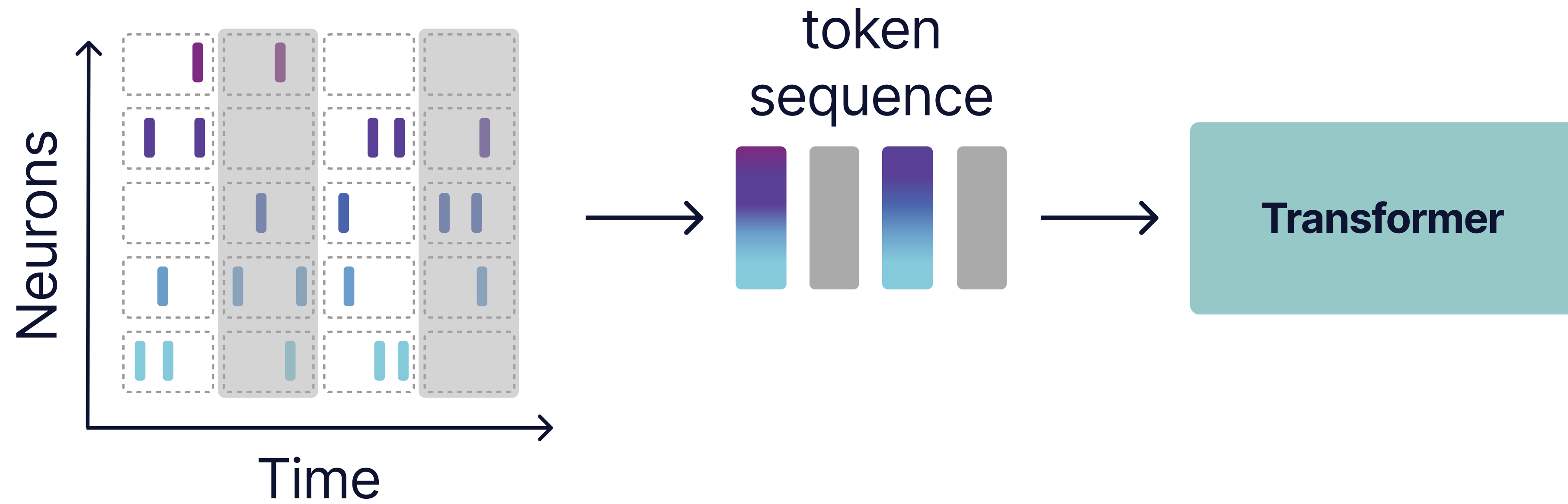
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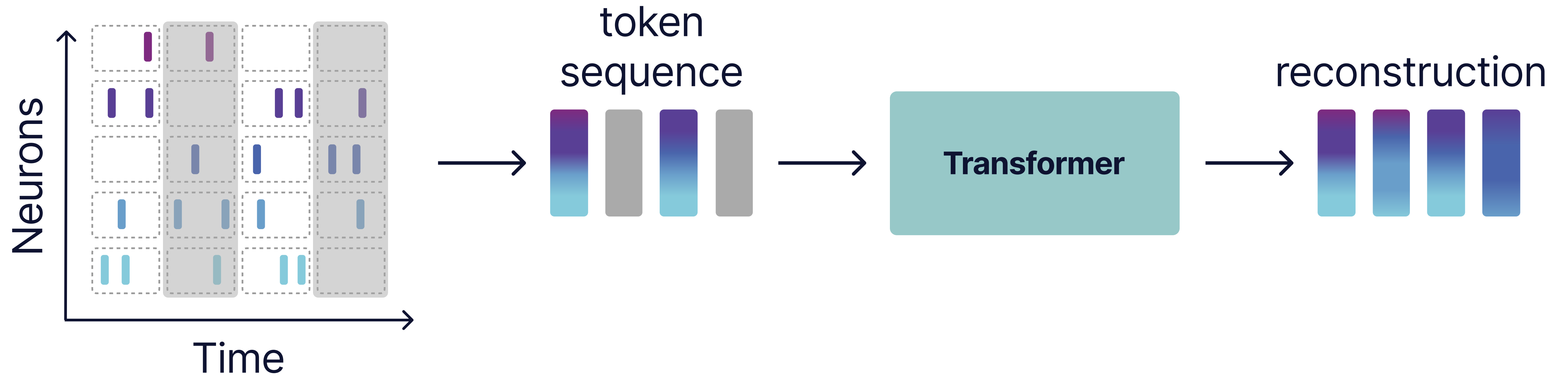
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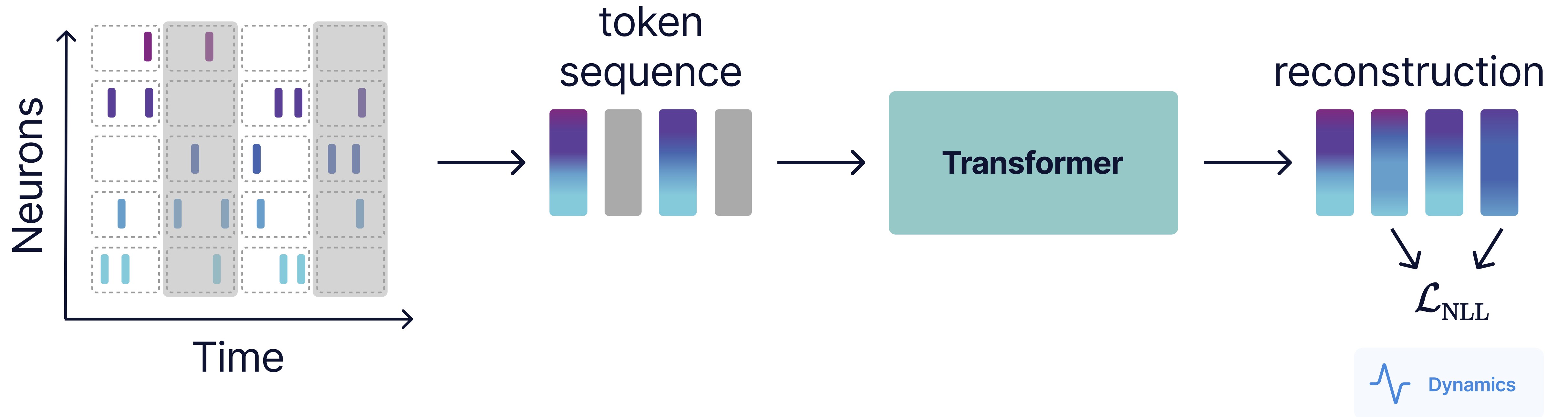
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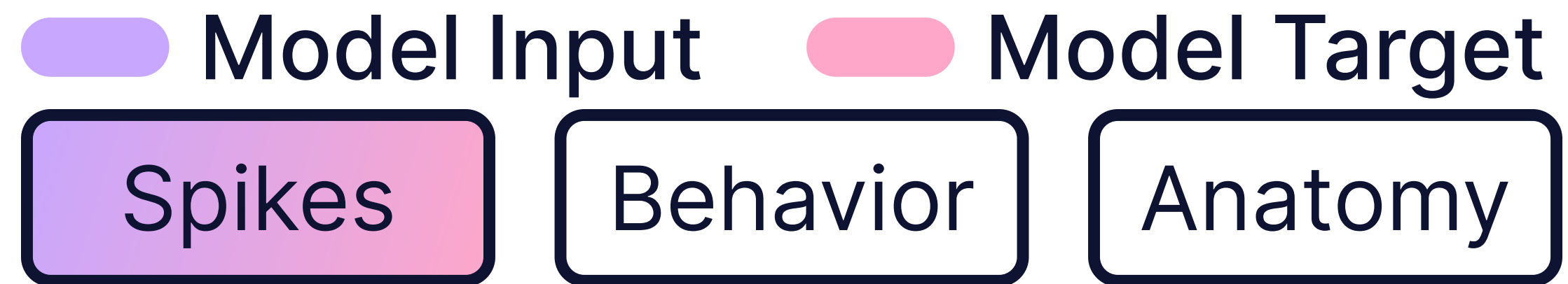
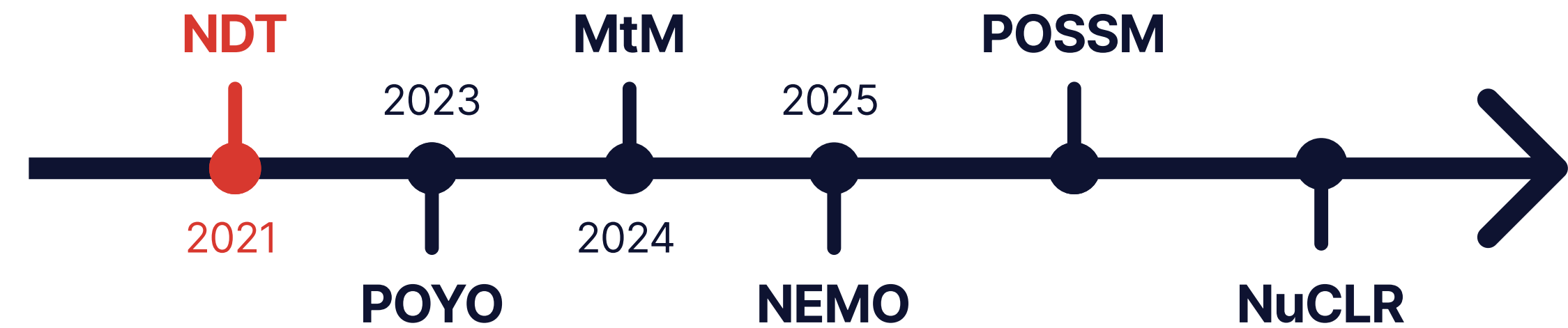


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Masked reconstruction (BERT-style)



# NDT (*Ye & Pandarinath, 2021*)



## Key features

- Transformer on binned spikes
- Per-session stitcher → multi-session pretraining
- Masked reconstruction (BERT-style)

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 Model Input     Model Target

NDT



POYO · POSSM



MtM



NuCLR · NEMO



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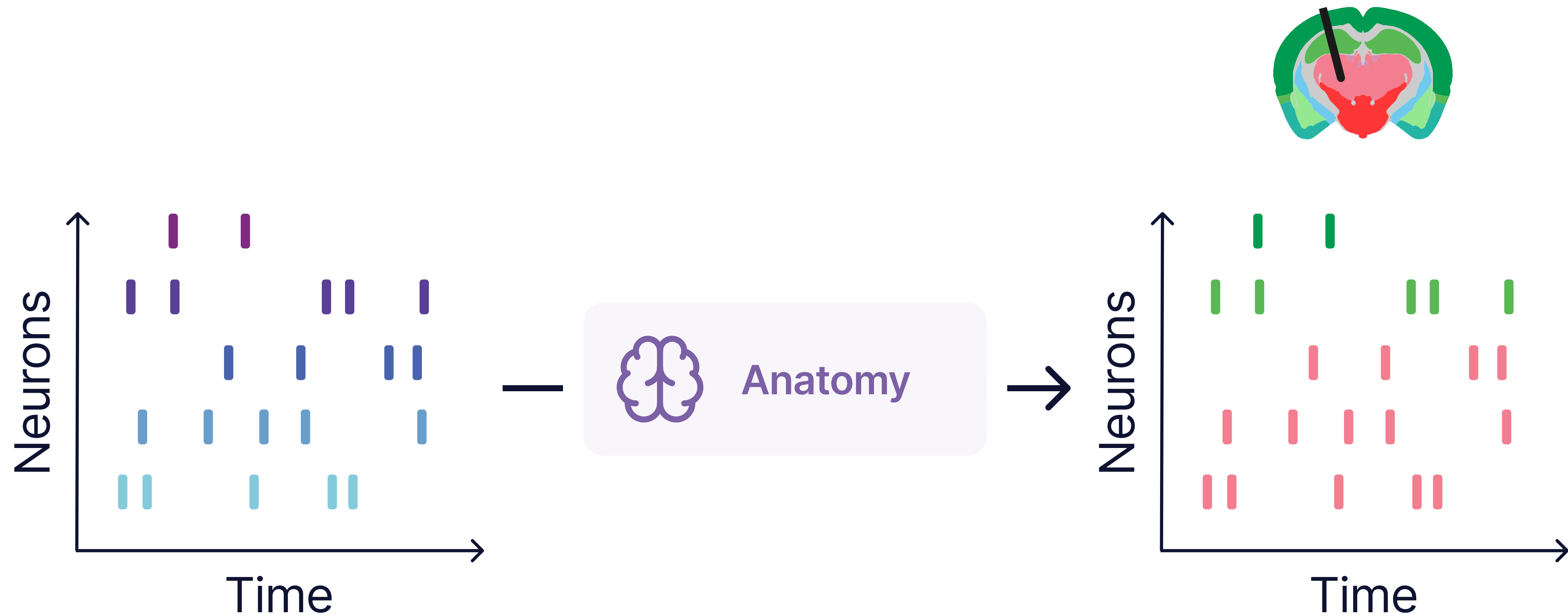


# **MtM** (Zhang et al., NeurIPS 2024)

Builds on NDT (transformer + per-session stitchers)

# MtM (Zhang et al., NeurIPS 2024)

Anatomy-aware (each neuron  $\Rightarrow$  brain region)



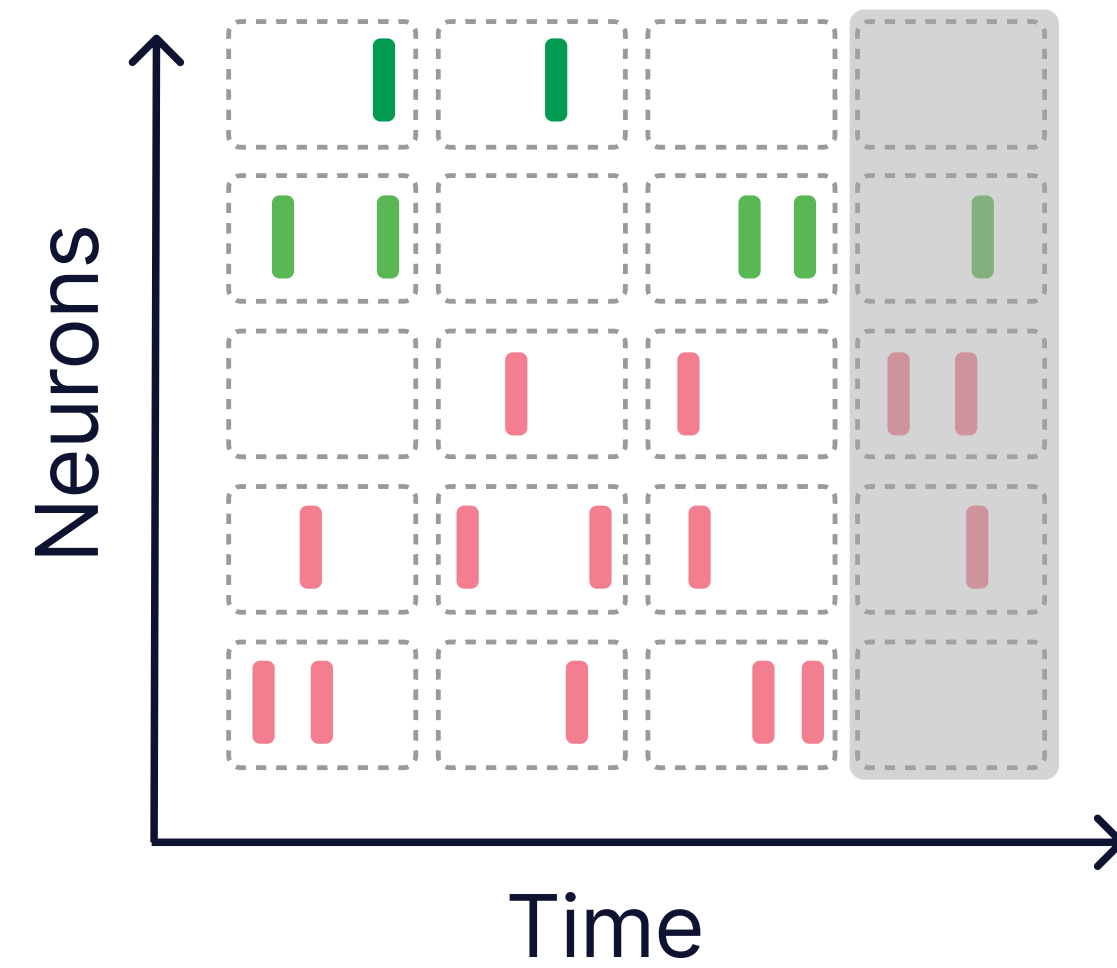
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Masks across time, neurons, and regions

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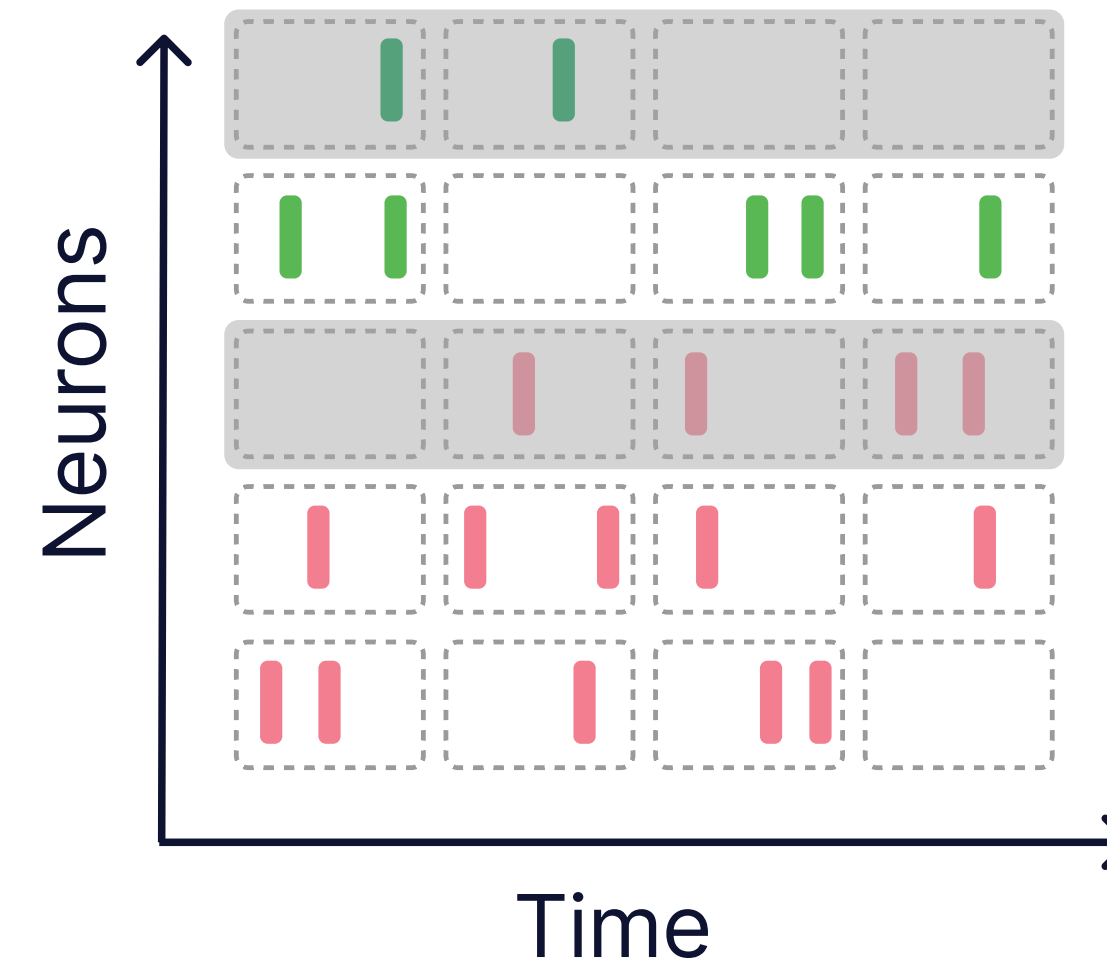
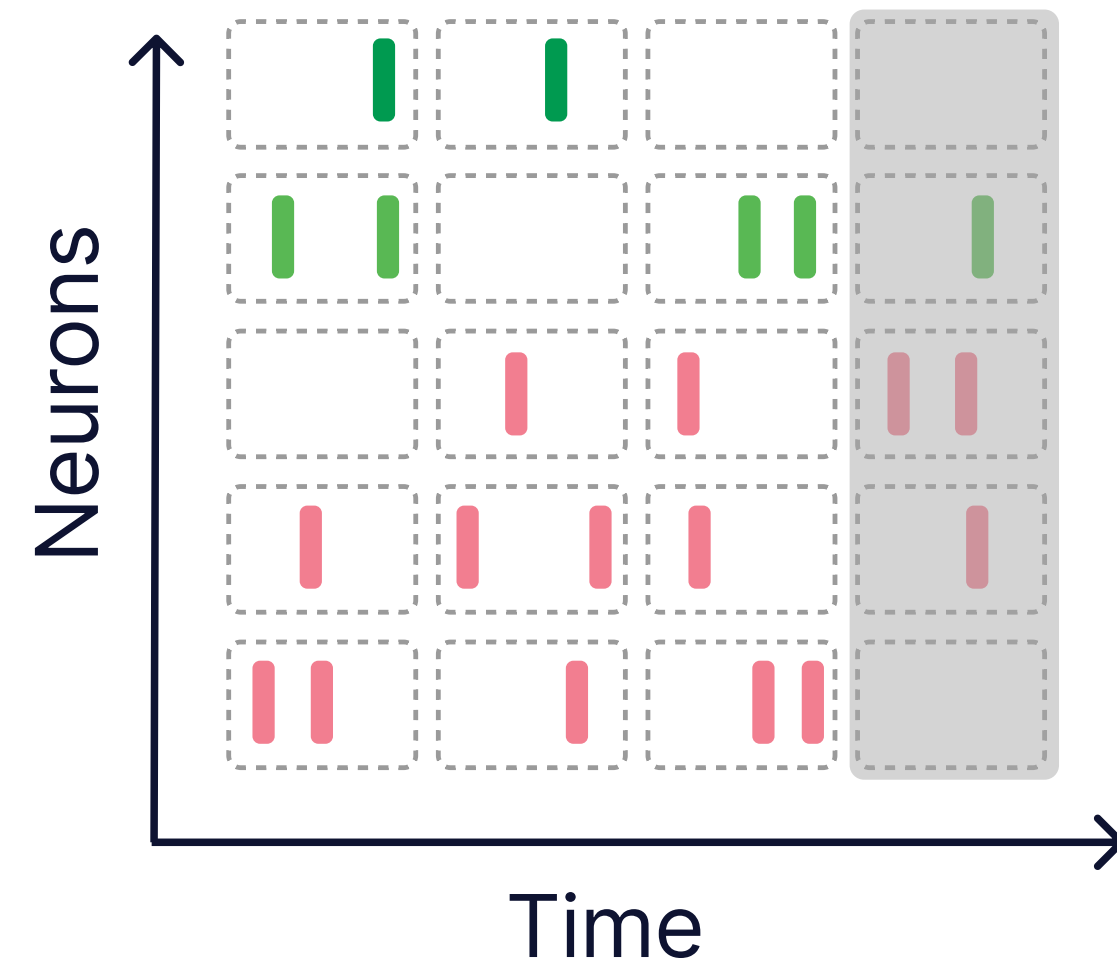
**forecasting**



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**forecasting**

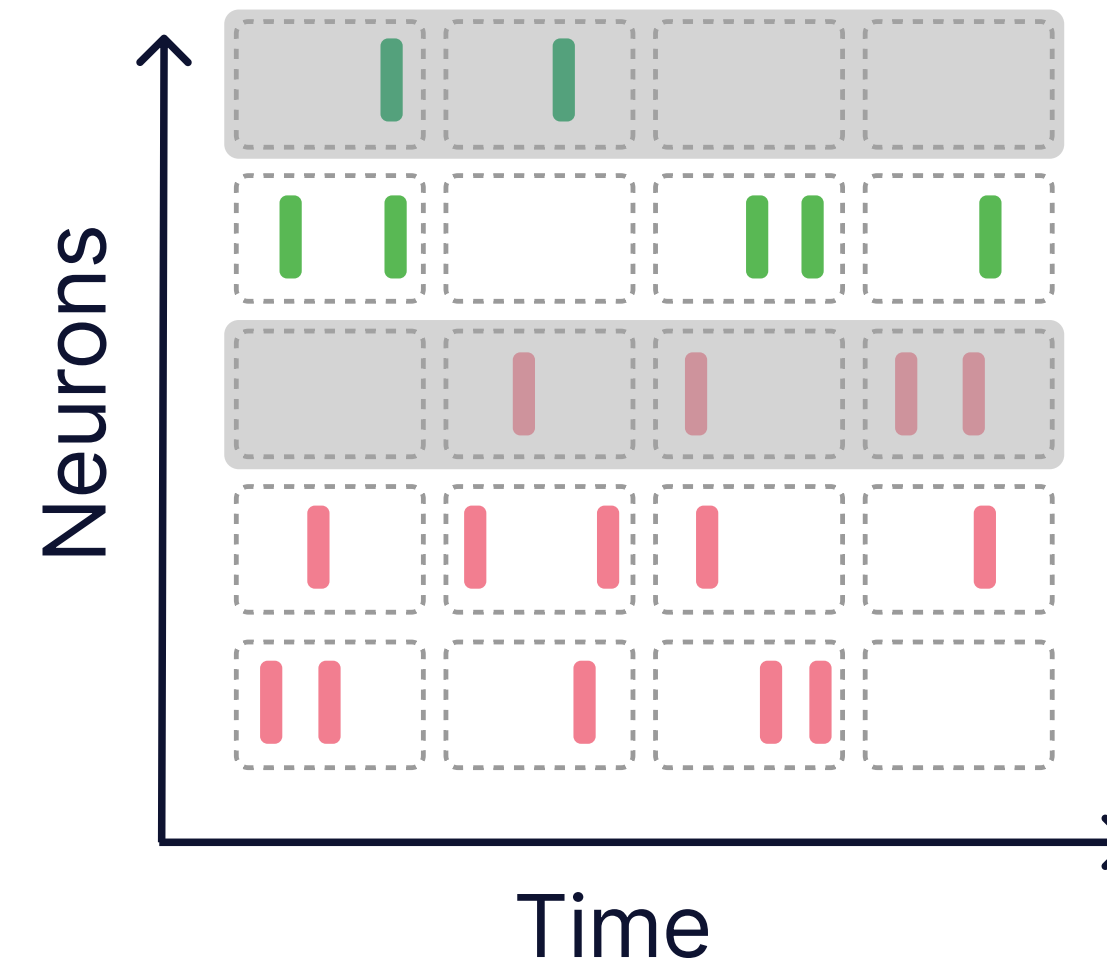
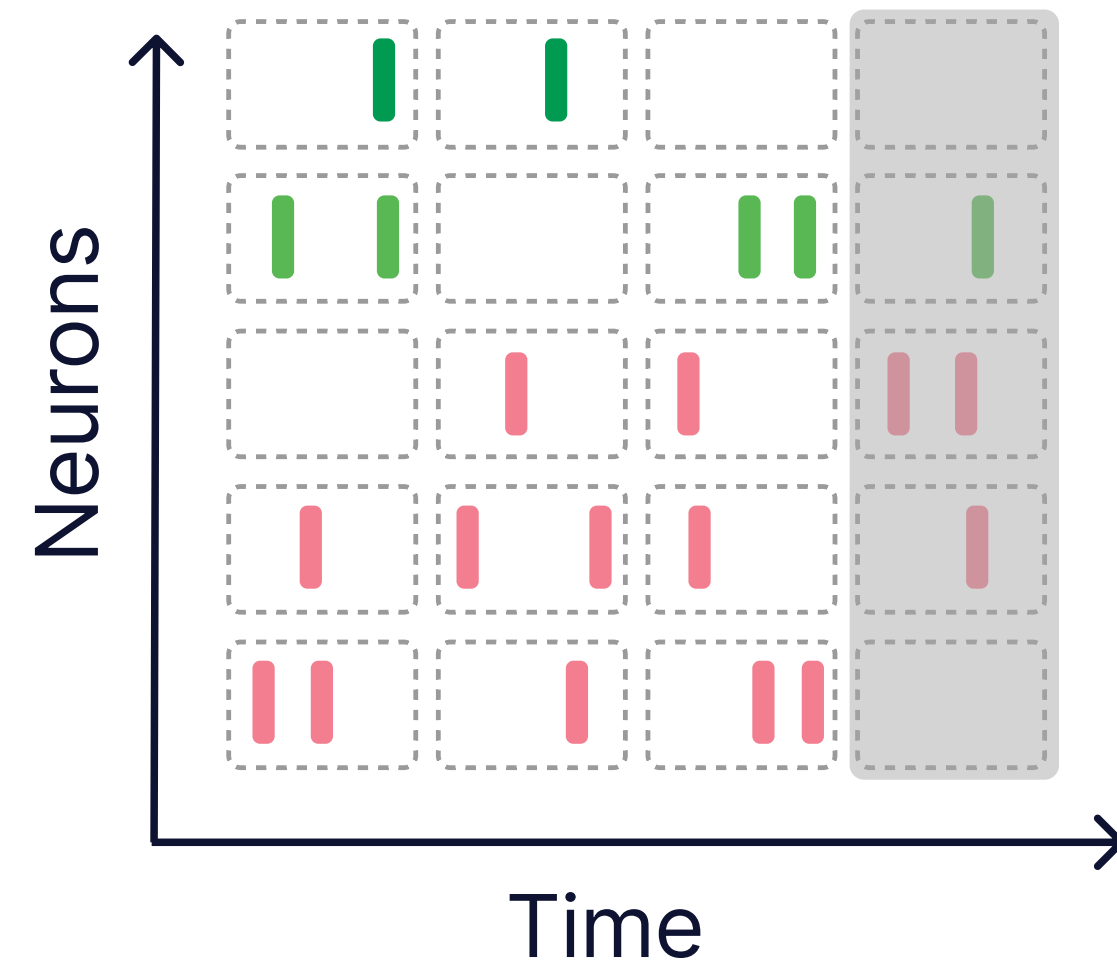


**co-  
smoothing**

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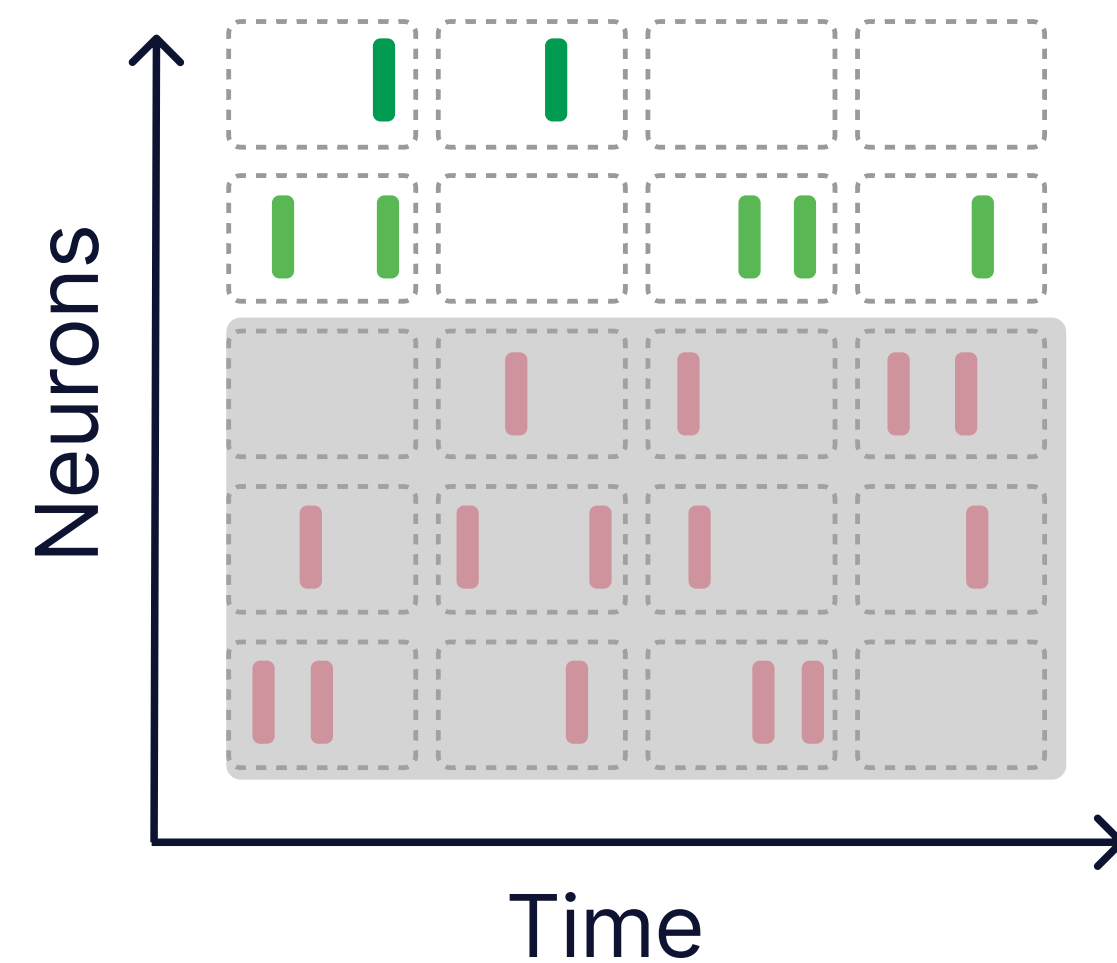
Masks across time, neurons, and regions

**forecasting**



**co-smoothing**

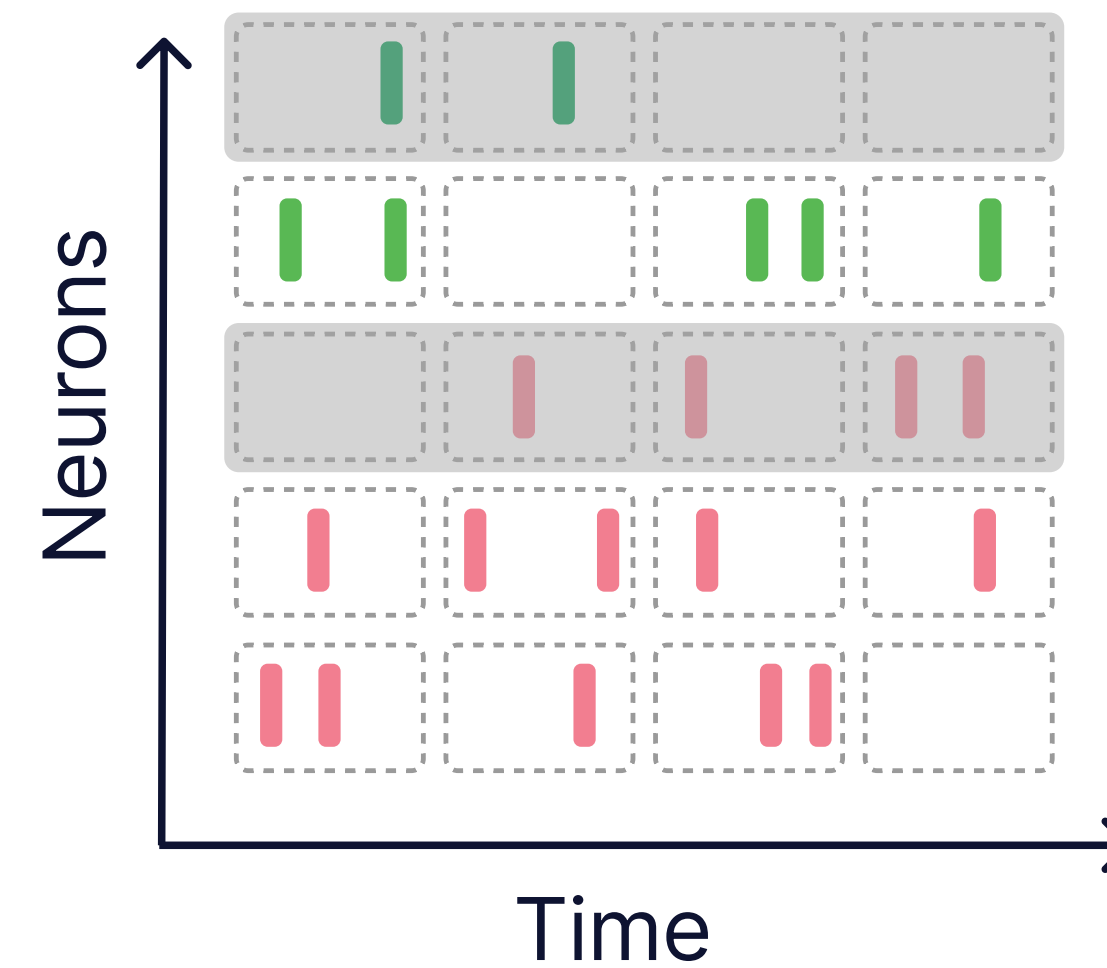
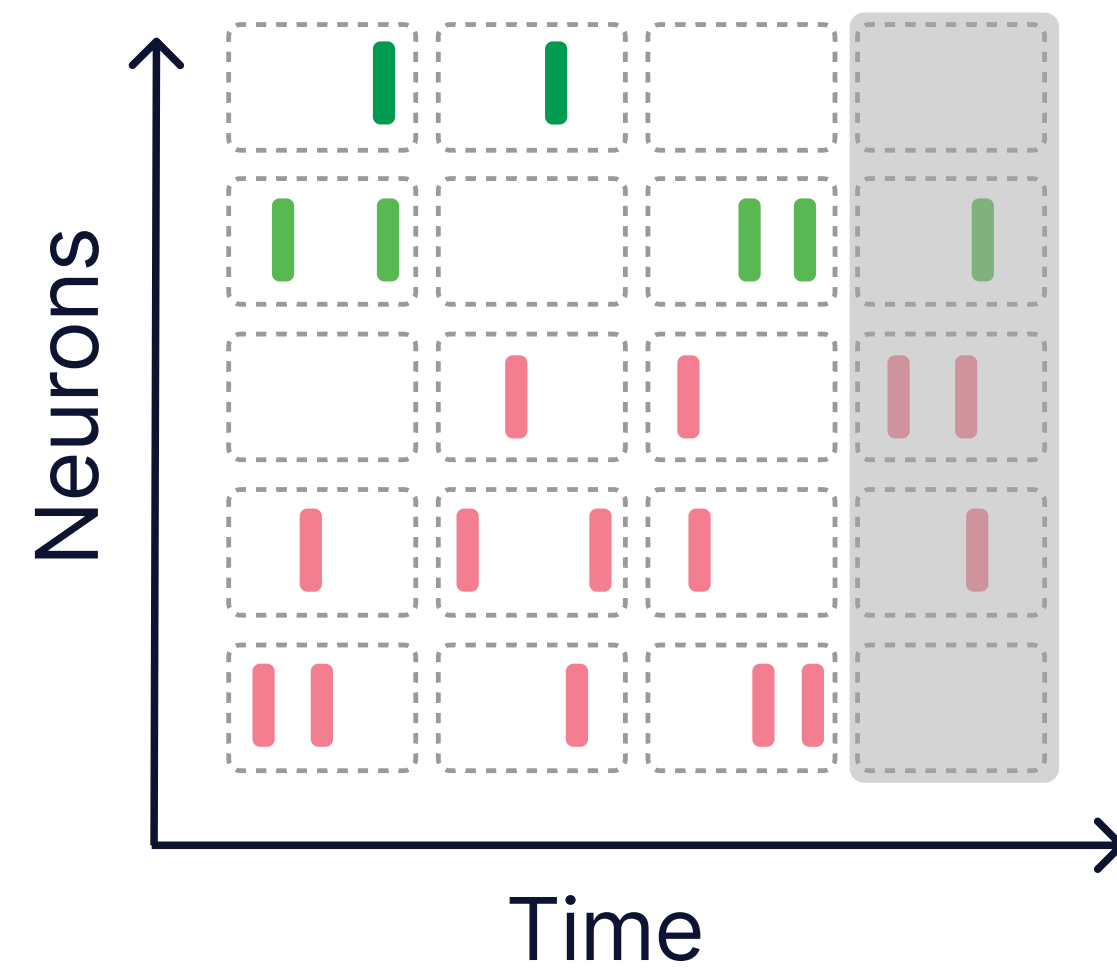
**inter region**



# MtM (Zhang et al., NeurIPS 2024)

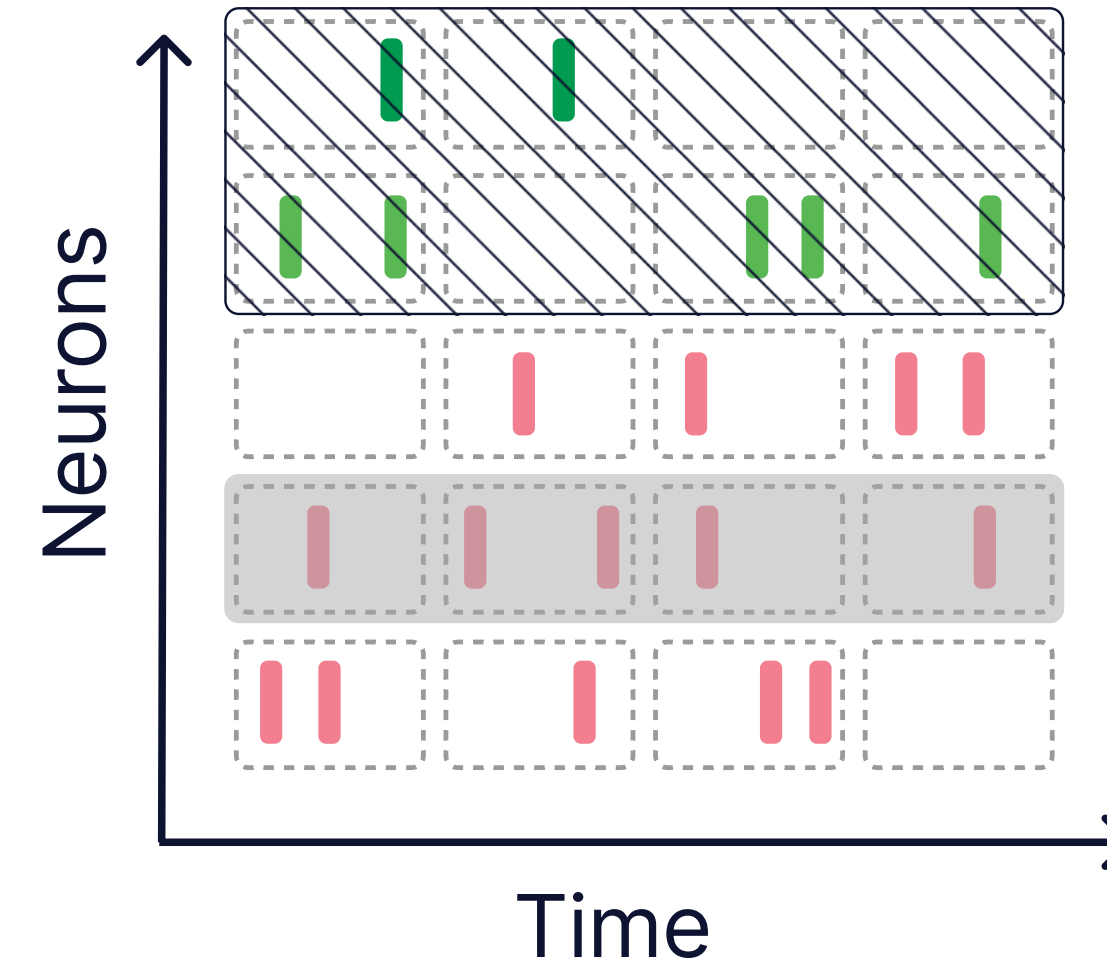
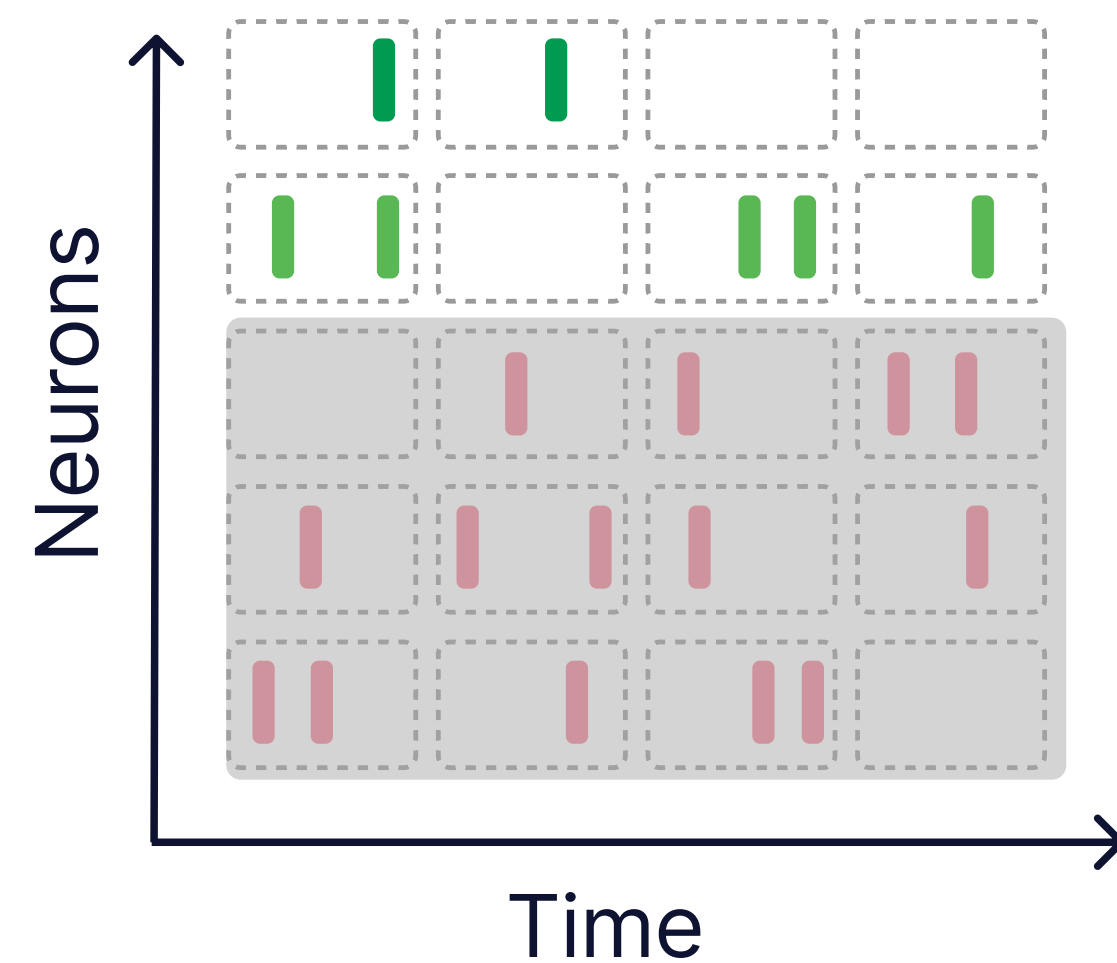
Masks across time, neurons, and regions

**forecasting**



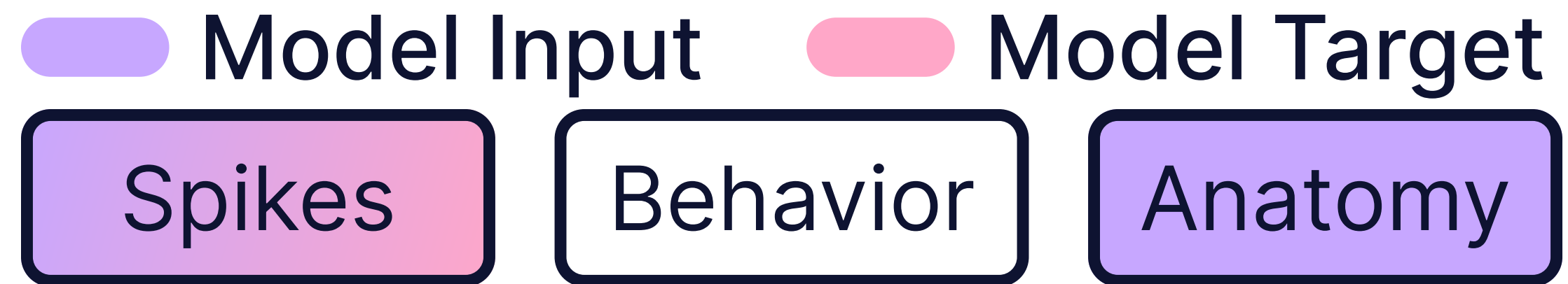
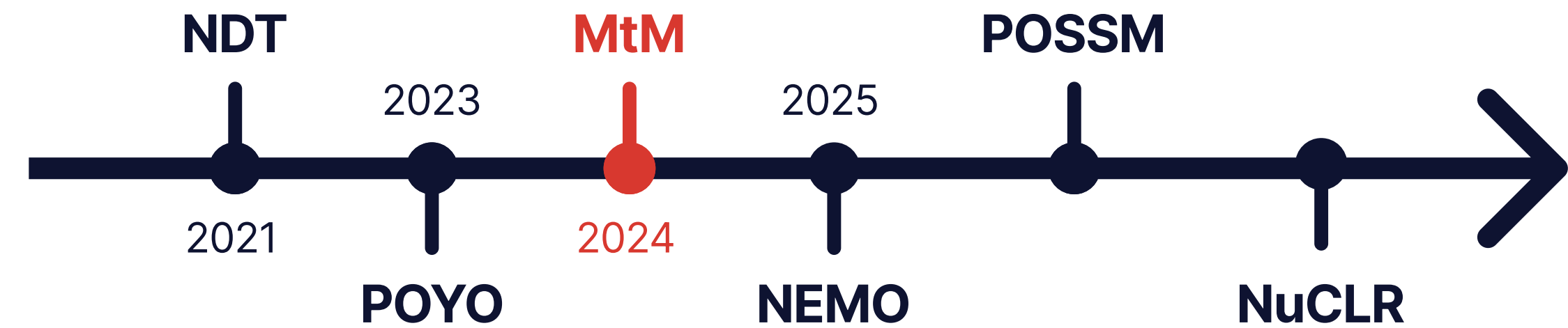
**co-smoothing**

**inter region**



**intra region**

# MtM (Zhang et al., NeurIPS 2024)



## Key features

- Builds on NDT (transformer + per-session stitchers)
- Anatomy-aware (each neuron  $\Rightarrow$  brain region)
- Masks across time, neurons, and regions

# Pretrained Baselines Within the Benchmark Framework

Model Input      Model Target

NDT



POYO · POSSM



MtM



NuCLR · NEMO



# Pretrained Baselines Within the Benchmark Framework

 Model Input     Model Target

NDT



POYO · POSSM



MtM

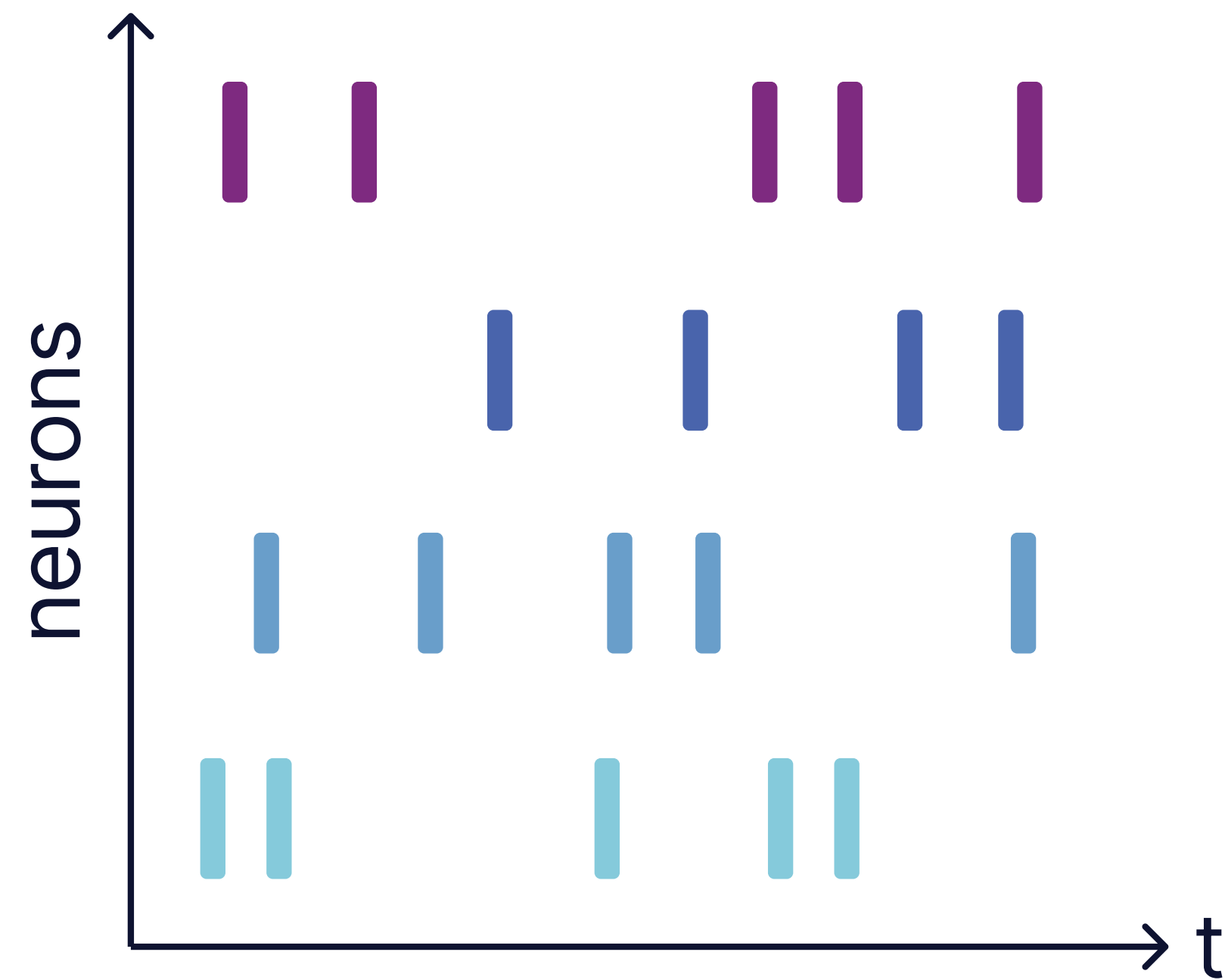


NuCLR · NEMO



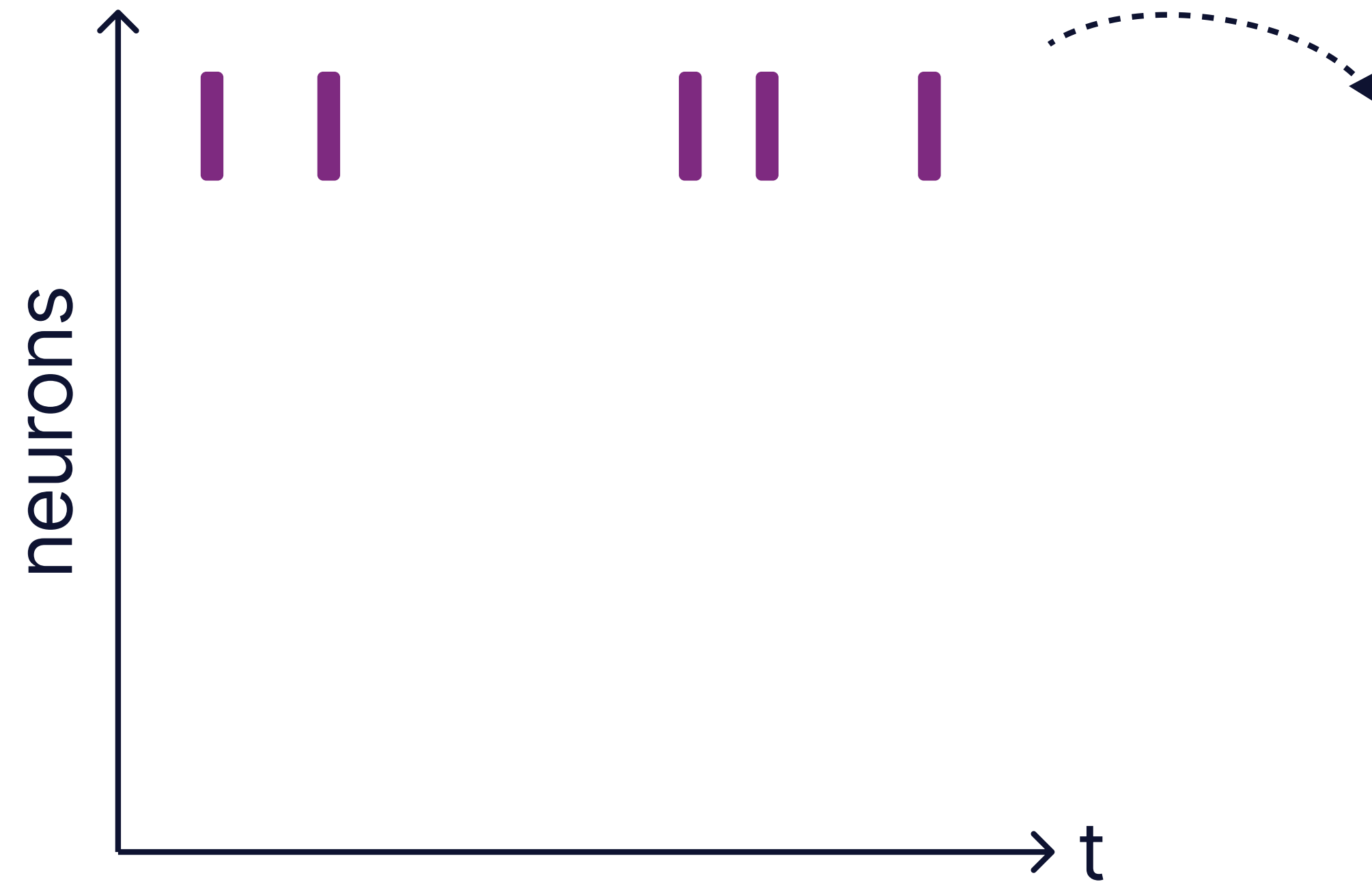
# POYO (Azabou et al., NeurIPS 2023)

Each spike is a token (unit identity + spike time)



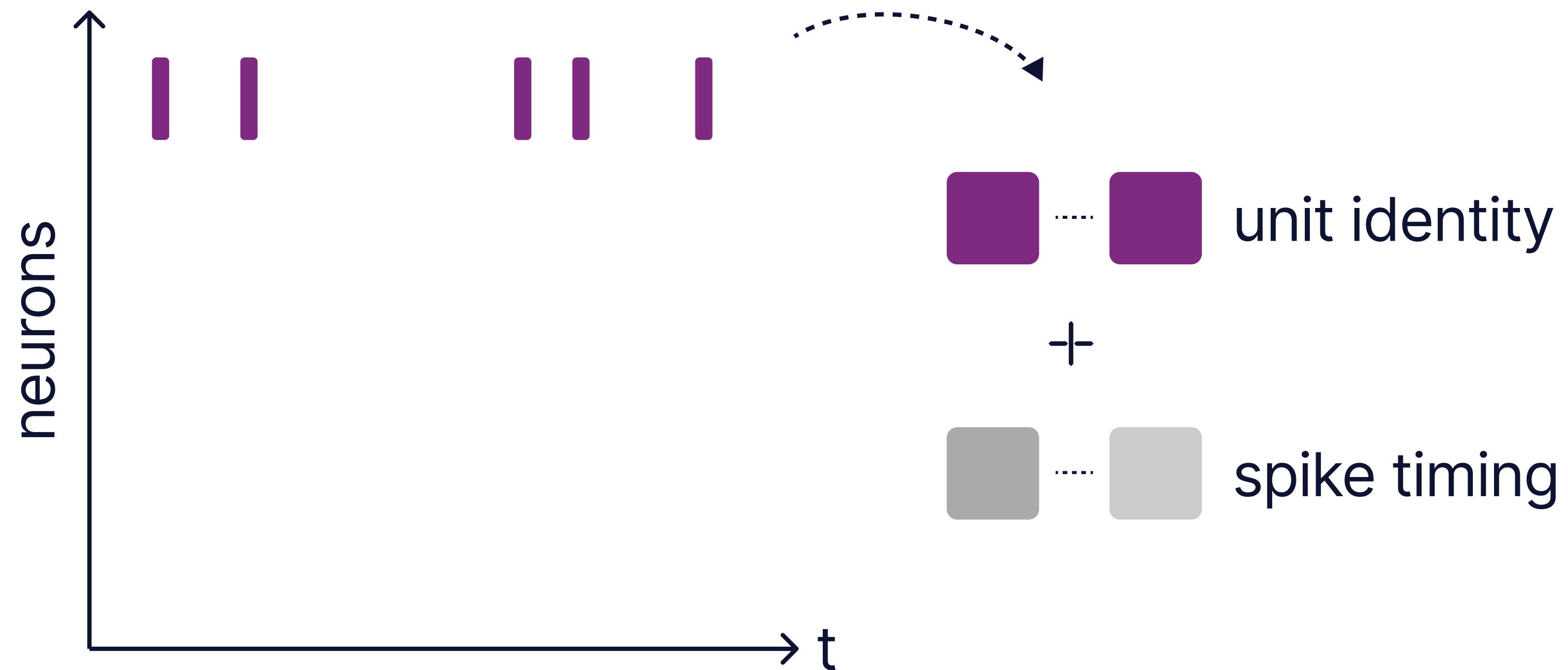
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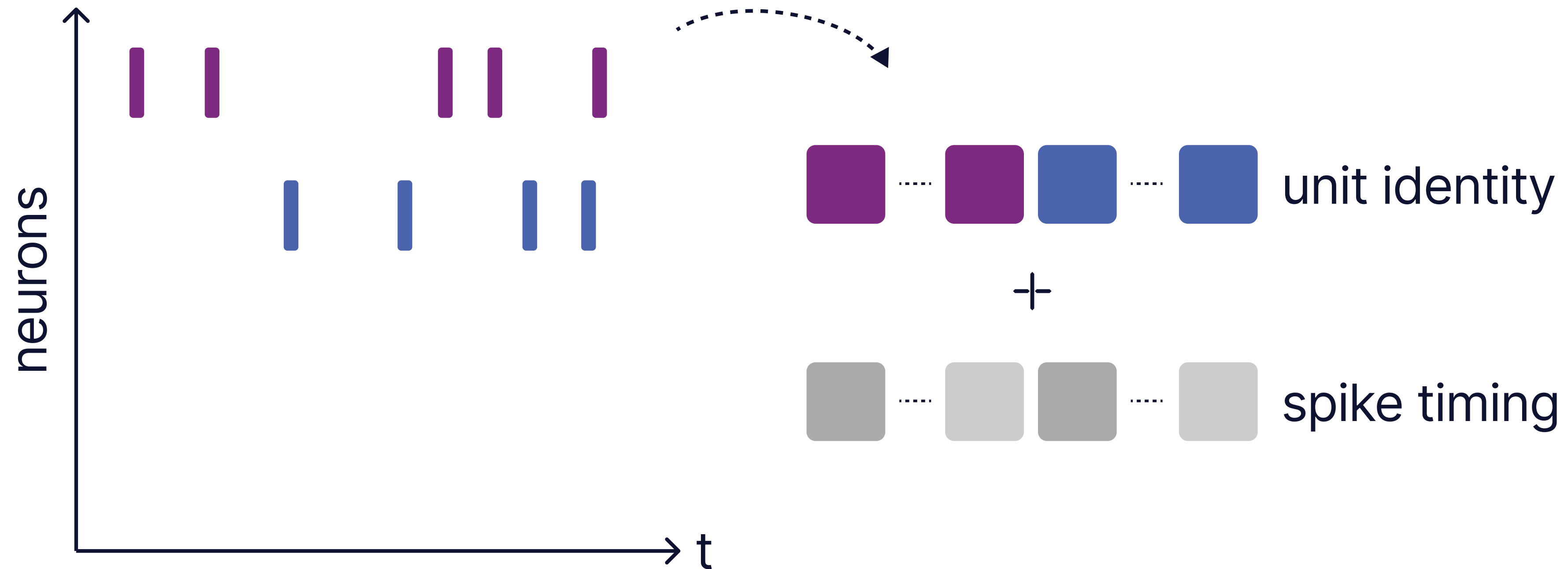
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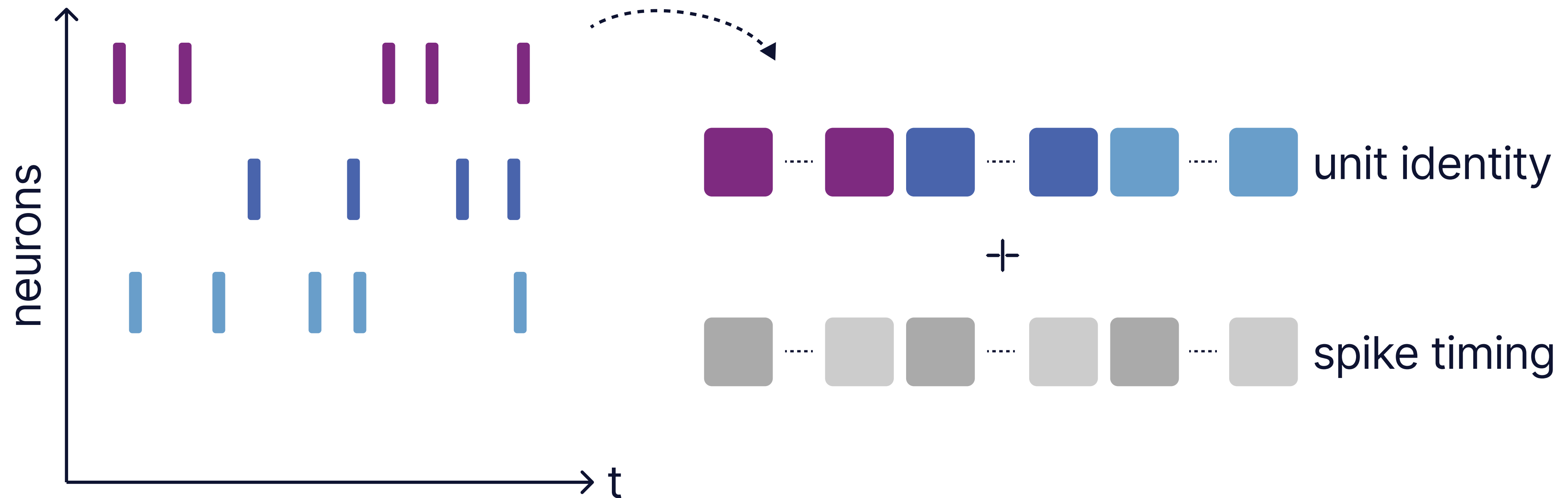
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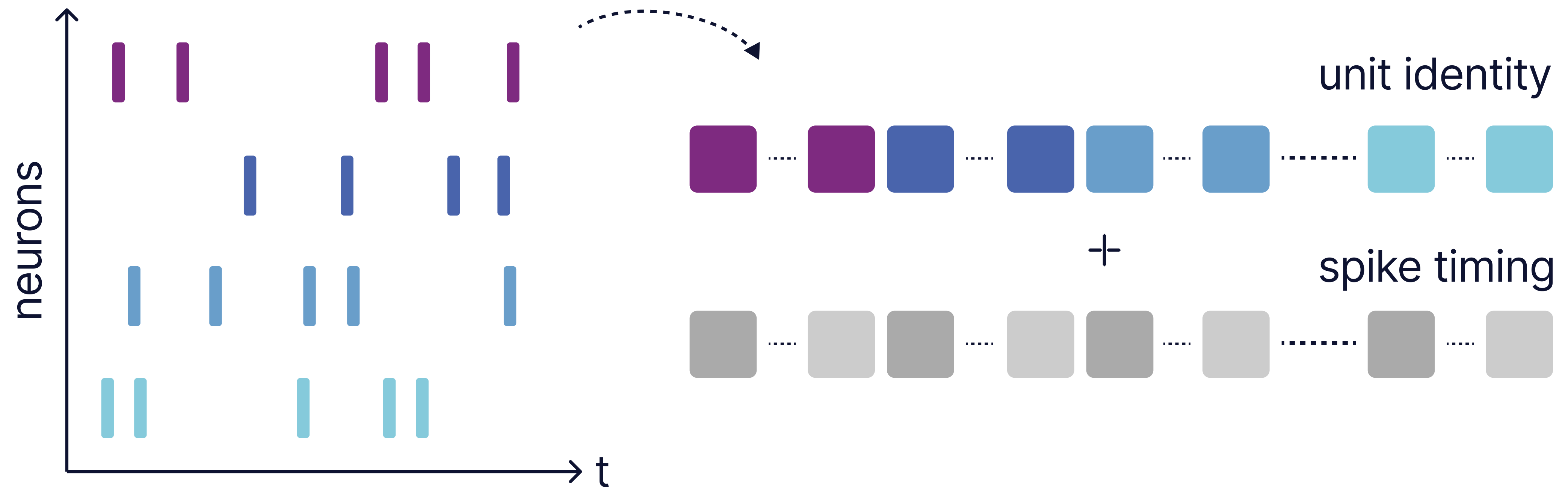
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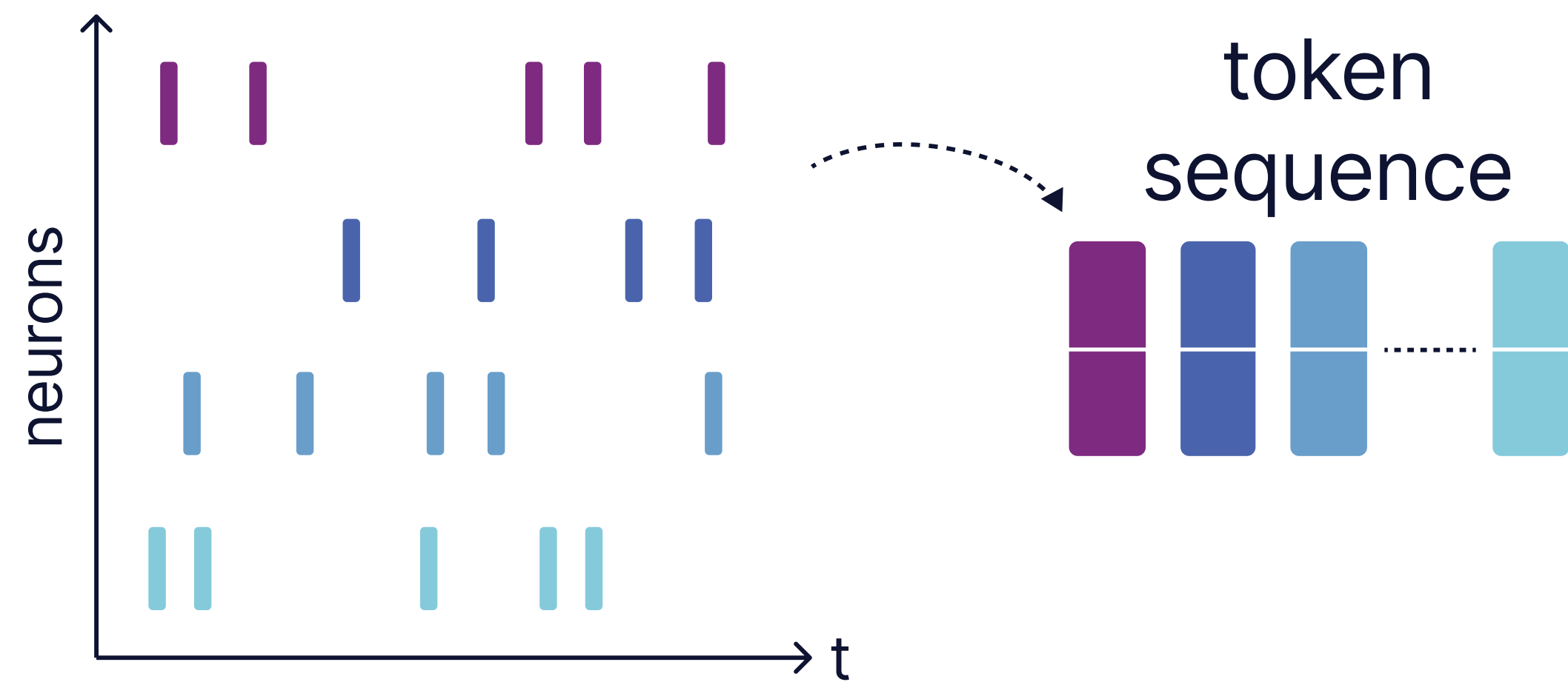
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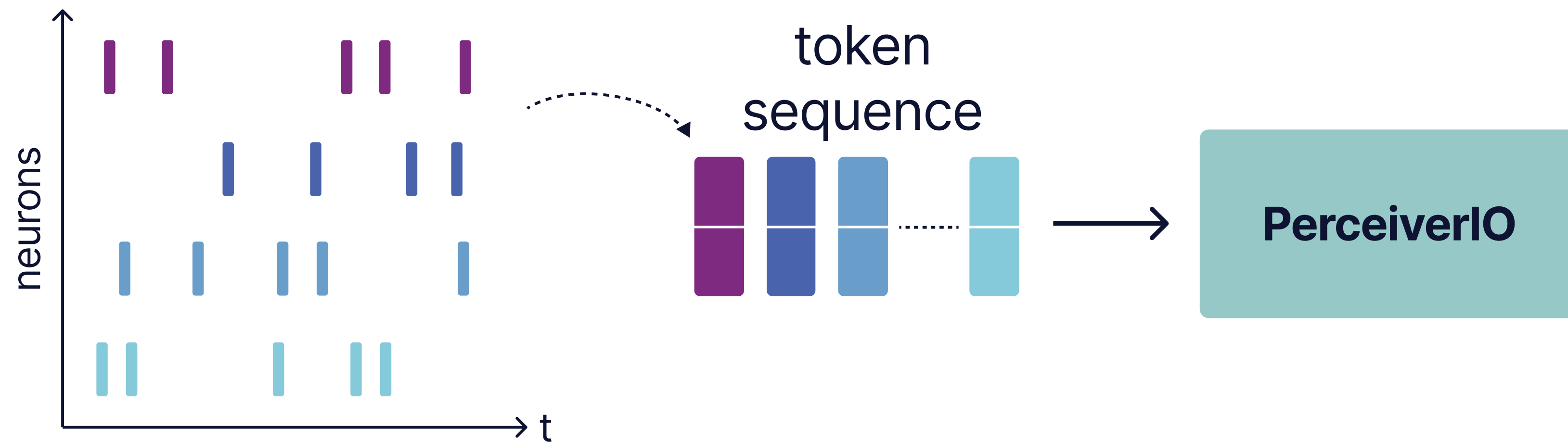
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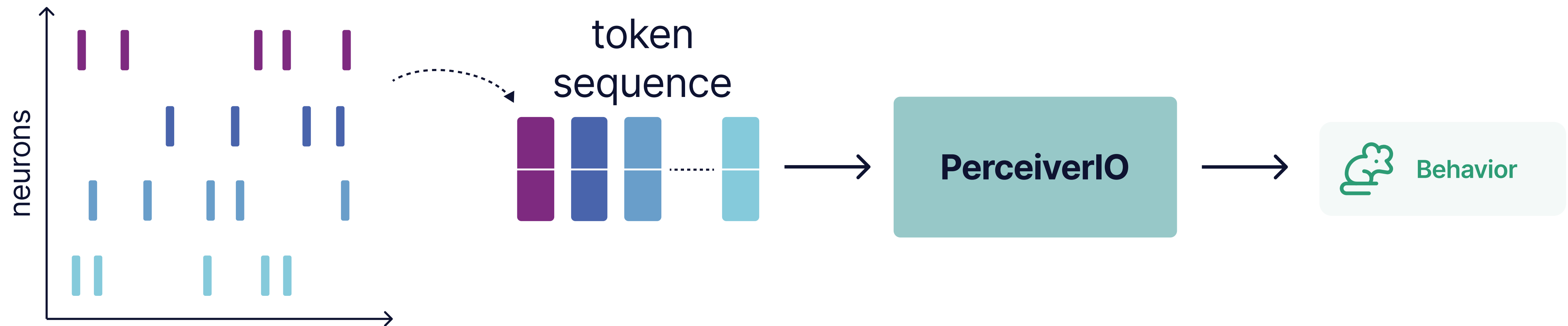
# POYO (Azabou et al., NeurIPS 2023)

PerceiverIO compresses spikes via a latent bottleneck



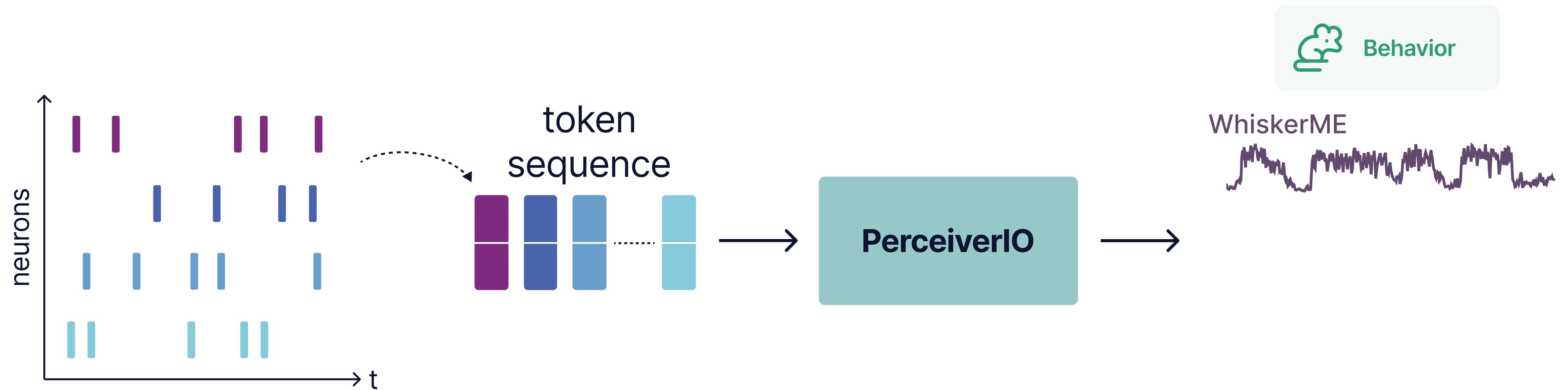
# POYO (Azabou et al., NeurIPS 2023)

Behavior queries read out any variable at any time



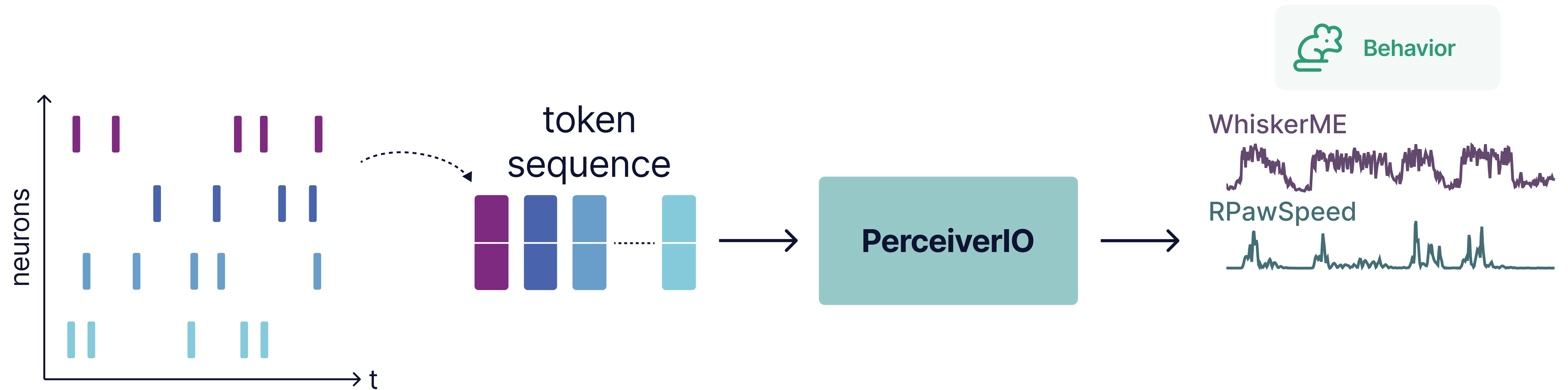
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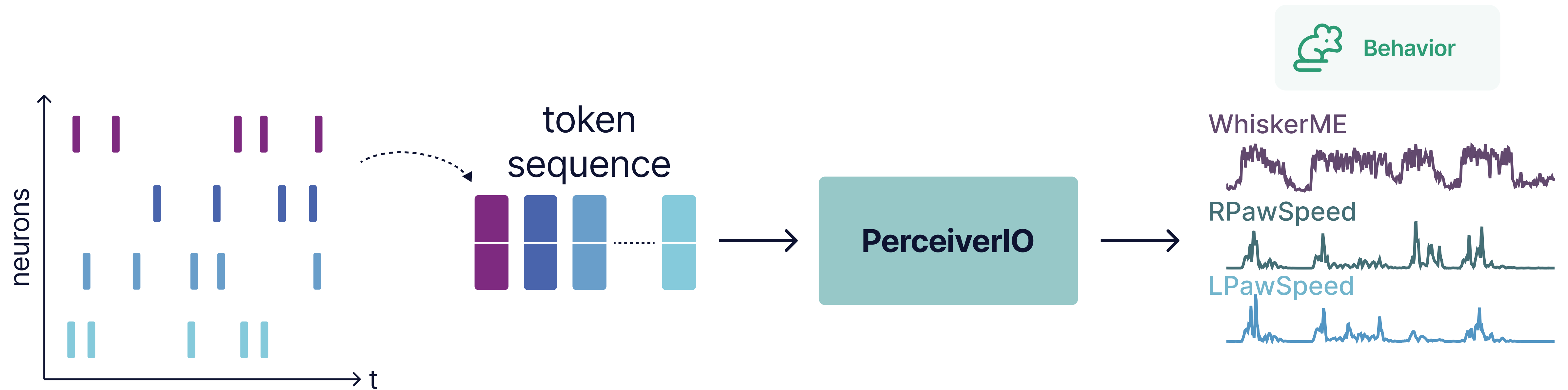
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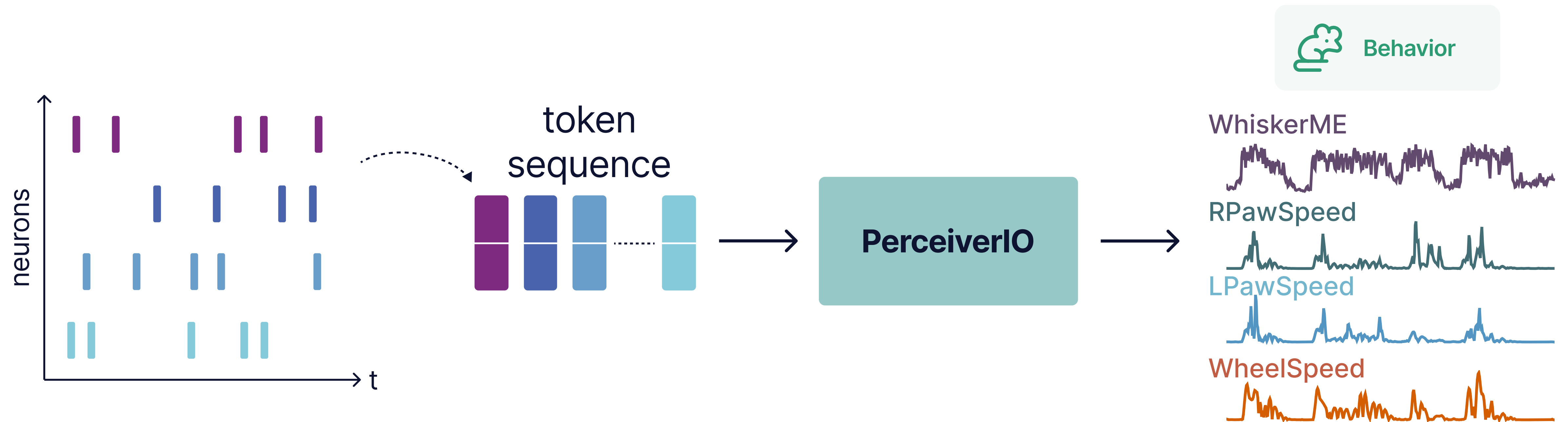
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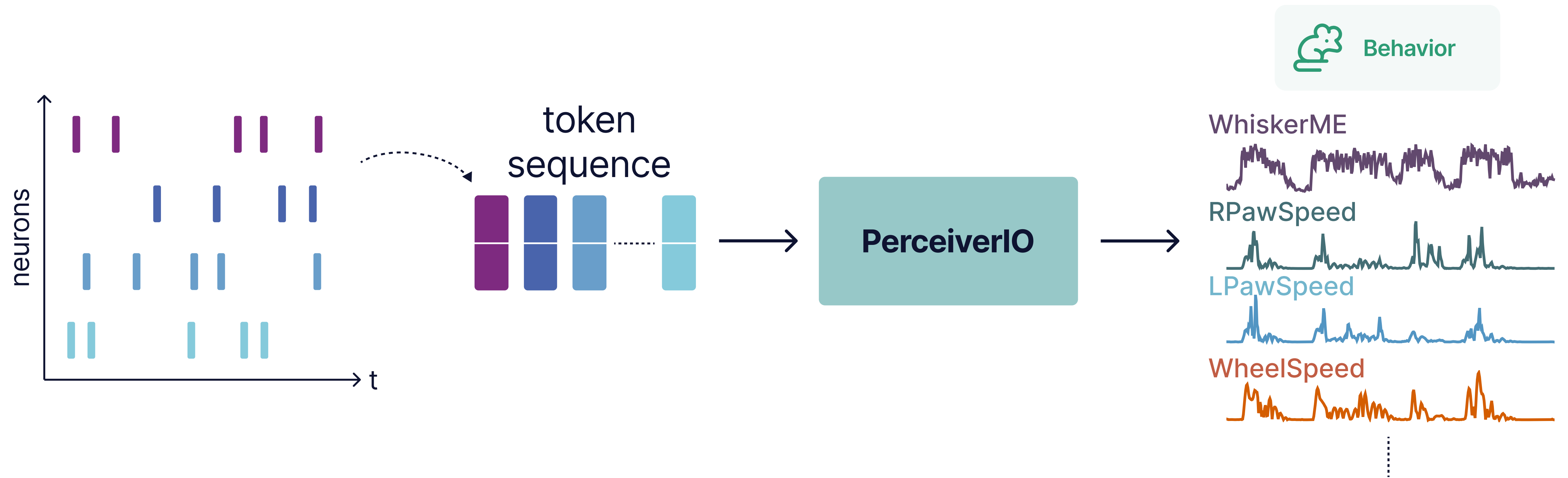
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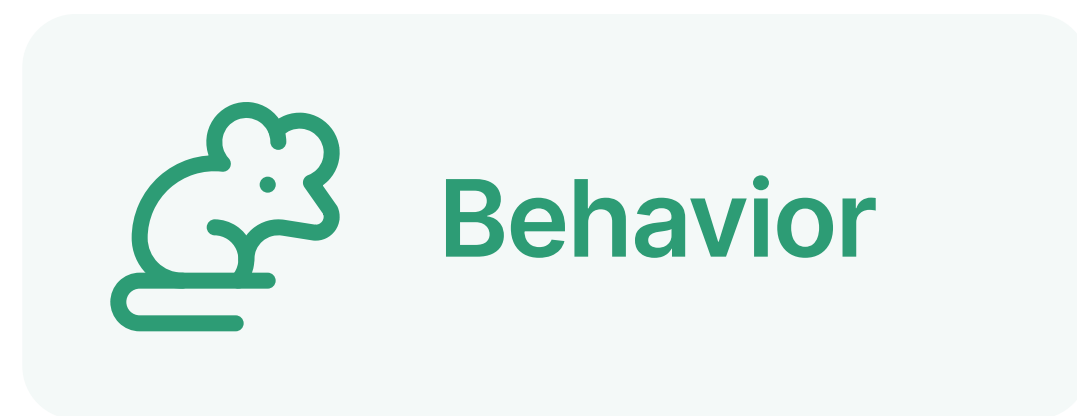
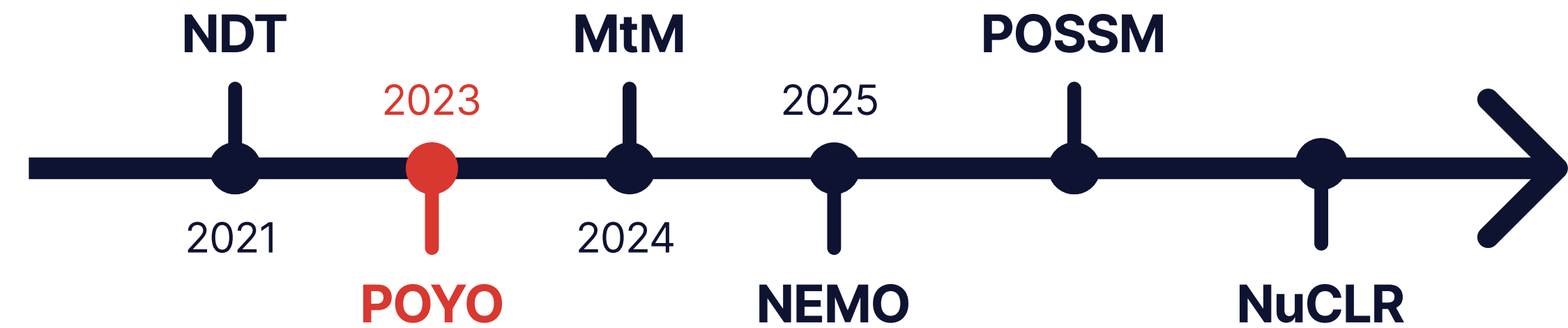


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# POYO (Azabou et al., NeurIPS 2023)

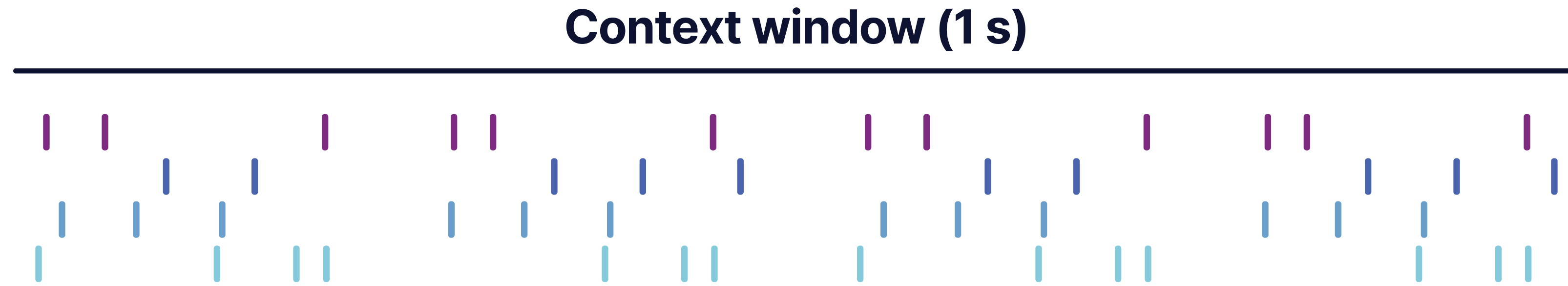


## Key features

- Each spike is a token (unit identity + spike time)
- PerceiverIO compresses spikes via a latent bottleneck
- Behavior queries read out any variable at any time

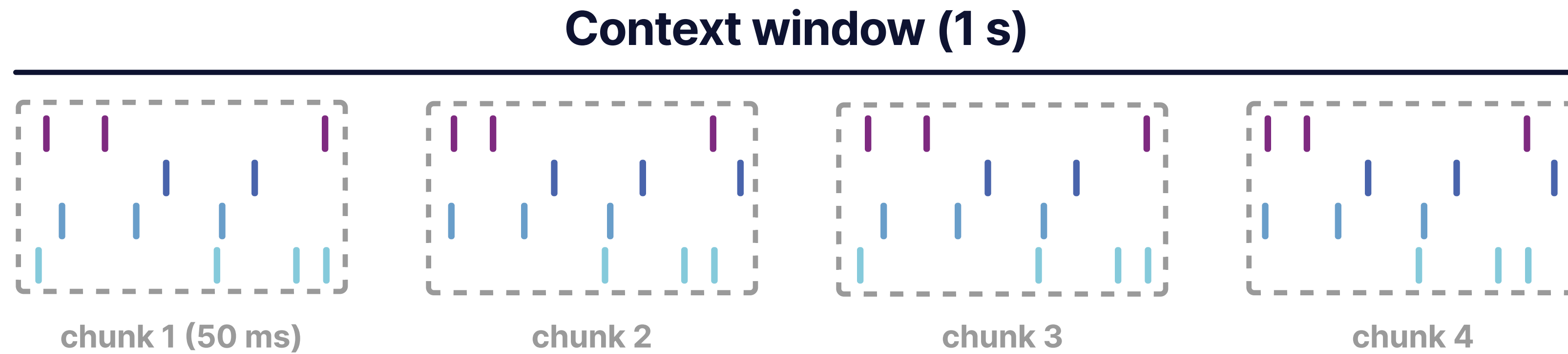
# POSSM (Ryoo et al., NeurIPS 2025)

POYO's spike tokens, compressed per 50 ms chunk



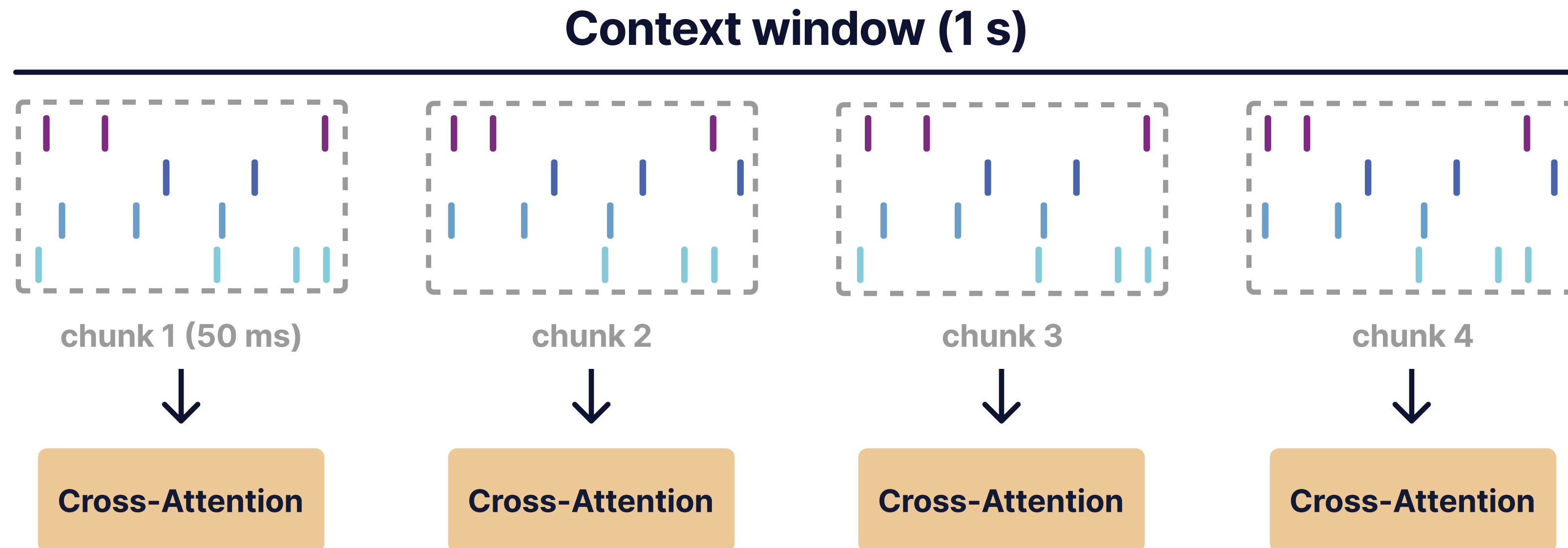
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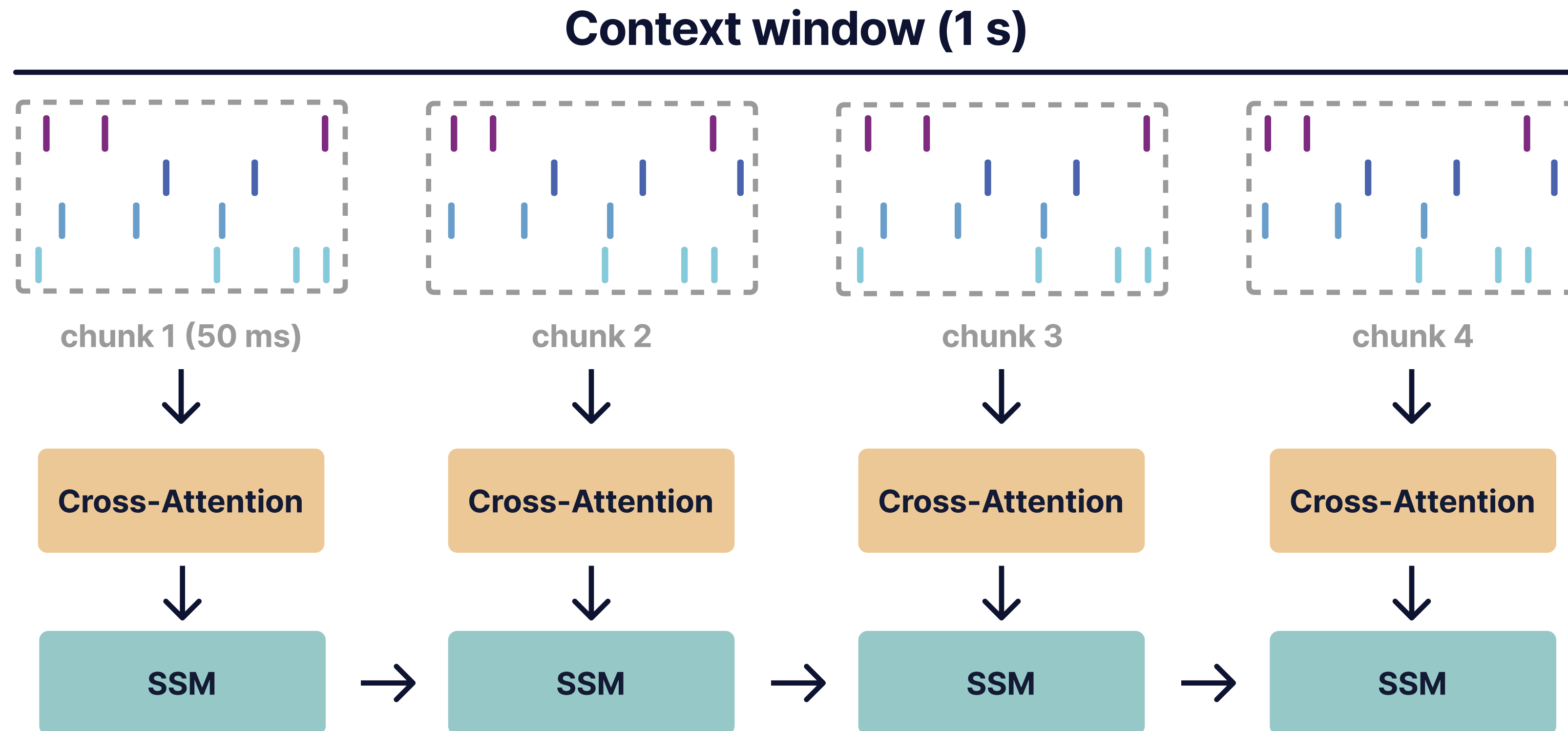
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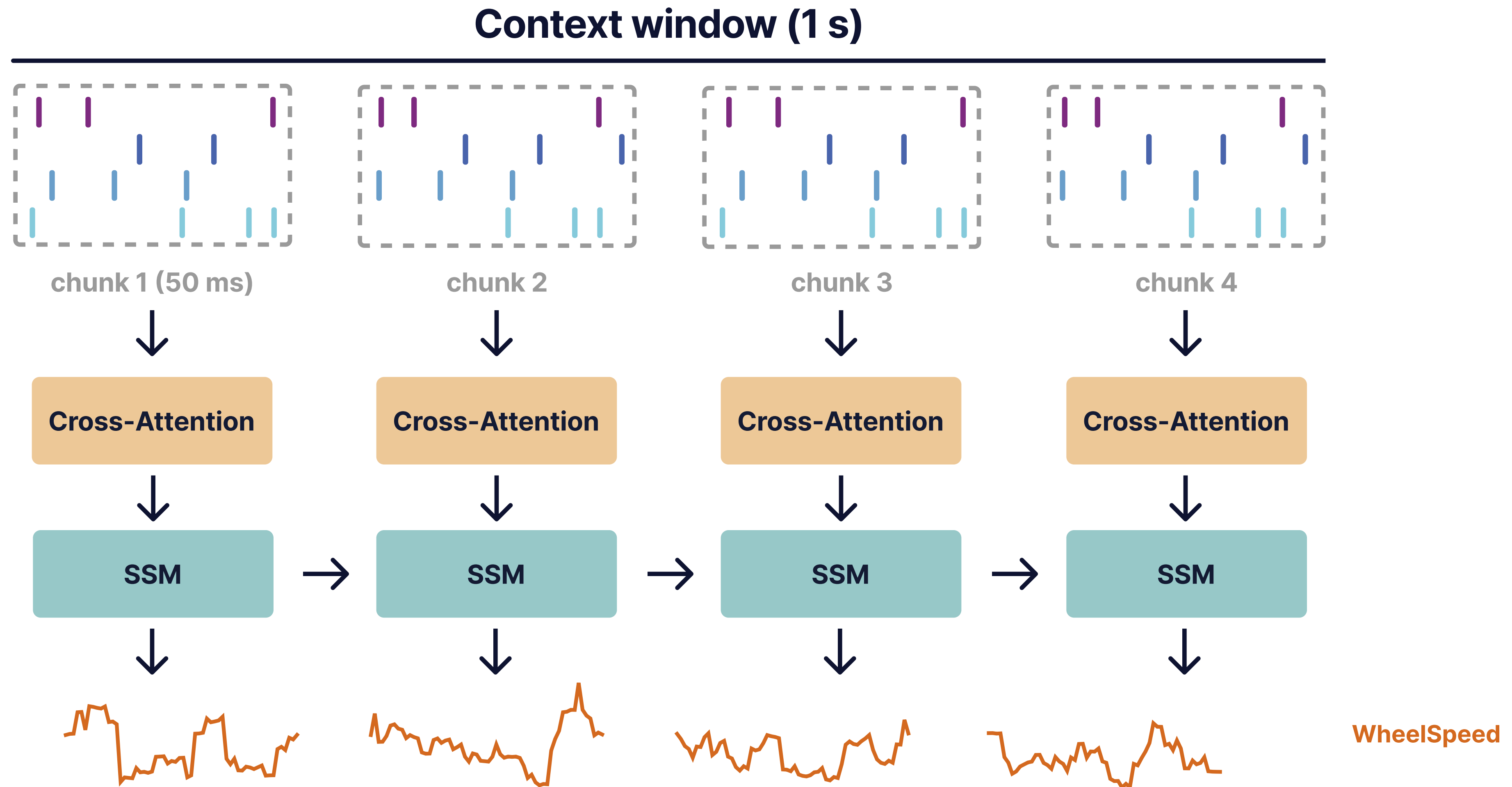
# POSSM (Ryoo et al., NeurIPS 2025)

Core backbone is an SSM that carries context across chunks

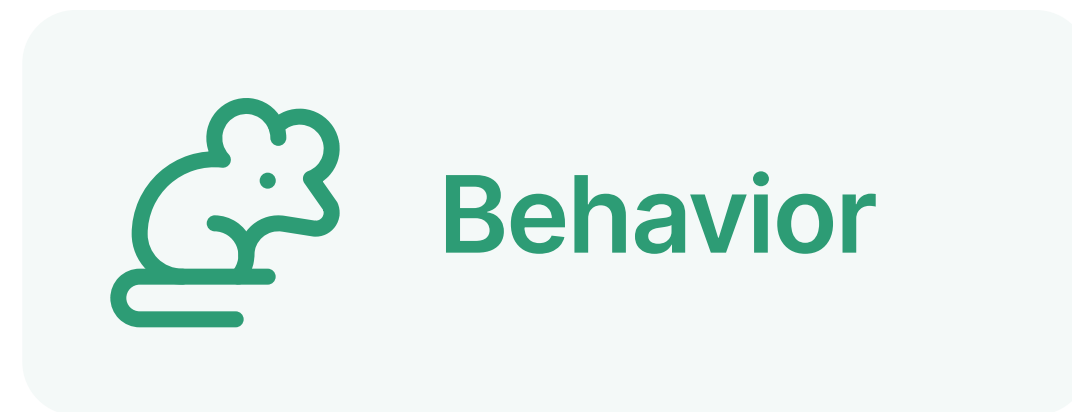
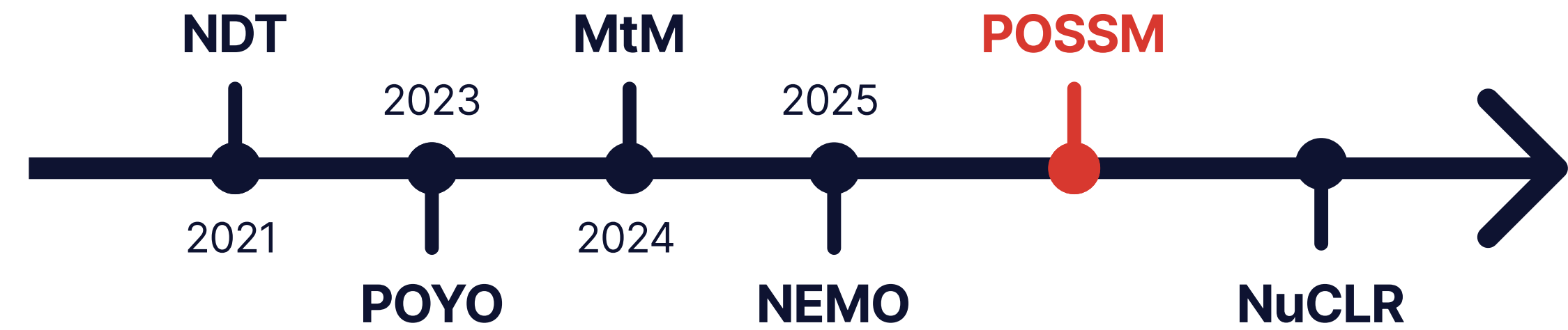


# POSSM (Ryoo et al., NeurIPS 2025)

Designed for real-time decoding (BCIs)



# POSSM (Ryoo et al., NeurIPS 2025)



## Key features

- POYO's spike tokens, compressed per 50 ms chunk
- Core backbone is an SSM that carries context across chunks
- Designed for real-time decoding (BCIs)

# Pretrained Baselines Within the Benchmark Framework

 Model Input     Model Target

NDT



POYO · POSSM



MtM



NuCLR · NEMO



# Pretrained Baselines Within the Benchmark Framework

 Model Input     Model Target

NDT



POYO · POSSM



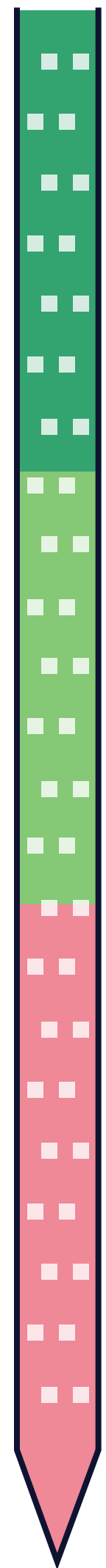
MtM



NuCLR · NEMO

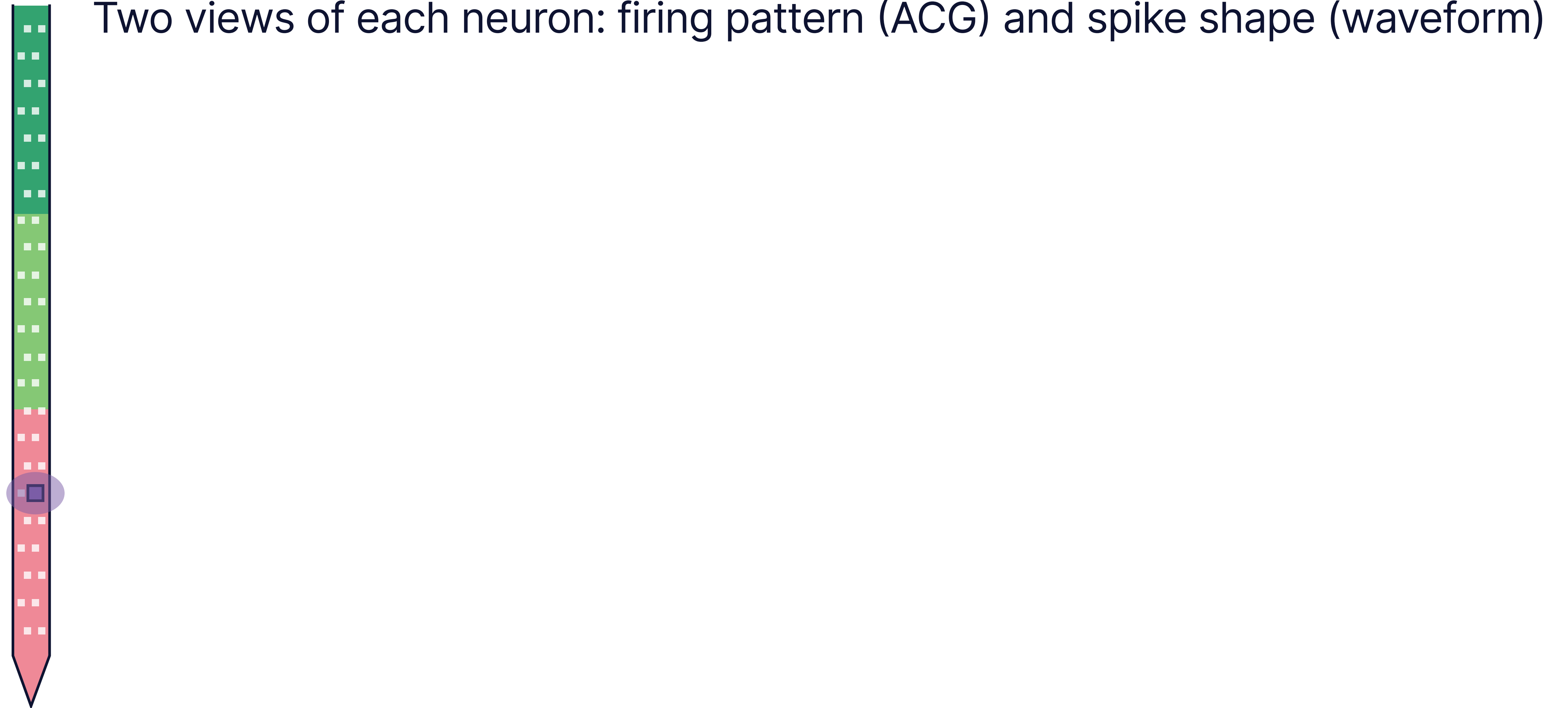


# NEMO (Yu et al., ICLR 2025)



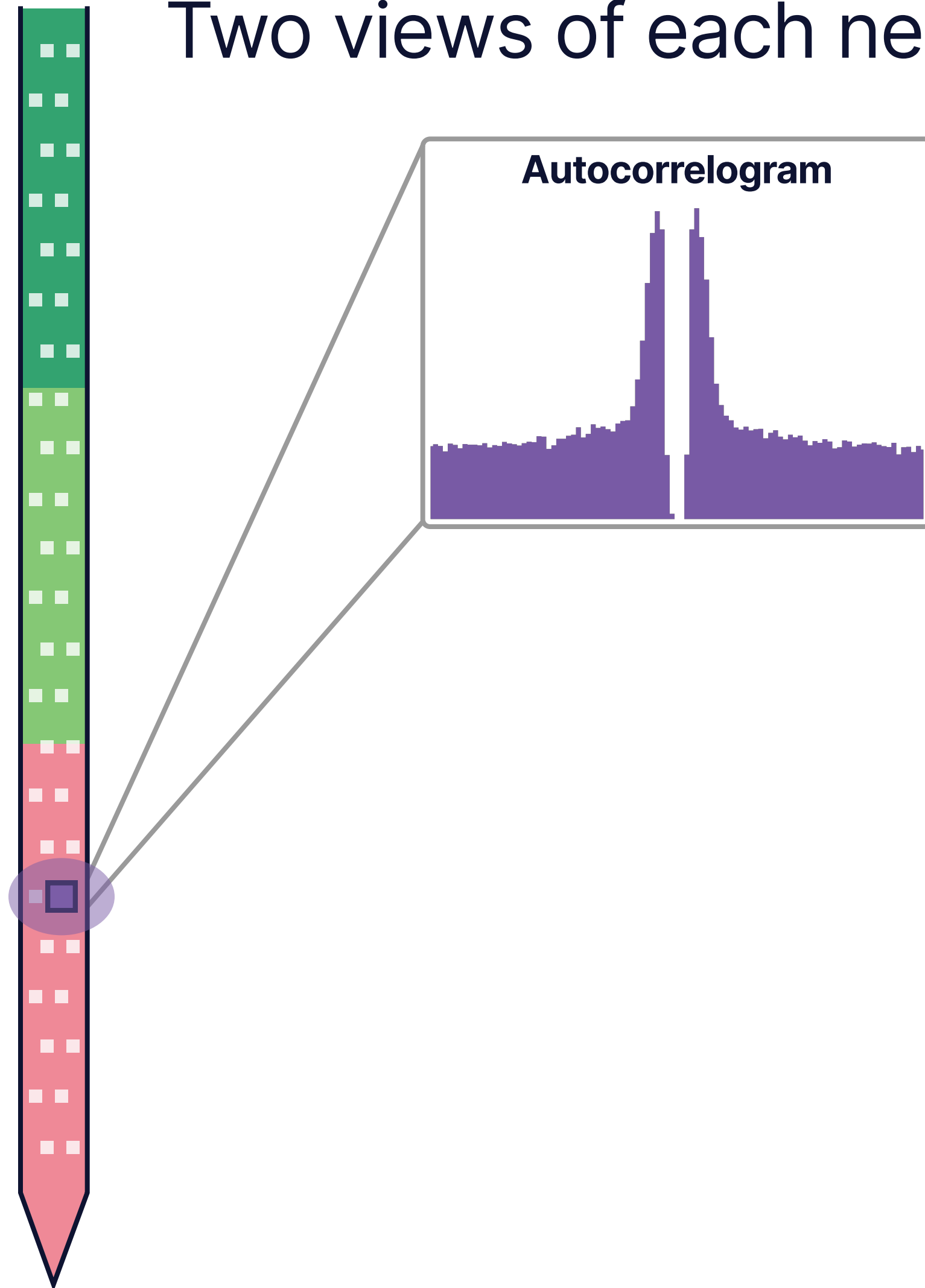
Two views of each neuron: firing pattern (ACG) and spike shape (waveform)

# NEMO (Yu et al., ICLR 2025)



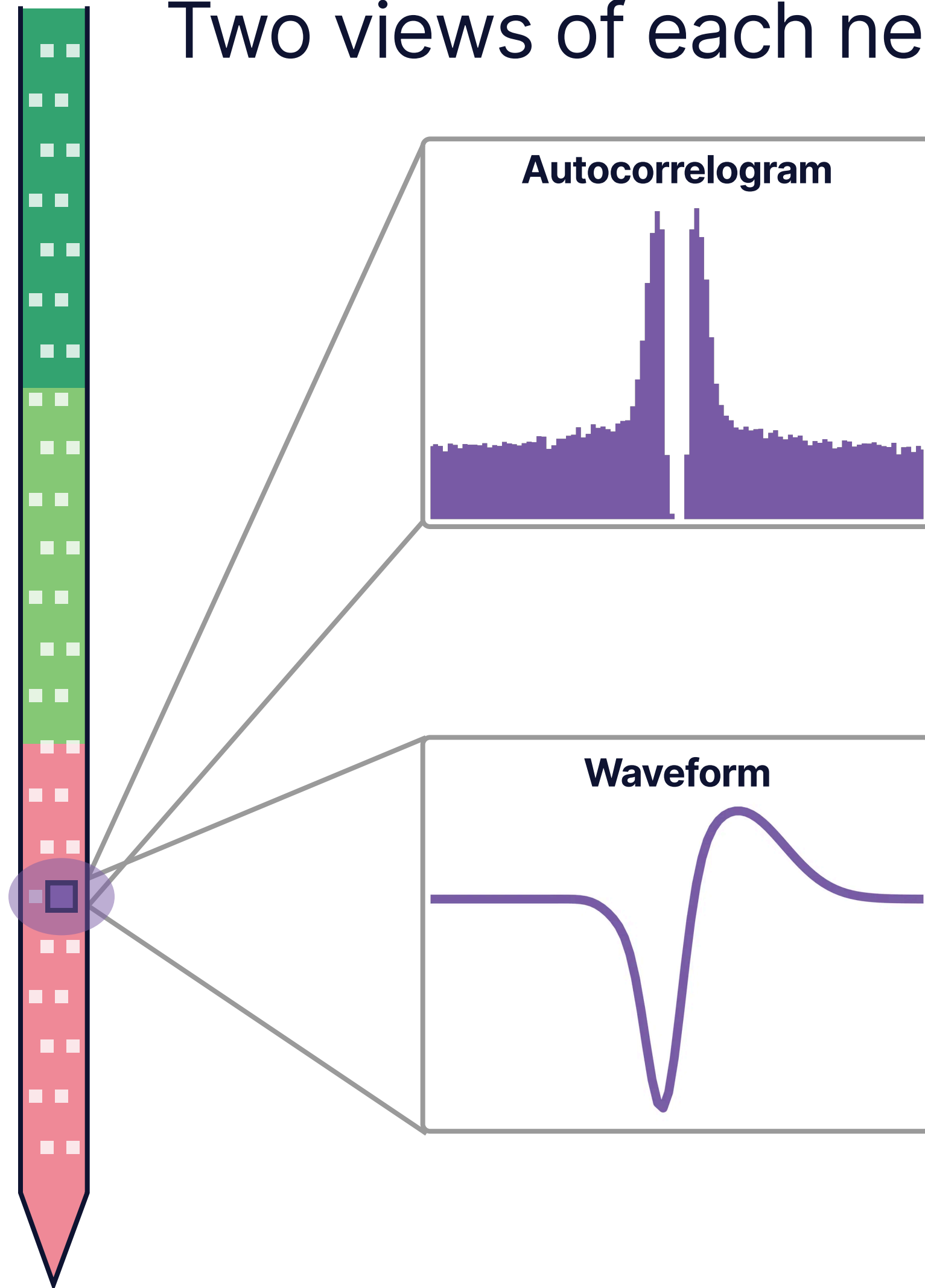
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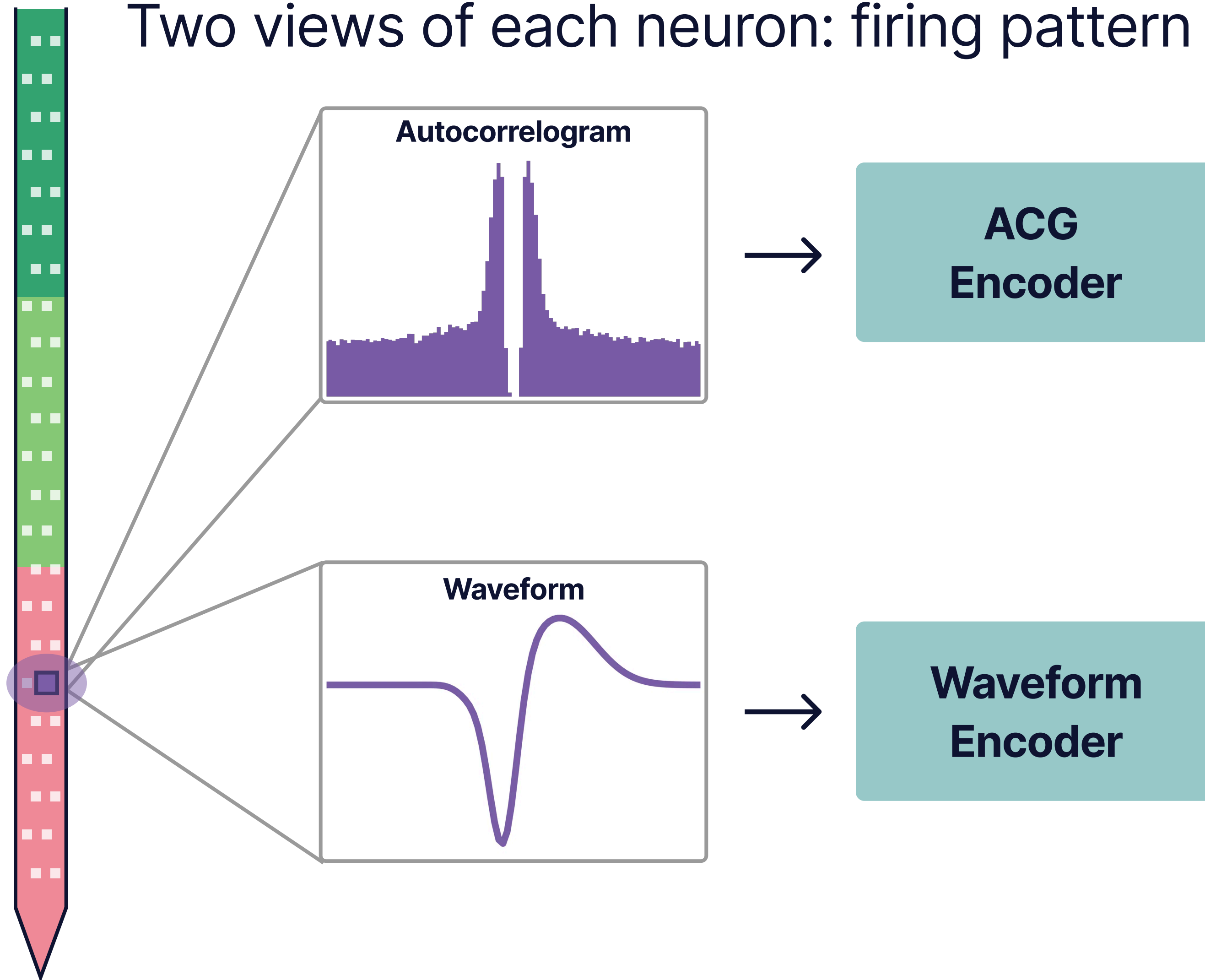
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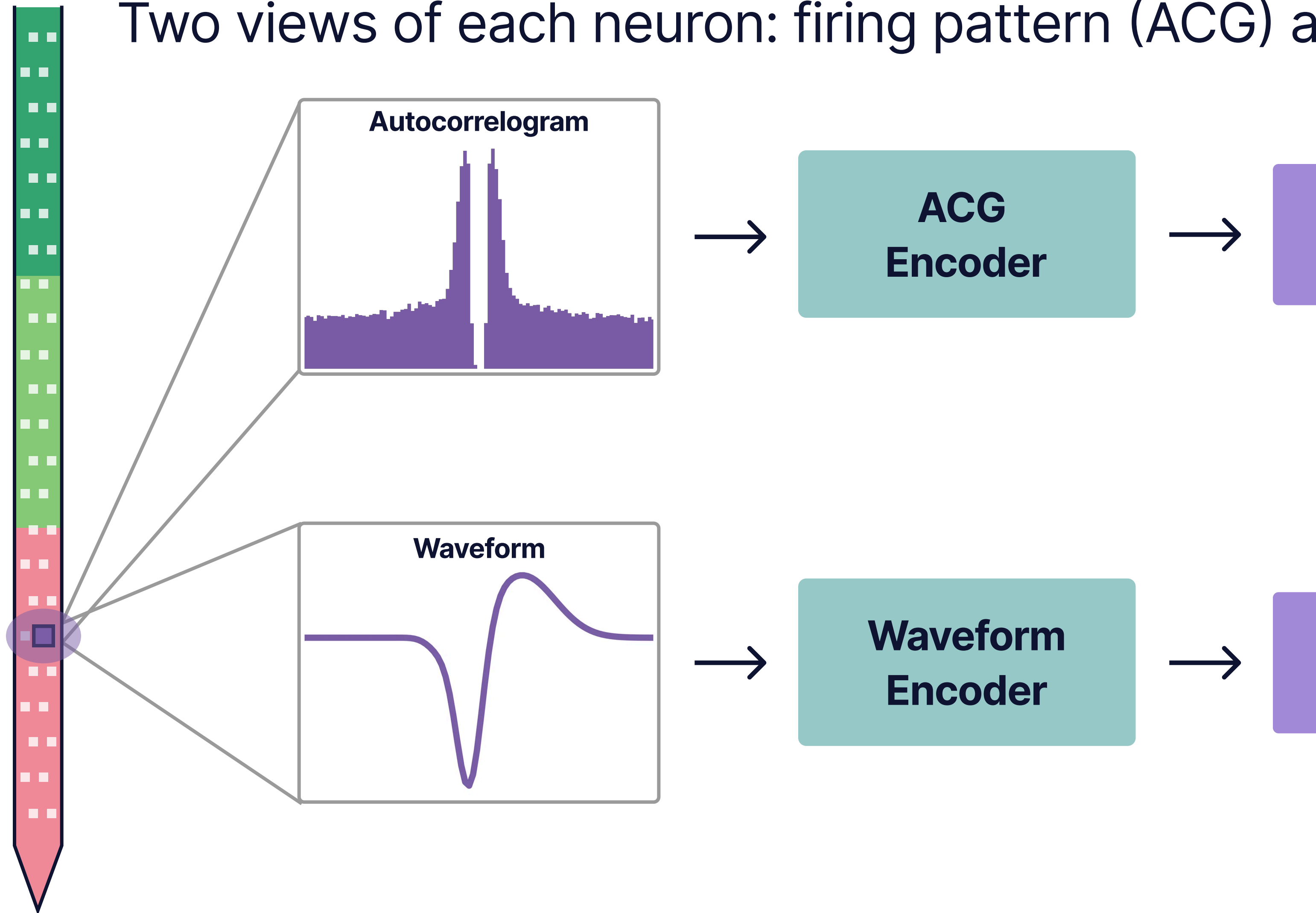
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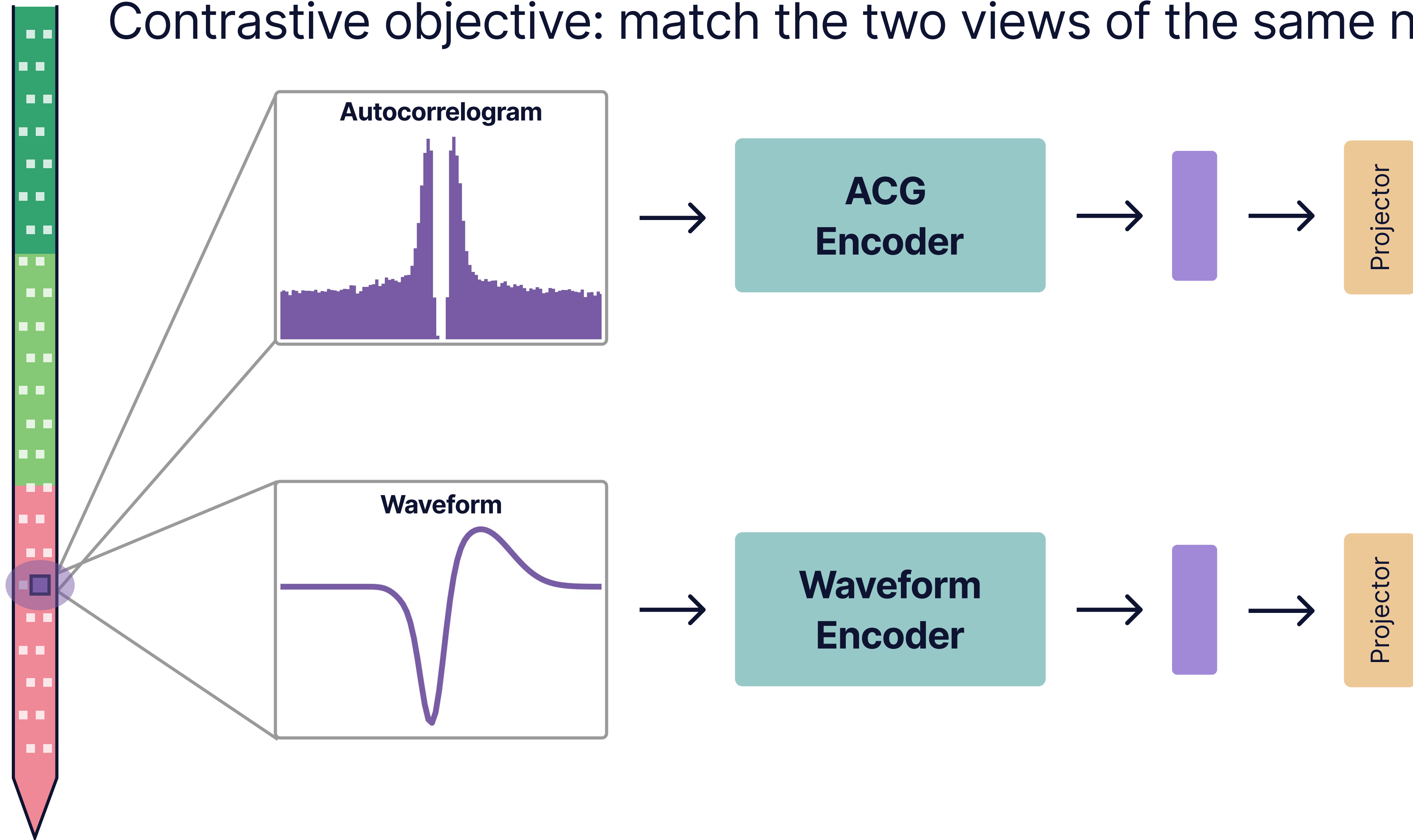
# NEMO (Yu et al., ICLR 2025)

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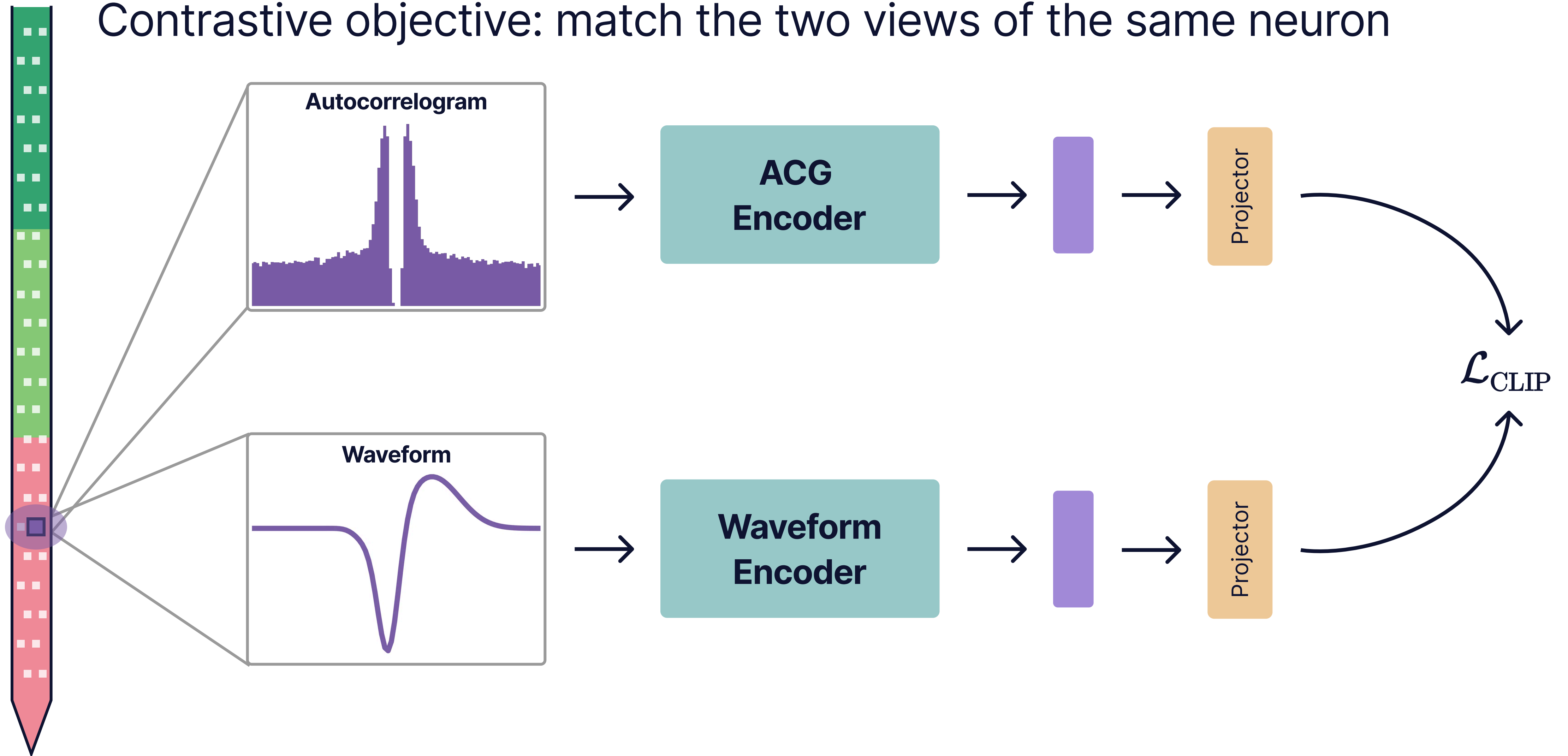
# NEMO (Yu et al., ICLR 2025)

Contrastive objective: match the two views of the same neuron



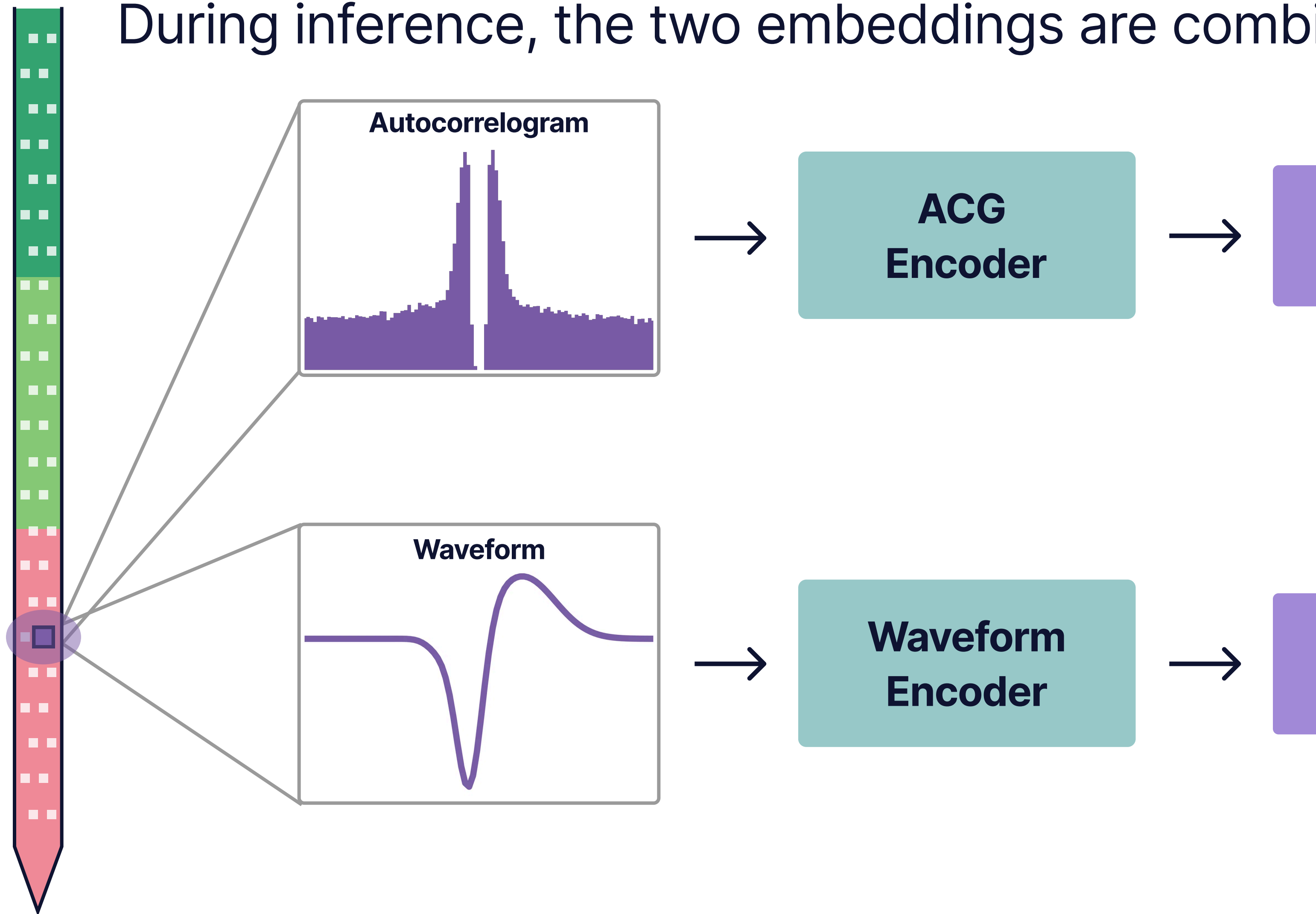
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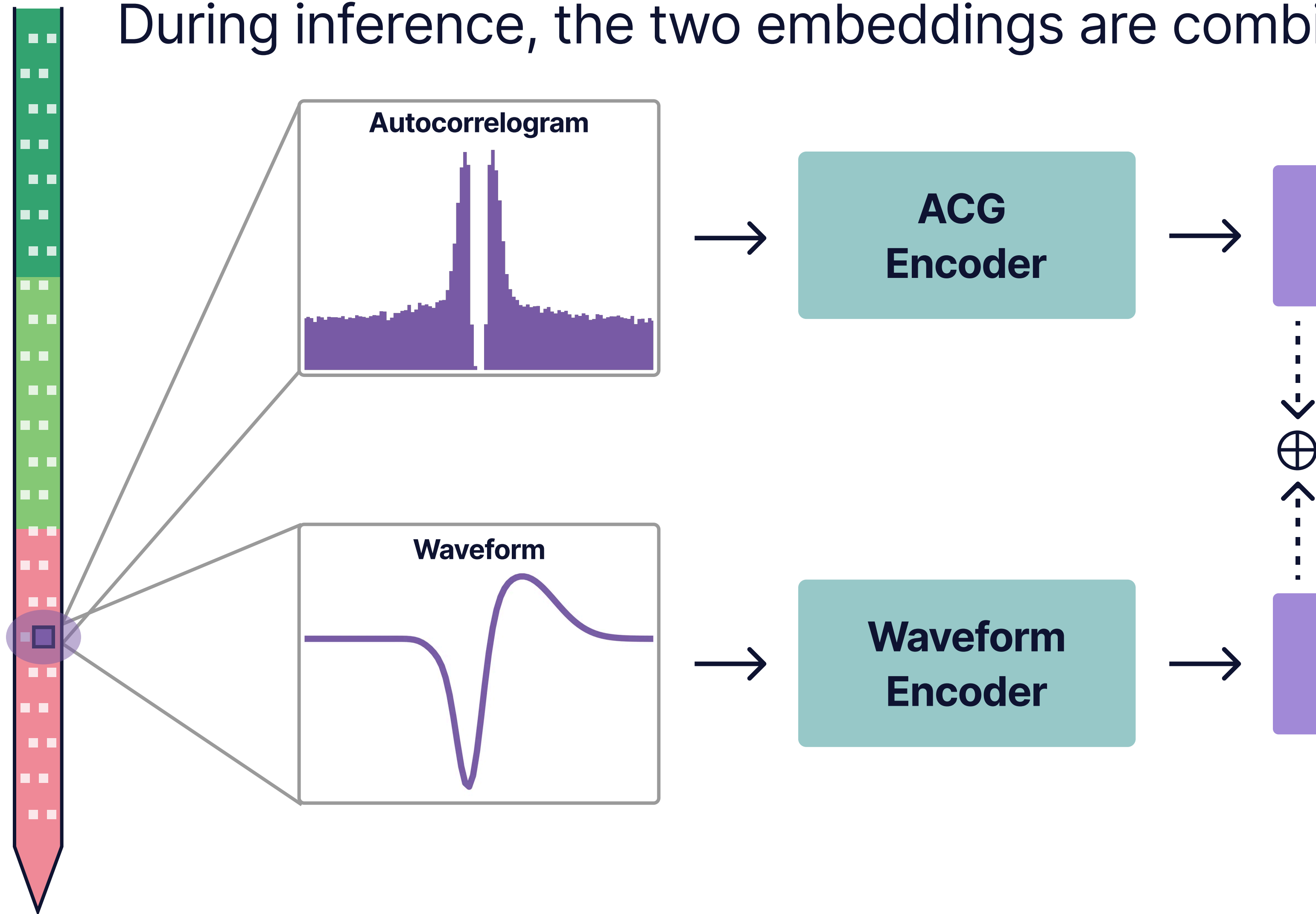
# NEMO (Yu et al., ICLR 2025)

During inference, the two embeddings are combined to represent units



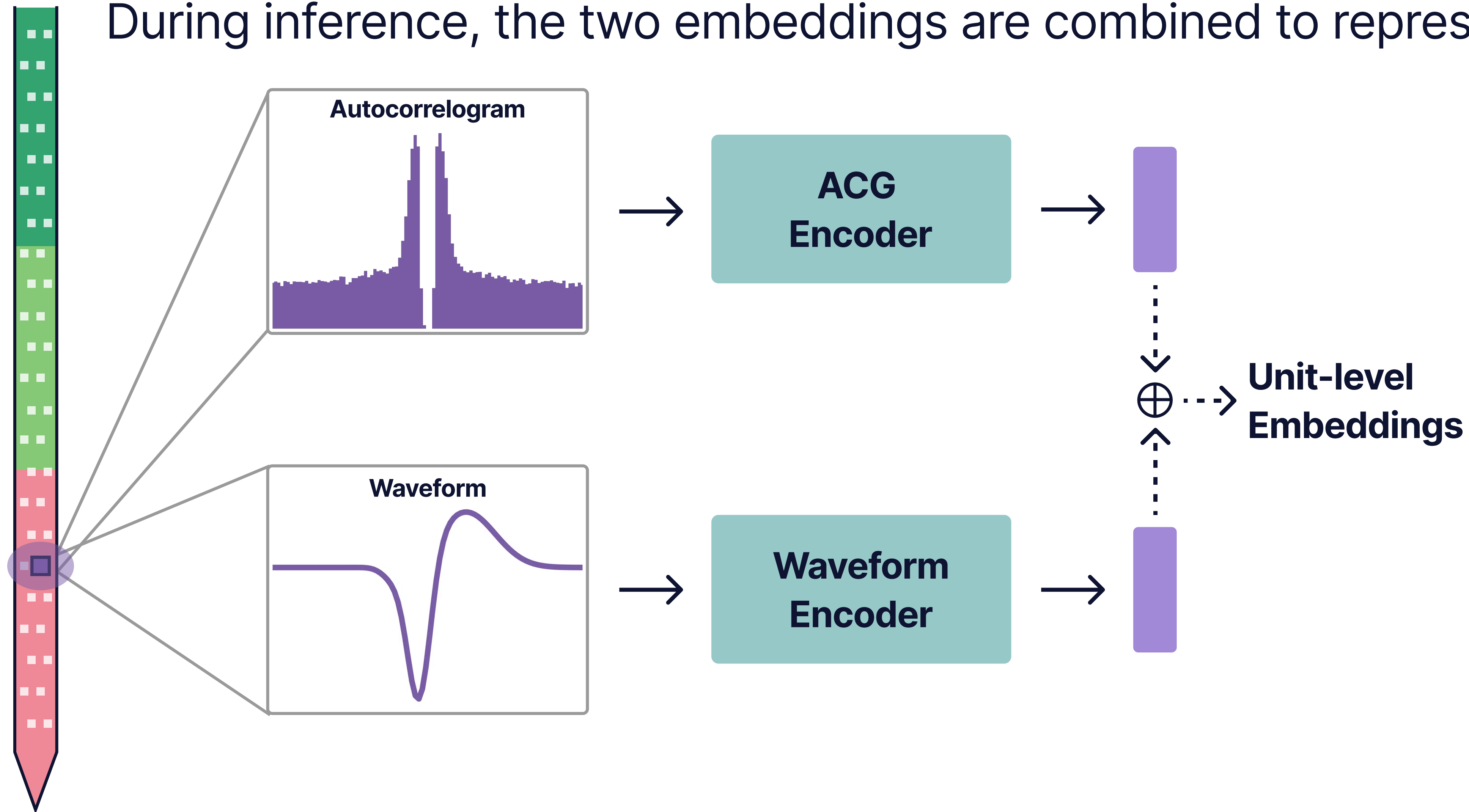
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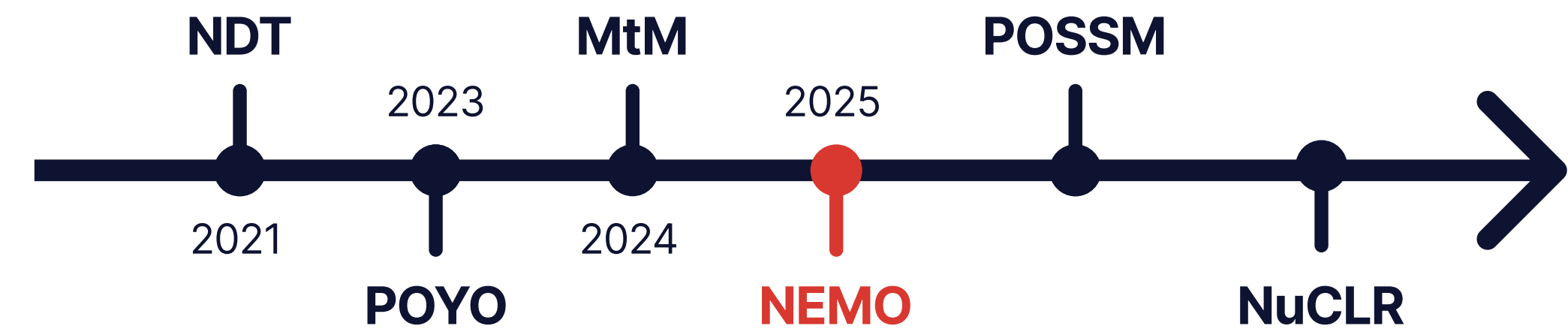


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# NEMO (Yu et al., ICLR 2025)



## Key features

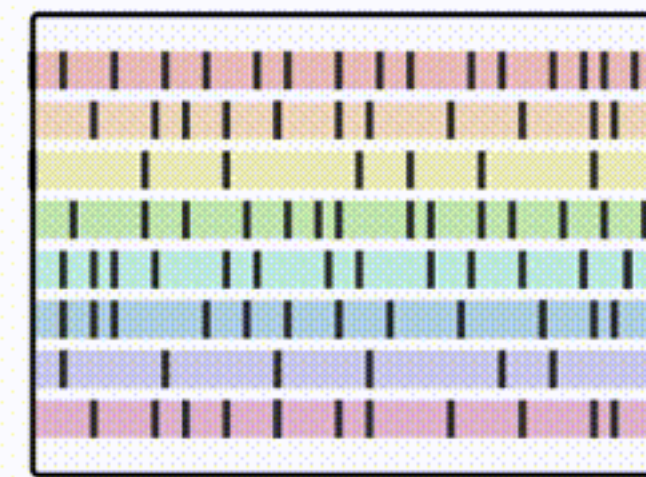
- Two views of each neuron: ACG and waveform
- Contrastive objective: match the two views of the same neuron
- During inference, the two embeddings are combined to represent units

# NuCLR (Arora et al., NeurIPS 2025)

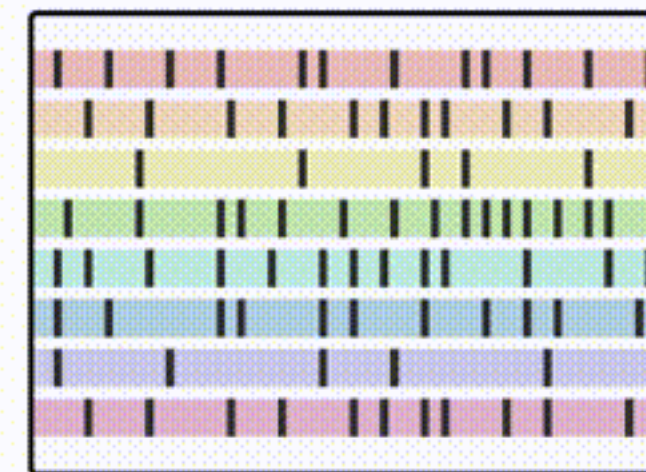
Two views: the same population of neurons in windows  $\Delta T$  apart

# NuCLR (Arora et al., NeurIPS 2025)

Two views: the same population of neurons in windows  $\Delta T$  apart



View 1



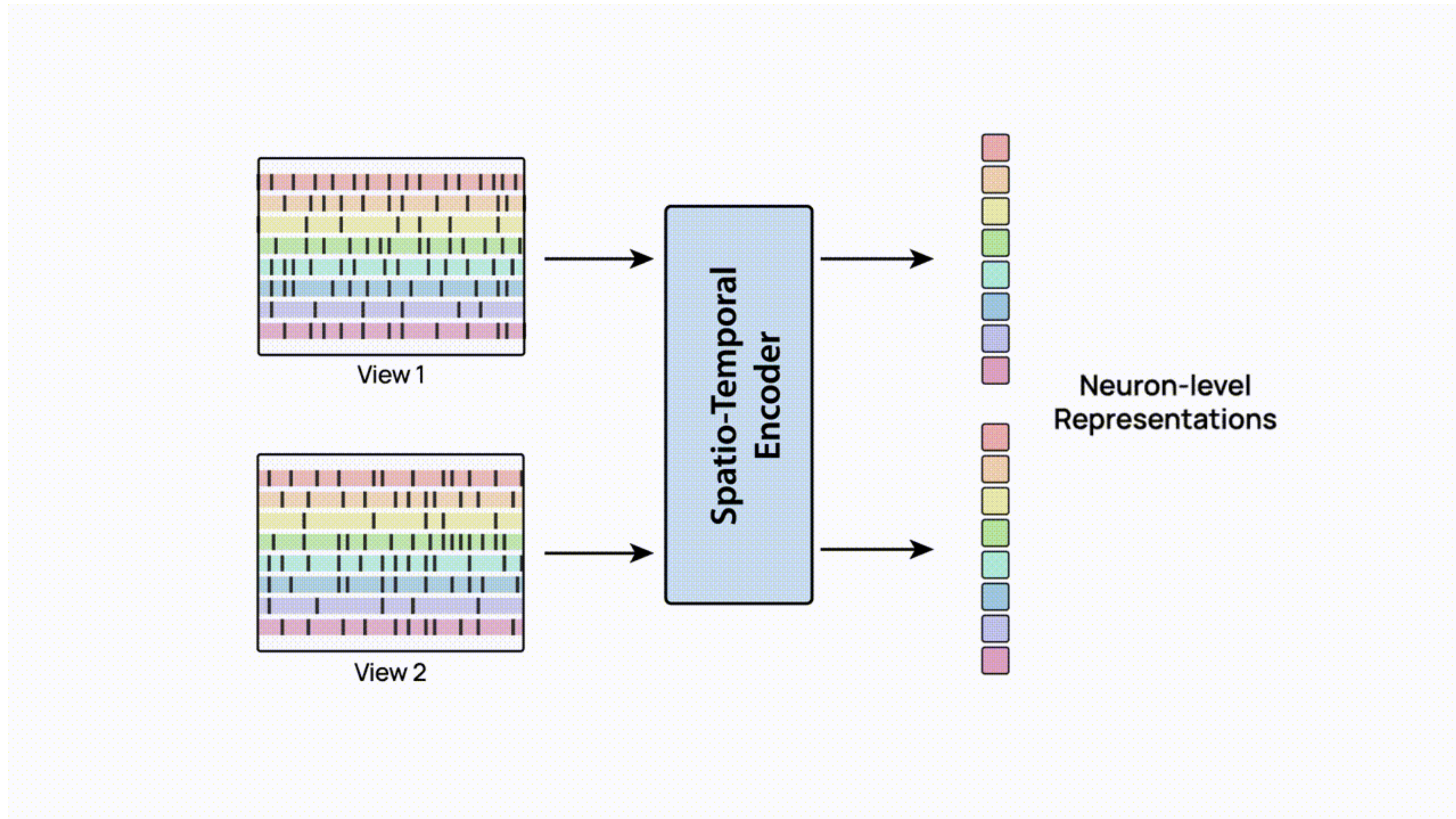
View 2

# NuCLR (Arora et al., NeurIPS 2025)

Each neuron is embedded in the context of its population

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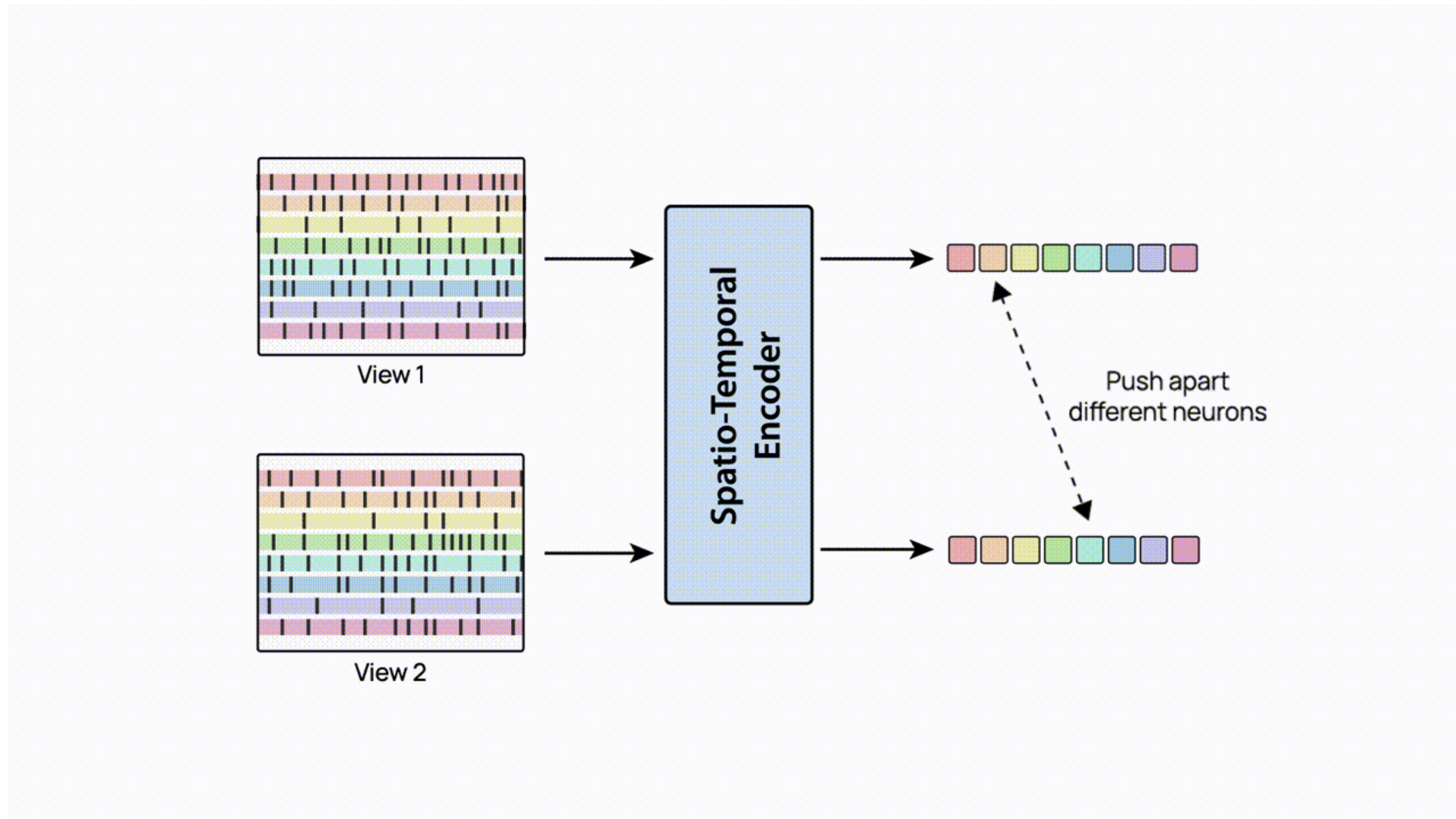


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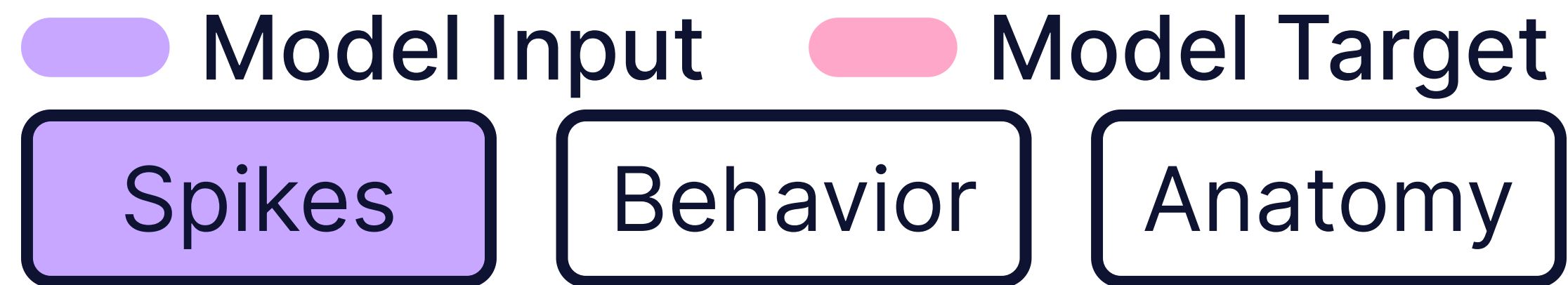
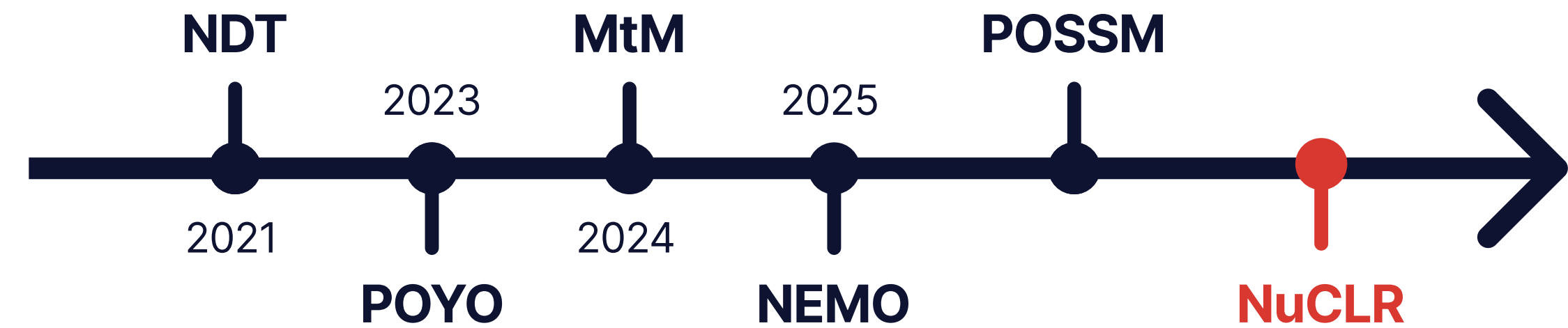
Contrastive objective: match each neuron to itself across the two views

# NuCLR (Arora et al., NeurIPS 2025)

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# NuCLR (Arora et al., NeurIPS 2025)



## Key features

- Two views: the same population of neurons in windows  $\Delta T$  apart
- Each neuron is embedded in the context of its population
- Contrastive objective: match each neuron to itself across the two views

# Pretrained Baselines Within the Benchmark Framework

 Model Input     Model Target

NDT



POYO · POSSM



MtM



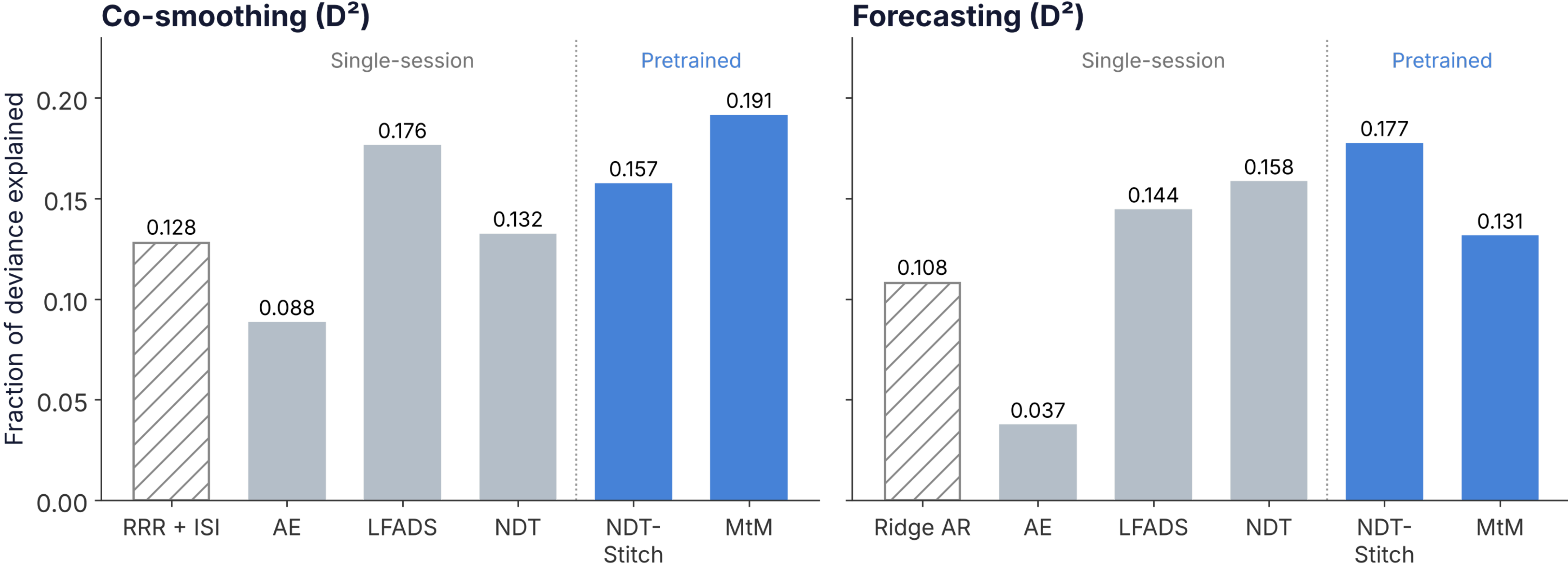
NuCLR · NEMO



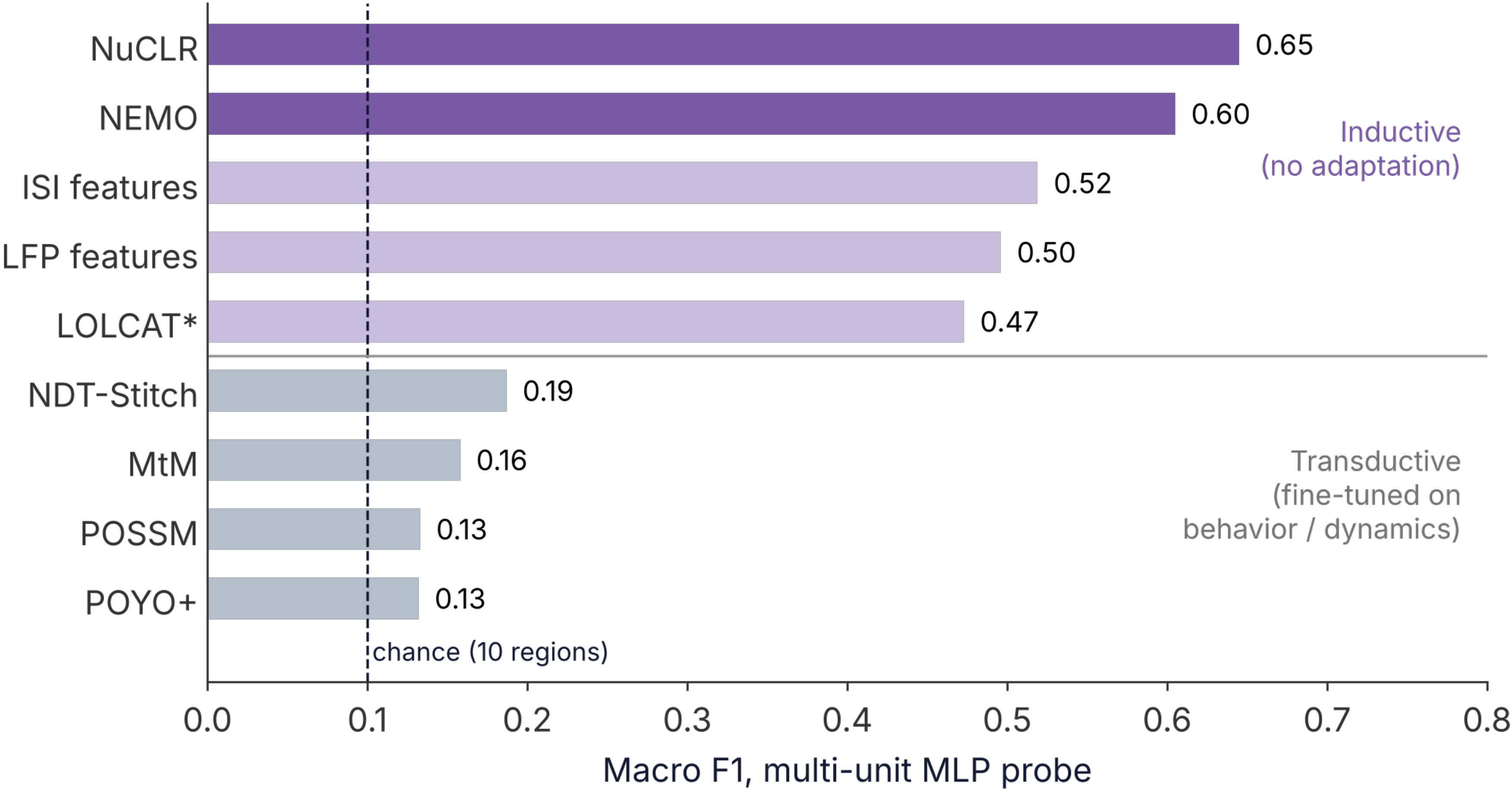
# Baseline results on TS1 (Decoding)

|            | Frame-level ( $R^2$ / $D^2$ ) |         |       |       |       | Sequence-level (bal. acc.) |        |          | Avg. rank ↓      |
|------------|-------------------------------|---------|-------|-------|-------|----------------------------|--------|----------|------------------|
|            | Licks                         | Whisker | Wheel | R paw | L paw | Reward                     | Choice | Contrast |                  |
| Linear     | 0.15                          | 0.17    | 0.22  | 0.09  | 0.08  | 0.81                       | 0.61   | 0.22     | <div></div> 7.73 |
| MLP        | 0.35                          | 0.27    | 0.33  | 0.16  | 0.14  | 0.88                       | 0.64   | 0.22     | <div></div> 5.46 |
| GRU        | 0.51                          | 0.36    | 0.30  | 0.21  | 0.18  | 0.89                       | 0.64   | 0.21     | <div></div> 3.96 |
| CNN        | 0.55                          | 0.39    | 0.33  | 0.23  | 0.19  | 0.89                       | 0.62   | 0.20     | <div></div> 3.74 |
| CEBRA      | 0.33                          | 0.28    | 0.20  | 0.16  | 0.12  | 0.78                       | 0.52   | 0.20     | <div></div> 7.44 |
| NDT        | 0.31                          | 0.30    | 0.21  | 0.16  | 0.13  | 0.87                       | 0.62   | 0.21     | <div></div> 6.80 |
| POYO       | 0.56                          | 0.29    | 0.30  | 0.20  | 0.14  | 0.88                       | 0.63   | 0.22     | <div></div> 4.80 |
| POYO+      | 0.68                          | 0.38    | 0.36  | 0.25  | 0.20  | 0.90                       | 0.69   | 0.27     | <div></div> 2.53 |
| POSSM      | 0.66                          | 0.35    | 0.35  | 0.24  | 0.17  | 0.89                       | 0.66   | 0.22     | <div></div> 3.18 |
| NDT-Stitch | 0.53                          | 0.38    | 0.31  | 0.22  | 0.17  | 0.86                       | 0.67   | 0.23     | <div></div> 3.58 |
| MtM        | 0.47                          | 0.36    | 0.32  | 0.22  | 0.17  | 0.87                       | 0.65   | 0.21     | <div></div> 4.05 |

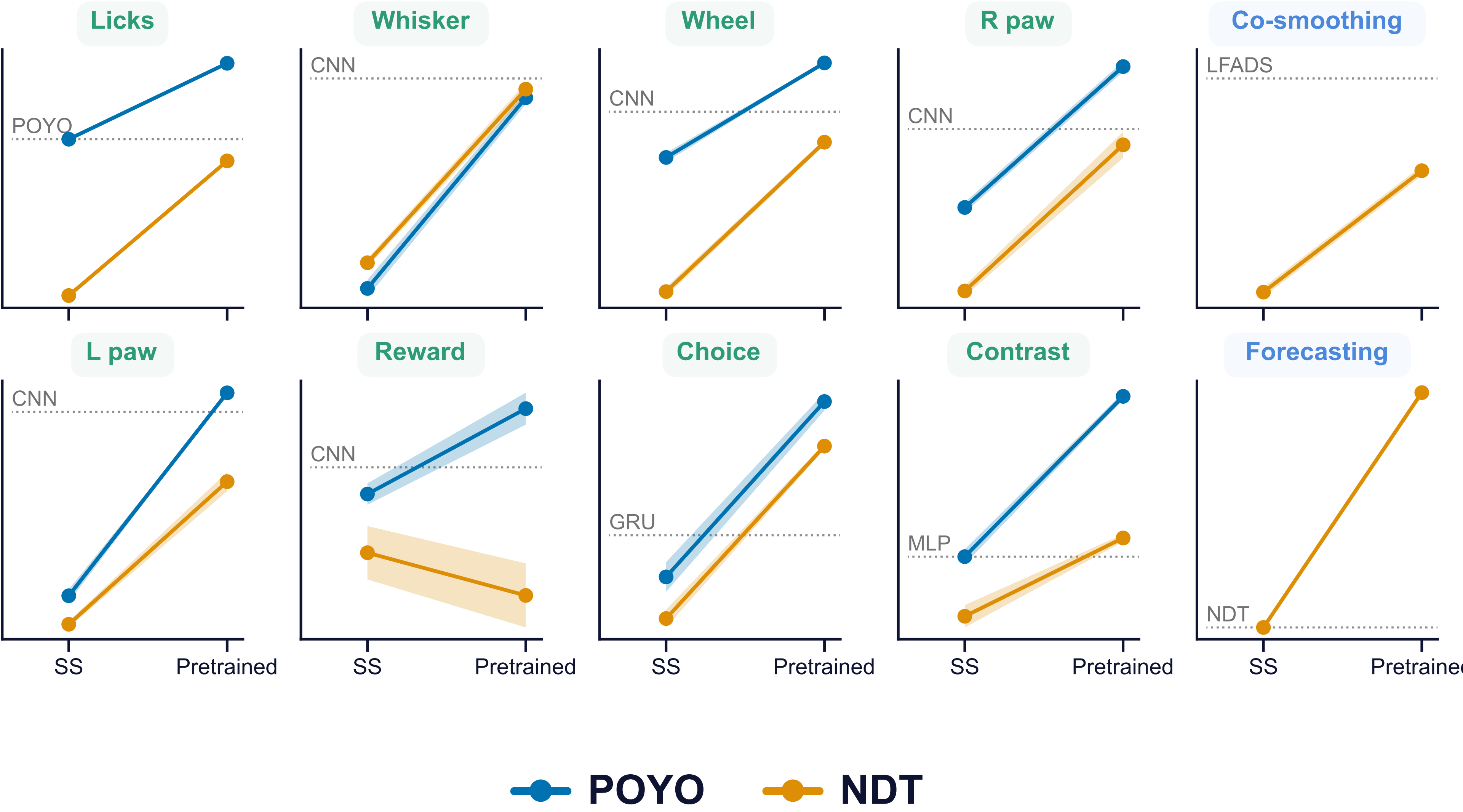
# Baseline results on TS2 (Activity Prediction)



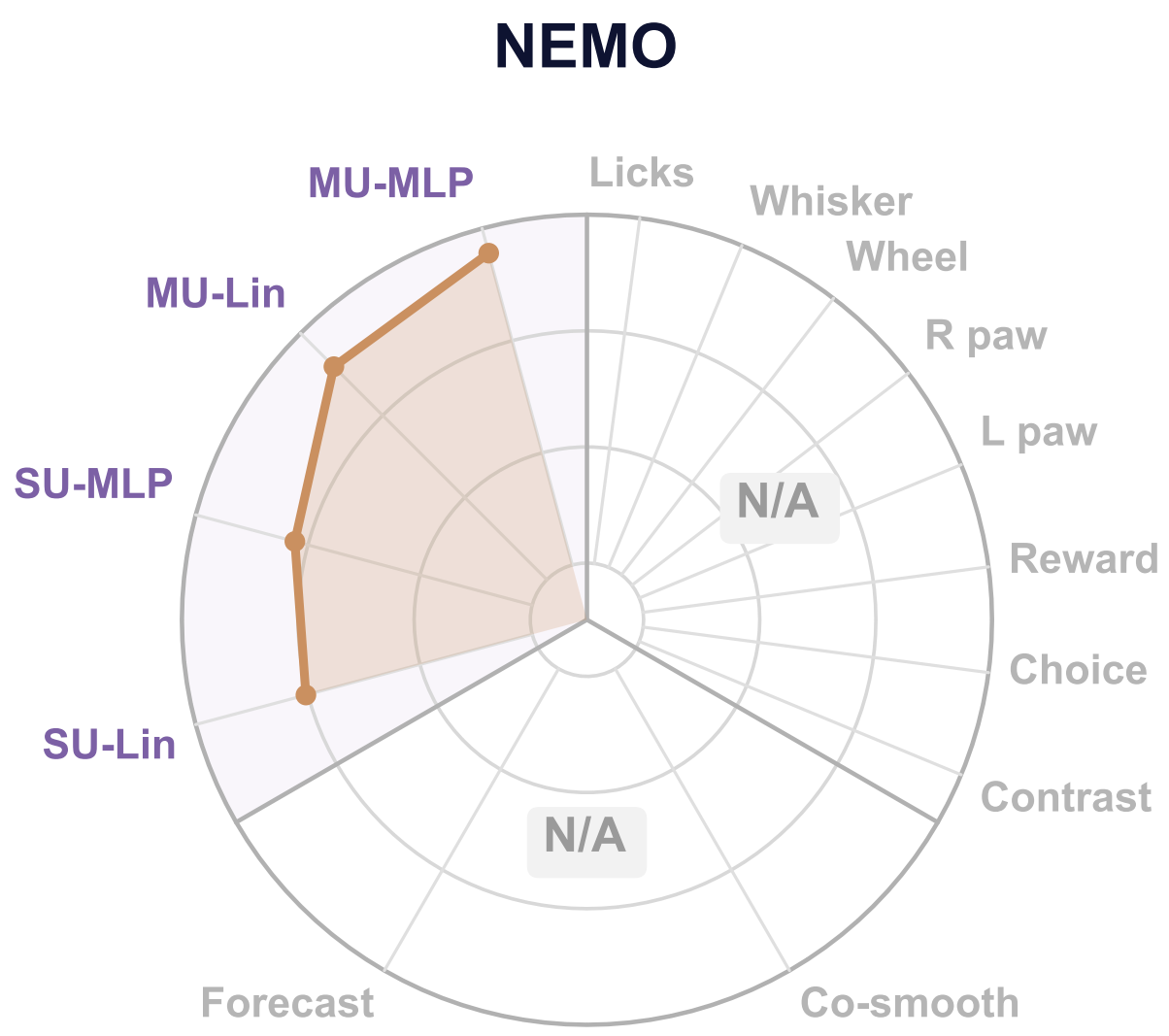
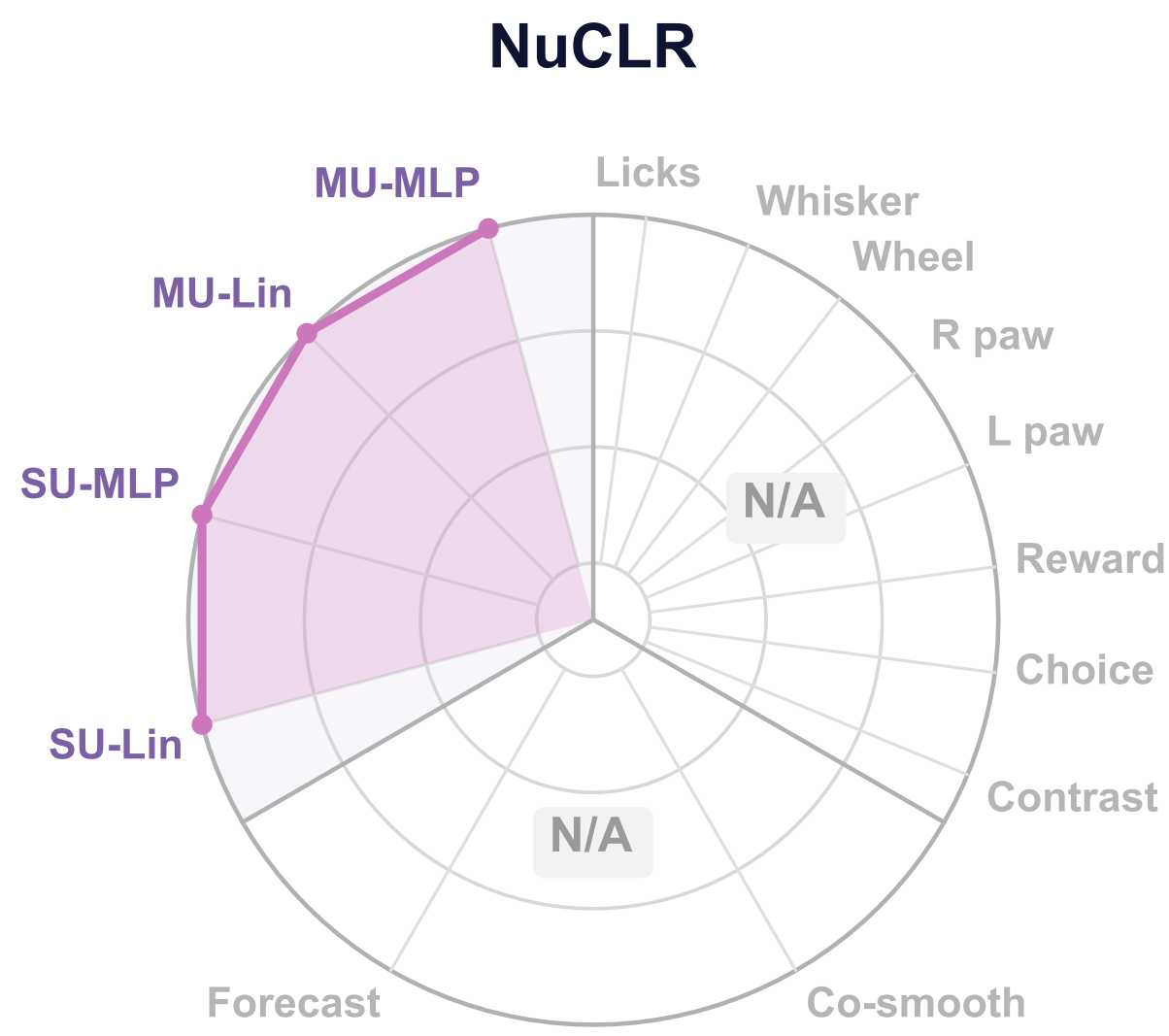
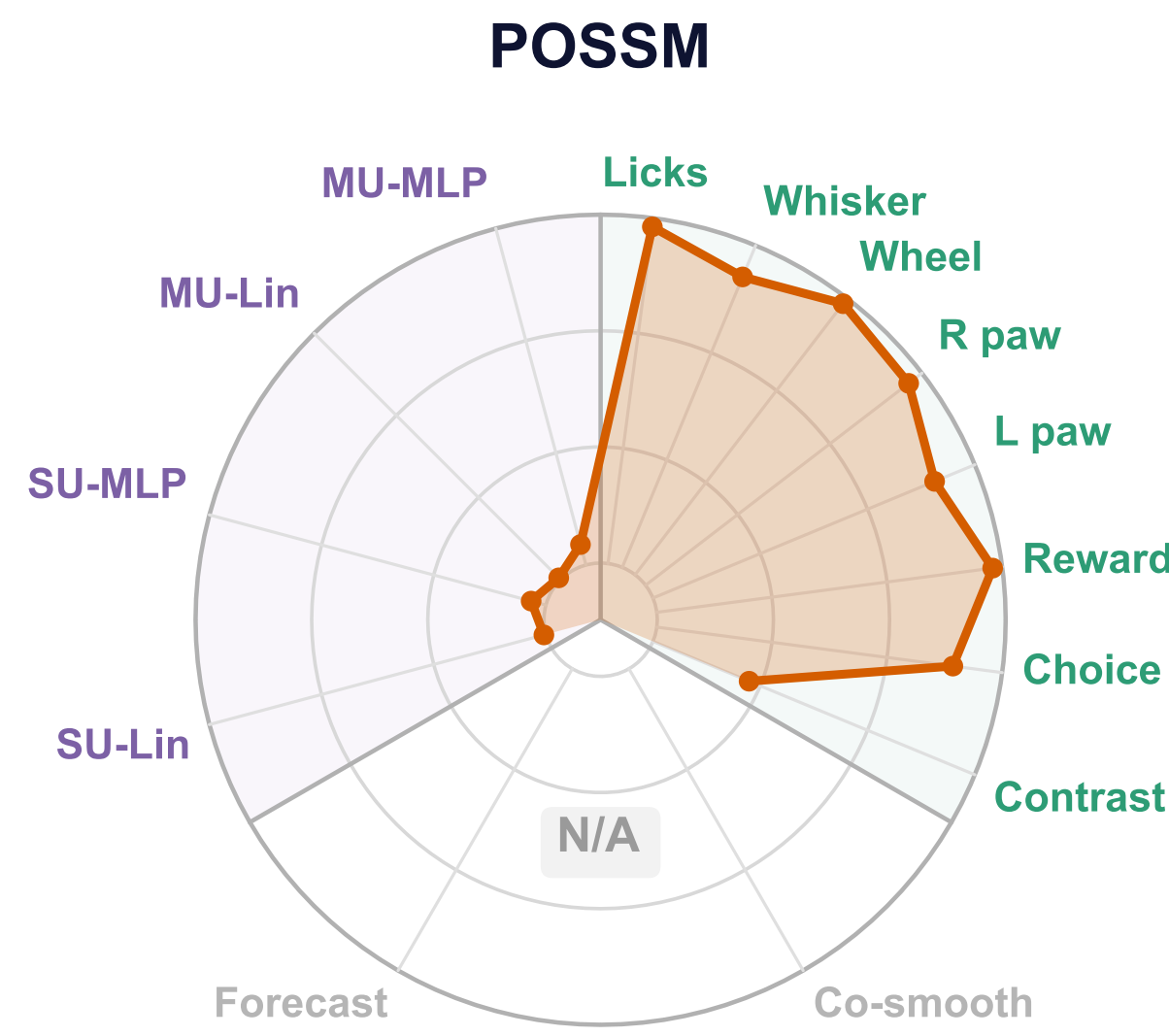
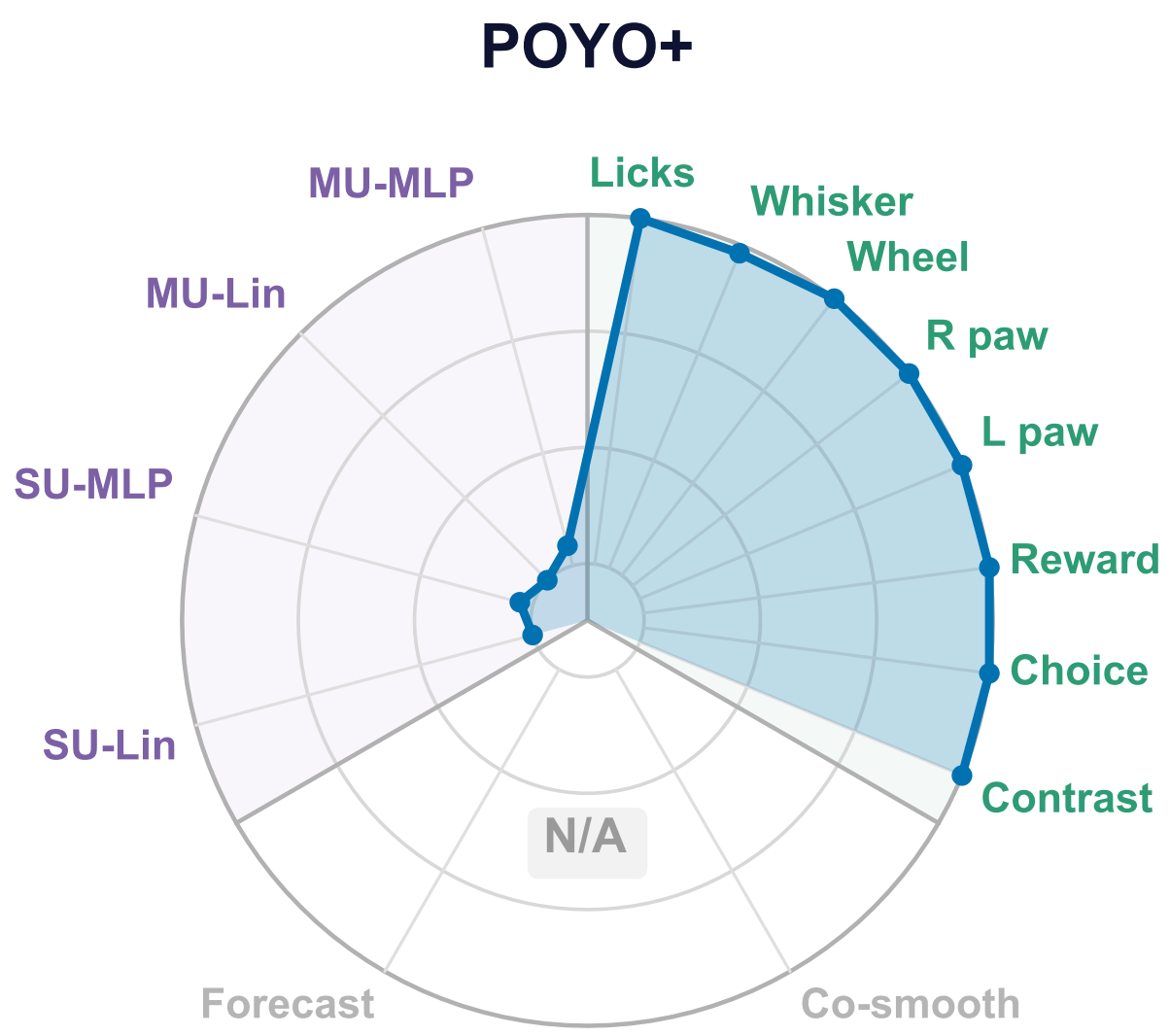
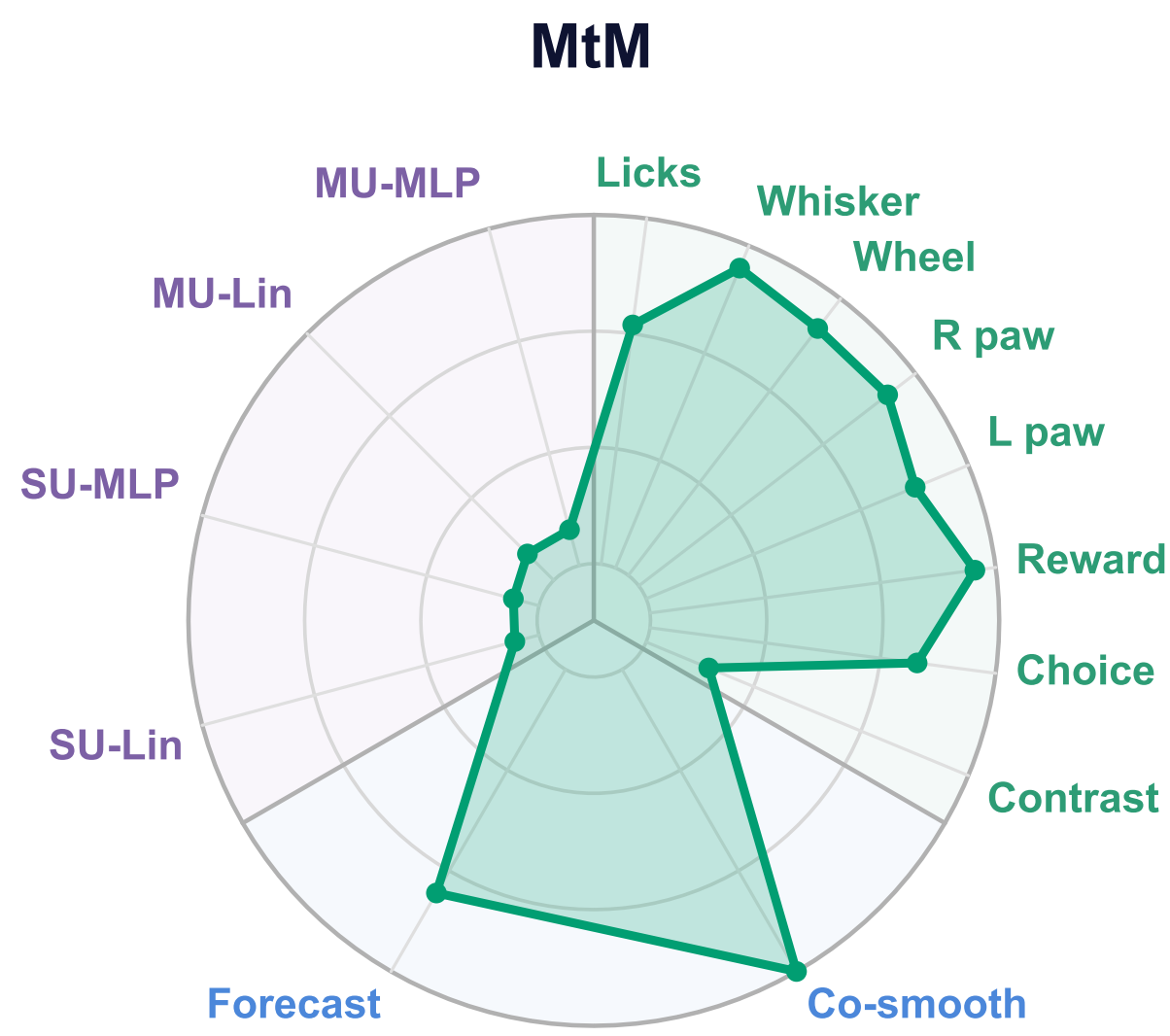
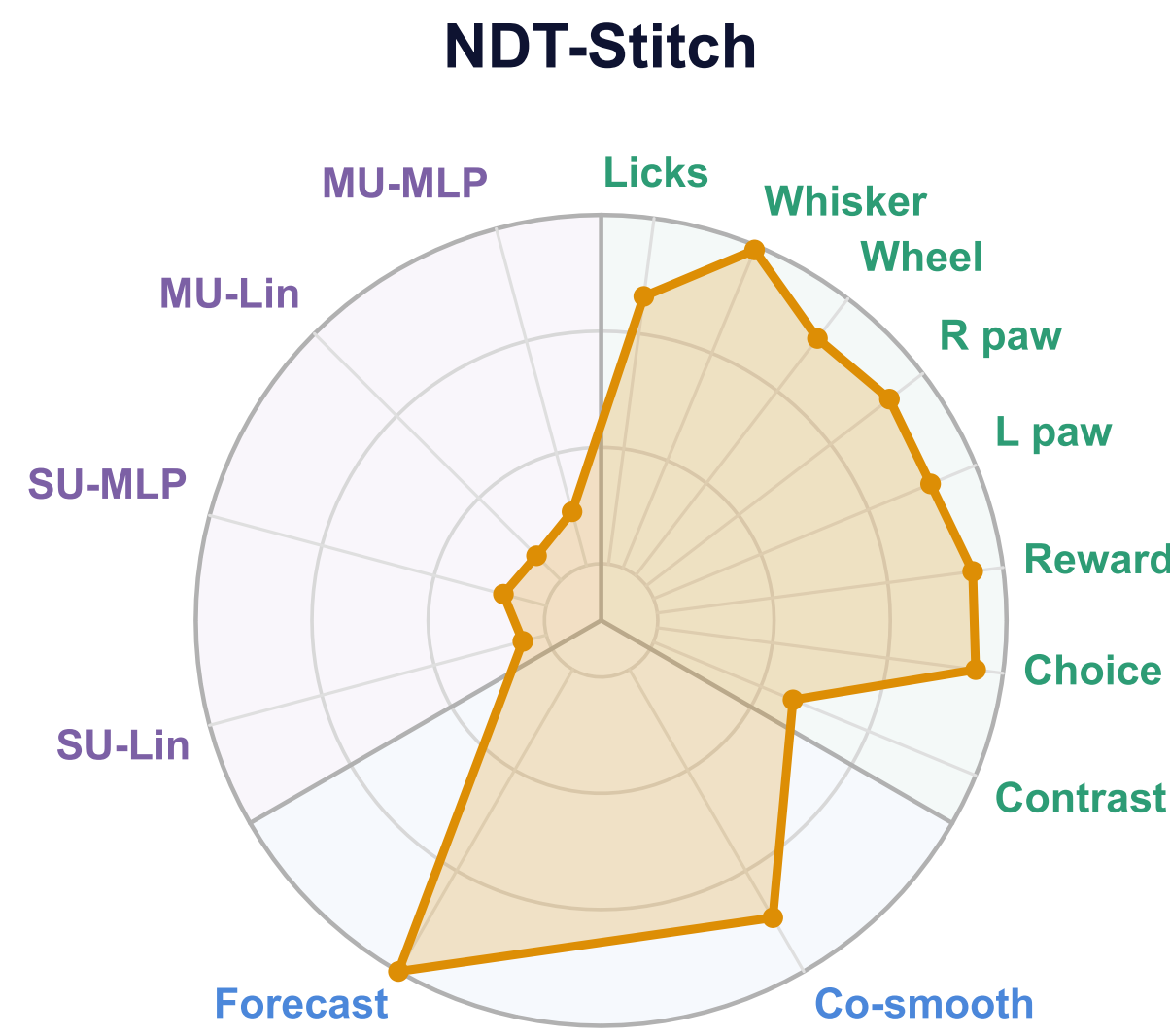
# Baseline results on TS3 (Region Identification)



# Pretraining Helps Across Tasks



# No free lunch (yet!)



# Question of Interest & Future Directions

## Questions to explore

- **Scale:** *Does adding more animals and sessions keep improving the model?*
- **Curriculum:** *Does the order of training data matter (e.g., simple tasks first)?*
- **Data quality:** *How much do QC choices (unit sorting, filtering) affect results?*
- **Generalization:** *Can one model generalize well across all task suites?*

## Future directions

- **Localization:** Finer brain-region classification and probe-location prediction
- **LFP:** Extend beyond spikes to local field potentials
- **Encoding:** Predict neural activity from stimuli and behavior

# Ways to participate

*Less time processing data and building evaluation, more time on modeling ideas*



Get started

[brainbench-org.github.io/ibl-bwb-project/](https://brainbench-org.github.io/ibl-bwb-project/)

**Leaderboard**  
Public, completed submissions scored against the benchmark's evaluation set.

**Suites** ⌵  
Select the suites or a combination of tasks to include in the ranking.

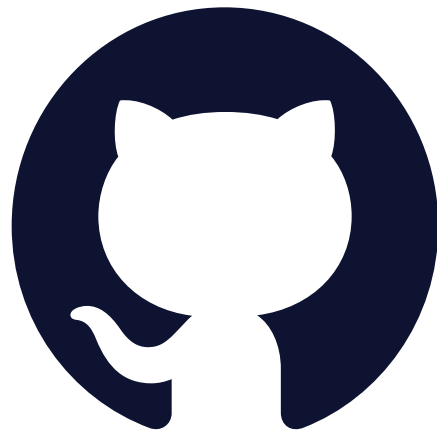
**Filters** ⌵  
Apply filters to restrict models or tasks included in the ranking. Click more filters to show all

2 models ranked across 11 tasks, 20 unranked

| # | Model                                                        | Choice<br>BAcc   | Left paw<br>R <sup>2</sup> | Licking<br>Poisson D <sup>2</sup> | Reward<br>BAcc   | Right paw<br>R <sup>2</sup> | Stimulus<br>BAcc | Wheel<br>R <sup>2</sup> | Whisker<br>R <sup>2</sup> | Co-smooth<br>Poisson D <sup>2</sup> | Forecast<br>Poisson D <sup>2</sup> | Cosmos<br>Macro F1 |
|---|--------------------------------------------------------------|------------------|----------------------------|-----------------------------------|------------------|-----------------------------|------------------|-------------------------|---------------------------|-------------------------------------|------------------------------------|--------------------|
| 1 | NDT-Stitch<br><small>Pretrained<br/>Brain Wide Bench</small> | 0.674<br>± 0.001 | 0.172<br>± 0.003           | 0.525<br>± 0.002                  | 0.863<br>± 0.006 | 0.219<br>± 0.005            | 0.229<br>± 0.001 | 0.310<br>± 0.002        | 0.384<br>± 0.003          | 0.157<br>± 0.001                    | 0.177<br>± 0.000                   | 0.187<br>± 0.008   |
| 2 | MtM<br><small>Pretrained<br/>Brain Wide Bench</small>        | 0.646<br>± 0.004 | 0.167<br>± 0.003           | 0.469<br>± 0.004                  | 0.874<br>± 0.003 | 0.224<br>± 0.003            | 0.213<br>± 0.002 | 0.323<br>± 0.006        | 0.362<br>± 0.004          | 0.191<br>± 0.001                    | 0.131<br>± 0.001                   | 0.159<br>± 0.009   |

Submit  
your model

## Open Science: Path to General Neural Models



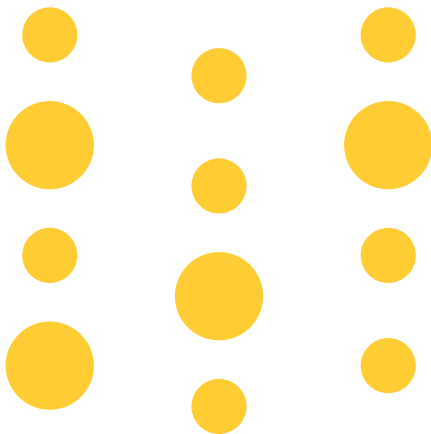
[github.com/brainbench-org/ibl-bwb](https://github.com/brainbench-org/ibl-bwb)



[torchbrain.org](https://torchbrain.org)



[hf.co/collections/nerdslab/ibl-bwb](https://hf.co/collections/nerdslab/ibl-bwb)



[wandb.ai/ibl-bwb/projects](https://wandb.ai/ibl-bwb/projects)

# Questions?

